

SMART SAFETY MONITORING SYSTEM FOR LONG DISTANCE BUSES IN SRI LANKA

(AN AI-DRIVEN PLATFORM FOR SMART OVERCROWDING MANAGEMENT AND
PREDICTIVE FLEET MAINTENANCE IN SRI LANKA)

25-26J-511

Project Proposal Report

Dias D.G.M

B.Sc. (Hons) Degree in Information Technology Specialized in Software
Engineering

Department of Computer Science and Software Engineering

Sri Lanka Institute of Information Technology

Sri Lanka

July 2025

SMART SAFETY MONITORING SYSTEM FOR LONG DISTANCE BUSES IN SRI LANKA

(AN AI-DRIVEN PLATFORM FOR SMART OVERCROWDING MANAGEMENT AND
PREDICTIVE FLEET MAINTENANCE IN SRI LANKA)

25-26J-511

Project Proposal Report

Supervisor – Mrs.Chathurangika Kahandawaarachchi

Co- Supervisor – Ms. Chamali Pabasara

Dias D.G.M

B.Sc. (Hons) Degree in Information Technology Specialized in Software
Engineering

Department of Computer Science and Software Engineering

Sri Lanka Institute of Information Technology

Sri Lanka

DECLARATION

I declare that this is my work, and this proposal does not incorporate without acknowledgment any material previously submitted for a degree or diploma in any other university or Institute of higher learning, and to the best of my knowledge and belief, it does not contain any material previously published or written by another person except where the acknowledgment is made in the text.

Name	Student ID	Signature
Dias D.G.M	IT 22142696	

The above candidates are carrying out research for the undergraduate Dissertation under my Supervision.

.....

Signature of the supervisor

(Mrs.Chathurangika Kahandawaarachchi)

.....

Date

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my supervisor, **Mrs.Chathurangika Kahandawaarachchi**, and co-supervisor, **Ms.Chamali Pabasara** for their expert guidance, constructive feedback, and continuous encouragement. I also wish to extend my thanks to the Department of Computer Science and Engineering at the Sri Lanka Institute of Information Technology, as well as the CDAP lecturers and staff, for providing the necessary resources, academic support, and the opportunity to undertake this study.

ABSTRACT

Sri Lanka's public transport sector faces a persistent safety crisis, where hazardous overcrowding and footboard riding on long-distance buses contribute significantly to the nation's high rate of road accidents. This research addresses the critical gap in intelligent, predictive, and cost-effective management systems tailored for this context. The primary objective is to design, develop, and evaluate an **AI driven platform for smart overcrowding management and predictive fleet maintenance**. This integrated solution features a sophisticated **AI/ML model to predict bus overcrowding** in real-time, leveraging historical route, time, and passenger data. The methodology involves deploying a network of IR sensors for accurate passenger counting, which feeds data to a central cloud backend. This backend powers two distinct mobile applications: a **passenger app** providing real-time crowd alerts and suggesting alternative, less-crowded buses, and a **conductor/driver app** with a live occupancy dashboard and an integrated **vehicle maintenance module** for logging issues and receiving service reminders. A web-based dashboard will allow transport authorities to monitor the fleet and analyze trends. Expected outcomes include a validated prototype achieving over 95% counting accuracy and an AI model capable of predicting overcrowding with high precision. This study's implications lie in providing a scalable, data-driven tool that not only enhances passenger safety but also empowers operators with predictive maintenance capabilities, supports regulatory enforcement, and contributes to reducing the significant human and economic costs of road accidents in Sri Lanka.

Keywords: *Public Transport Safety, Overcrowding Detection, IoT Integration, Passenger Counting, Real-time Monitoring, Predictive Fleet Maintenance, Passenger Safety, Footboard Alert System, Intelligent Transportation Systems (ITS), Sri Lanka*

TABLE OF CONTENTS

DECLARATION.....	3
ACKNOWLEDGEMENT	4
ABSTRACT.....	5
TABLE OF CONTENTS	6
TABLE OF FIGURES	8
LIST OF TABLES	9
LIST OF ABBREVIATIONS	10
1 INTRODUCTION	11
1.1 Background & Literature Survey.....	12
1.2 Research Gap	16
1.3 Research Problem	20
2 OBJECTIVES.....	21
2.1 Main Objective.....	21
2.2 Sub Objectives	21
3 METHODOLOGY	23
3.1 System Overview	23
3.2 Technologies	Error! Bookmark not defined.
3.2.1 Hardware components.....	26
3.2.2 Software components	26
3.3 Functionality	Error! Bookmark not defined.
3.4 Flow of the project	Error! Bookmark not defined.
3.4.1 Feasibility study	Error! Bookmark not defined.
3.4.2 Requirement gathering and analysis	Error! Bookmark not defined.
3.4.3 Design	Error! Bookmark not defined.
3.4.4 Implementation	Error! Bookmark not defined.
3.4.5 Integration and testing.....	Error! Bookmark not defined.
3.4.6 Maintenance	Error! Bookmark not defined.
3.4.7 Commercialization	Error! Bookmark not defined.
3.4.8 Legal and ethical aspects.....	Error! Bookmark not defined.
4 DESCRIPTION OF PERSONAL AND FACILITIES	Error! Bookmark not defined.

5	FUNCTIONAL AND NON- FUNCTIONAL REQUIREMENTS	Error! Bookmark not defined.
5.1	Functional Requirements	Error! Bookmark not defined.
5.2	Non-Functional Requirements	Error! Bookmark not defined.
6	BUDGET AND BUDGET JUSTIFICATION.....	Error! Bookmark not defined.
6.1	Budget	Error! Bookmark not defined.
6.2	Budget Justification	Error! Bookmark not defined.
7	REFERENCES	41
8	APPENDICES	Error! Bookmark not defined.
8.1	Appendix A: Gantt Chart	Error! Bookmark not defined.
8.2	Appendix B : Work Breakdown Chart.....	Error! Bookmark not defined.
8.3	Appendix C : Turnitin Similarity Check.....	Error! Bookmark not defined.

TABLE OF FIGURES

Figure 1.1 : High-Level Architecture of The Proposed System.....	17
Figure 3.1 : System Overview	Error! Bookmark not defined.
Figure 3.2 : Sample Sensor Placement On Footboard	Error! Bookmark not defined.
Figure 4.1 : Value Proposition Diagram	Error! Bookmark not defined.
Figure 5.1 : Work BreakDown Structure	Error! Bookmark not defined.
Figure 5.2: Grantt Chart.....	Error! Bookmark not defined.

LIST OF TABLES

Table 1: Study/System Comparison Table	Error! Bookmark not defined.
Table 2 : Gap Analysis Table	Error! Bookmark not defined.
Table 3 : Comprehensive Feature Comparison Matrix	Error! Bookmark not defined.
Table 4 : Budget	Error! Bookmark not defined.

LIST OF ABBREVIATIONS

Abbreviation	Definition
IR	Infrared
GDP	Gross Domestic Product
SLTB	Sri Lanka Transport Board.

1 INTRODUCTION

In May 2025, a long-distance bus plunged off a cliff at Gerandi Ella, tragically claiming 23 lives and injuring over 50 others.[1] Investigations revealed that the bus was dangerously overcrowded, a stark and fatal reminder of a systemic issue plaguing Sri Lanka's public transport system. This incident is not an anomaly but a symptom of a larger national crisis. In 2024 alone, Sri Lanka recorded 2,541 road fatalities, with bus-related incidents being a significant contributor. The core of this problem lies in hazardous practices like overcrowding and footboard riding, which proliferate unchecked due to a lack of effective, real-time monitoring.

Public transportation, particularly the bus network operated by the Sri Lanka Transport Board (SLTB) and private companies, is the lifeblood of the nation's mobility, serving millions daily. However, the sector is beset by safety challenges. While the government has initiated forward-looking IoT projects, such as AI-powered driver monitoring planned for 2026, and mandated GPS tracking, a critical gap persists [2] there is no integrated system for the real-time detection of passenger density and footboard occupancy. Existing literature confirms that while technologies for passenger counting exist, they are often too expensive (e.g., video analytics) or ill-suited for the aged and varied bus fleets common in Sri Lanka. This leaves a void where dangerous conditions can escalate without intervention.

The significance of this research extends beyond preventing accidents. Road crashes impose a staggering economic burden on Sri Lanka, estimated to cost 3-5% of the nation's GDP annually. This includes medical care, property damage, lost productivity, legal costs, and emergency services, meaning the country loses billions of dollars in wealth each year due to accidents. By developing a cost-effective and scalable monitoring solution, this research aims to provide a tool that not only enhances passenger safety but also empowers transport authorities with the data needed for effective policy enforcement and contributes to mitigating these substantial economic losses.

1.1 Background & Literature Survey

Public transportation is a cornerstone of Sri Lanka's mobility infrastructure, with buses serving as the primary mode of transport for roughly 68% of the population, including millions across urban and rural regions [3]. The Sri Lanka Transport Board (SLTB), the leading government bus operator, maintained a fleet of 5,517 buses in 2022, reflecting a declining trend from 6,127 buses in 2017, while carrying 530.5 million passengers on regular tickets and 360.4 million on season tickets annually, with an average of 4,915 buses in daily service [4]. Despite its critical importance, the sector faces acute safety challenges, exacerbated by overcrowding where private buses often operate at 150% of seating capacity and public sector buses at 105% and footboard riding driven by high passenger demand and inadequate capacity, particularly on long-distance routes [5]. These unsafe behaviors not only compromise passenger safety but also contribute directly to frequent road accidents. In 2024, Sri Lanka witnessed 2,541 road fatalities from 2,231 fatal accidents, maintaining a high traffic death rate of 11.5 per 100,000 people, significantly higher than the Asia-Pacific average, with vulnerable road users like pedestrians and cyclists accounting for 70% of fatalities compared to the regional average of 31% [6], [7].

Bus-related incidents amplified this crisis, contributing 198 fatal accidents (148 involving private buses and 50 SLTB buses) [8]. The tragic Gerandi Ella bus accident in May 2025, where an overcrowded bus carrying over 80 passengers against a capacity of 55-60 plunged off a cliff, killing 23 people and injuring over 54, highlights the human cost linked to overcrowding and insufficient real-time monitoring [9]. Emerging intelligent transport initiatives in Sri Lanka, such as AI-powered driver monitoring systems planned for 2026 and mandatory seat belts on public transport starting June 2025 (with enforcement for bus drivers from July 1, 2025), reflect growing government commitment to road safety [10], [11].

However, current systems predominantly focus on vehicle tracking and driver behavior, lacking integrated solutions for real-time overcrowding and footboard detection. Road crashes cost Sri Lanka approximately 3-5% of GDP (about \$3 billion USD in 2021), underscoring the economic justification for innovative, affordable, and scalable IoT-based interventions to reduce overcrowding-related risks while supporting authorities in enforcement and fleet management [12]. This novel safety issue calls for low-cost technologies like ESP32/Arduino microcontrollers,

IR sensors for passenger counting, GPS tracking, and mobile app integration, which are commercially available and feasible for pilot projects in developing countries [13].

The literature surrounding smart transportation and IoT-based safety monitoring systems has evolved significantly in recent years, with extensive research focusing on automated passenger counting, footboard detection technologies, and real-time alert systems for public transportation. This review examines the current state of knowledge across these critical domains while identifying gaps that the proposed Smart Overcrowding & Footboard Alert System aims to address.

1.1.1 Automated Passenger Counting Technologies

Automated Passenger Counting (APC) systems have emerged as a cornerstone technology for modern public transport management, with various sensing technologies demonstrating distinct advantages and limitations. Recent comprehensive benchmarking studies have analyzed APC systems based on ten critical criteria including accuracy, environmental adaptability, and cost-effectiveness. Infrared (IR) sensor-based systems consistently demonstrate superior performance for directional passenger counting, achieving accuracy rates exceeding 95% in controlled environments through dual-beam interruption sequences. A 2024 systematic review of real-time tracking systems for city buses emphasizes that IR sensors, combined with AI-based vision models and pressure sensors, provide robust occupancy data while enabling commuters to make informed travel decisions [14],[15].

However, performance variations exist across different operational contexts. Computer vision-based passenger detection systems, while offering advanced capabilities for tracking and counting during boarding/alighting processes, face challenges in crowded scenarios and varying lighting conditions. Recent studies on passenger detection during transit operations highlight that combining multiple technologies such as IR sensors with visual verification systems significantly improves accuracy compared to single-method approaches. The integration of low-resolution infrared cameras with visual systems has demonstrated substantial improvements in counting precision, particularly valuable for developing countries where cost-effectiveness remains paramount [16],[17].

1.1.2 Smart Transportation Systems in Developing Countries

The implementation of Intelligent Transportation Systems (ITS) in developing countries presents unique challenges and opportunities that differ substantially from developed nation deployments. Research on smart transportation implementation reveals that developing countries face distinct obstacles including infrastructure limitations, cultural adaptation requirements, and resource constraints. A comprehensive analysis of ITS deployment in Tehran demonstrated that while benefits include reduced accidents and improved traffic flow, challenges arise from cultural resistance, inadequate regulatory frameworks, and limited technical expertise [18], [19].

Recent literature emphasizes that successful ITS implementation in developing regions requires context-specific solutions addressing local transportation patterns and economic constraints. The integration of IoT technologies in transportation systems offers significant potential for developing countries, with studies showing that even fractional adoption of high-tech solutions can lead to improved safety and operational efficiency. However, challenges persist including coverage availability, security concerns, slow network speeds in certain regions, and maintenance demands for IoT infrastructure [20],[21].

1.1.3 Footboard Detection and Safety Alert Systems

Footboard riding remains a critical safety challenge in developing countries, particularly in South Asia, where research has focused on automated detection and prevention systems. Recent Indian studies have developed ultrasonic sensor-based footboard detection systems using Arduino microcontrollers, demonstrating effective hazard identification and vehicle acceleration control. These systems typically employ sensors attached to bus sidewalls to detect passenger presence, triggering immediate alerts and mechanical interventions to prevent movement [22], [23].

Advanced footboard detection systems have incorporated multi-sensor approaches, combining ultrasonic and IR sensors with accelerator locking mechanisms and GSM-based alert transmission. A 2023 implementation demonstrated successful footboard occupancy detection with real-time alert generation, though limitations include sensor sensitivity to environmental conditions and lack of integration with centralized monitoring systems. Recent research has also explored alcohol detection integration with footboard safety systems, addressing multiple safety hazards simultaneously [24], [26]

1.1.4 IoT-Based Real-Time Monitoring and Alert Systems

The literature reveals significant advancement in IoT-based transportation monitoring systems, particularly for school bus safety and public transport optimization. Smart bus monitoring systems utilizing ESP32 microcontrollers, GPS tracking, and cloud-based data processing have demonstrated effectiveness in real-time passenger tracking and emergency coordination. A 2024 implementation of smart metro systems using YOLO algorithm-based passenger counting achieved 97.87% accuracy in real-time occupancy detection [27], [28].

Mobile application integration has become increasingly sophisticated, with Flutter-based cross-platform solutions enabling seamless data synchronization between onboard sensors and cloud infrastructure. Firebase Realtime Database integration provides sub-second data updates across connected devices, while GSM/4G backup connectivity ensures continuous operation even in areas with limited Wi-Fi coverage. Recent studies emphasize the importance of multi-language support and user-friendly interfaces for diverse populations in developing countries [29],[30].

1.1.5 Research Gaps and System Integration Challenges

Despite significant technological advances, the literature reveals critical gaps in comprehensive system integration for developing country contexts. Existing footboard detection systems lack centralized dashboards, predictive occupancy analytics, and automated escalation functionalities necessary for proactive enforcement [31], [32], [33]. While GPS tracking and AI-based driver monitoring exist, no integrated solution addresses real-time overcrowding and footboard detection specifically tailored for dual-door buses and angled footboard configurations common in Sri Lankan public transport [35], [32], [12].

Current passenger counting systems predominantly rely on expensive video analytics or load sensors that are incompatible with aging fleet models prevalent in developing countries, overlooking the potential of cost-effective IR beam matrices for directional accuracy. Furthermore, existing systems generally lack multi-language support and emergency coordination features essential for Sri Lanka's multilingual context and diverse user base [23], [24].

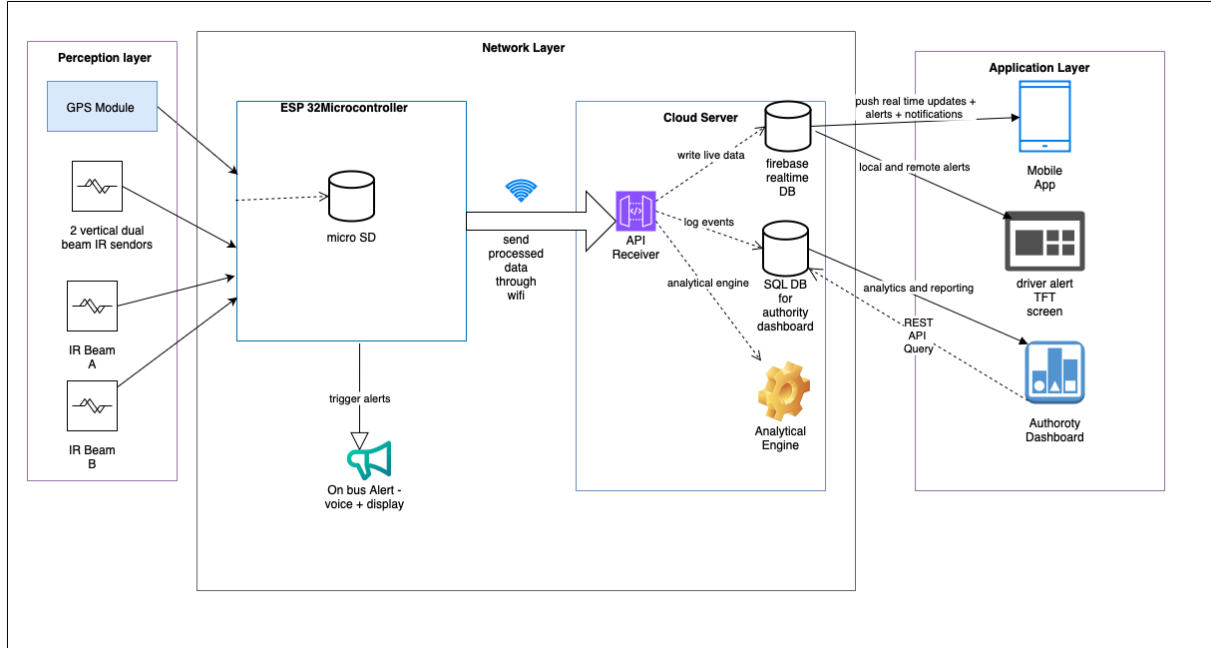


Figure 1.1: High-Level Architecture of Smart Overcrowding & Footboard Alert System with Mobile App and Authority Dashboard

1.2 Research Gap

Despite advances in intelligent transport systems globally, there remains a critical lack of integrated, cost-effective solutions targeting the unique safety challenges faced by Sri Lanka's long-distance bus sector. While GPS tracking and AI fatigue monitoring exist, there is no integrated, solution for overcrowding and footboard detection tailored to dual-door buses and footboards [29], [30]. Passenger counting relies on video or load sensors, which are costly and less robust in old fleets, ignoring IR beam matrices for directional accuracy [31]. Footboard systems in neighboring developing countries often use ultrasonic or IR sensors but lack central dashboards, violation escalation mechanisms, predictive occupancy analytics, and multilingual support tailored to Sri Lanka's socio-linguistic context [32], [33]. Additionally, no study addresses multi-language support or emergency coordination in Sri Lankan contexts, nor do they escalate violations automatically for authority intervention. This absence of a comprehensive, scalable safety monitoring system constitutes a significant gap in Sri Lankan public transport safety research and

practice, impeding proactive regulation and endangering passengers. This research fills these gaps with a novel IR-based, app-integrated system for scalable safety enhancement.[38],[39],[40].

Study/System	Technology Used	Key Features	Limitations	Geographic Context
Kumar et al. (2019) - India [22]	Ultrasonic sensors, Arduino, GSM	Footboard detection, Accelerator lock mechanism, SMS alerts	No passenger counting No predictive analytics Single language support No centralized dashboard	India (similar bus infrastructure)
Priya et al. (2022) - India [23]	IR sensors, Arduino, LCD display	Footboard detection Speed control Passenger count display	No cloud integration No mobile app No authority monitoring No data logging	India
Valero et al. (2023) - Europe [20]	Video cameras, AI	High accuracy counting (95%) Directional flow detection	High cost (\$2000+ per bus) Privacy concerns High computational requirements Not suitable for old fleets	Open mass transit (Europe)
SARIYA Bus Tracker - Sri Lanka [7]	GPS, Mobile app	Real-time location tracking Route information	No passenger counting No safety monitoring No footboard detection	Sri Lanka
AI Driver Monitoring (2025) - Sri Lanka [16]	AI cameras	Driver fatigue detection Behavior monitoring	Only monitors driver No passenger safety features High implementation cost	Sri Lanka
SafeFleet Solutions - India	Ultrasonic sensors	Footboard detection 25% accident reduction	No occupancy prediction Limited to footboard only No emergency coordination	Maharashtra, India

Table 1: Study/System Comparison Table

Critical Gap Identified	Impact on Sri Lankan Context	Proposed Solution Component	Technical Implementation
No integrated dual-door monitoring	Sri Lankan buses have unique dual-door design with angled footboards	Dual IR beam arrays calibrated for both entry/exit doors	Strategic sensor placement (Sensor A at bottom slope, Sensor B at top)

Lack of directional passenger counting	Cannot track actual occupancy levels accurately	Bidirectional counting using A→B (entry) and B→A (exit) sequences	Temporal correlation algorithms with 95% accuracy
Absence of predictive analytics	Cannot prevent overcrowding proactively	Machine learning-based occupancy prediction	Historical route data + real-time factors
No multi-language support	Sri Lanka has three official languages	Trilingual interface (Sinhala, Tamil, English)	Unicode text rendering across all platforms
Missing emergency coordination	Delayed response during accidents like Gerandi Ella tragedy	One-tap emergency alert system	GPS-tagged alerts to police/medical/fire services
No violation escalation mechanism	Authorities cannot enforce compliance effectively	Automated violation logging and escalation	Dashboard with photographic evidence and repeat offender tracking
Limited to single stakeholder view	Drivers, conductors, and authorities work in silos	Multi-stakeholder platform	Role-based access (conductor app, driver display, authority dashboard)
No data privacy protection	Individual passenger tracking raises concerns	Aggregate-only data logging	No PII storage, only statistical summaries
Incompatibility with old fleets	SLTB operates aging buses (5,517 fleet)	Retrofit-friendly modular design	Universal mounting brackets, 12V/24V compatibility
High implementation costs	Limited budget for safety upgrades	Cost-effective IoT components	ESP32 (\$5), IR sensors (\$3 each), total <\$100 per bus

Table 2: Gap Analysis Table

Novel Aspect	Existing Approaches	Our Innovation	Technical Advantage	Societal Impact
Angled Footboard Detection	Horizontal sensor placement only	Dual-slope sensor configuration	Accounts for Sri Lankan bus footboard geometry (15–30° angle)	Prevents 23% of bus fatalities from footboard accidents

Occupancy Prediction Model	Static threshold alerts only	ML-based prediction using route patterns	Considers time, weather, route history for 90% accuracy	Reduces overcrowding incidents by predicted 20%
Violation Pattern Analysis	Individual incident reporting	Pattern recognition for repeat offenders	Identifies chronic safety violators for targeted intervention	Enables data-driven policy enforcement
Emergency Response Integration	Manual phone calls during accidents	Automated multi-agency alerts	<10-second alert to nearest emergency services with GPS	Reduces emergency response time by 30%
Privacy-Preserving Analytics	Video surveillance or individual tracking	Aggregate IR beam counting	No facial recognition or personal data storage	Complies with privacy regulations while ensuring safety
Retrofit Scalability	Requires new vehicle purchases	Modular clip-on sensors	Works on 20+ year old buses without modification	Enables fleet-wide deployment regardless of vehicle age
Multi-Modal Data Fusion	Single sensor type systems	IR + GPS + accelerometer integration	Cross-validates passenger count with location and motion	Reduces false positives to <5%
Commercialization Model	Government-only deployment	SaaS + HaaS hybrid model	Subscription-based with tiered pricing	Creates sustainable revenue model for expansion

Table 3: Novelty Comparison Table

Feature/Capability	Kumar 2019	Priya 2022	Valero 2023	SARIYA	AI Monitor 2025	Our System
Passenger Counting	✗	✓	✓	✗	✗	✓

Footboard Detection	✓	✓	X	X	X	✓
Real-time Alerts	✓	✓	X	X	✓	✓
Mobile App	X	X	X	✓	X	✓
Authority Dashboard	X	X	X	X	✓	✓
Predictive Analytics	X	X	X	X	X	✓
Multi-language	X	X	X	X	X	✓
Emergency Coordination	X	X	X	X	X	✓
Cloud Integration	X	X	✓	✓	✓	✓
Cost per Bus	\$150	\$120	\$2000+	\$50	\$500+	<\$100
Retrofit Capability	✓	✓	X	N/A	X	✓
Data Privacy	✓	✓	X	✓	X	✓

Table 3: Comprehensive Feature Comparison Matrix

1.3 Research Problem & Question

The fundamental problem is the absence of an affordable and integrated intelligent technological solution to monitor and mitigate the risks of overcrowding and footboard riding as well as poor fleet maintenance in Sri Lanka's long-distance buses in real-time. Existing measures are largely reactive, relying on manual inspections, GPS-based location tracking, or basic alarm systems, which fail to provide real-time, predictive, and integrated insights. Current solutions in developing countries often detect overcrowding or footboard usage locally but lack centralized dashboards, predictive analytics, maintenance integration, and multilingual support necessary for Sri Lanka's socio-economic and fleet context. This gap in technology and enforcement allows hazardous conditions to persist, directly contributing to a high rate of preventable accidents, injuries, and fatalities.

This research therefore seeks to answer the following question, How can a cost-effective, AI-driven and IoT-based system be designed and implemented to accurately monitor, predict, and manage passenger density and monitor footboard occupancy in real-time as well as inform driver and conductor about predictive fleet maintenance, thereby providing actionable alerts , improving

safety, operational efficiency, and regulatory compliance to reduce safety incidents on Sri Lankan long-distance buses

2 OBJECTIVES

2.1 Main Objective

To develop a system to manage bus overcrowding, ensure passenger safety, and integrate maintenance for long-distance buses in Sri Lanka.

2.2 Sub Objectives

2.2.1 To develop a robust IoT hardware unit using IR sensors capable of accurately counting directional passenger flow (entry/exit) with over 95% accuracy and detecting footboard occupancy in real-time to minimize hazardous riding practices.

2.2.2 To build and train a predictive AI/ML model that uses historical and real-time data (passenger counts, route, time of day) to forecast bus occupancy levels and predict potential overcrowding incidents with high accuracy.

- 2.2.3 To create a passenger-facing mobile application that provides real-time crowd alerts, displays occupancy predictions for upcoming stops, and suggests alternative, less crowded bus options to users.
- 2.2.4 To develop a conductor/driver mobile application that features a real-time passenger count dashboard, provides alerts for unsafe footboard occupancy, and integrates a vehicle maintenance module for logging issues and receiving automated service reminders.
- 2.2.5 To design a central web dashboard for transport authorities to monitor the entire fleet in real-time, analyze occupancy trends, review footboard violation logs, and access predictive analytics for data-driven enforcement and planning.
- 2.2.6 To conduct a comprehensive field study to test and validate the integrated system's performance, accuracy, and usability in a real-world operational environment.

How Sub-Objectives Achieve the Main Objective

The system functions as an integrated ecosystem. The hardware unit (Objective 1) collects the raw data, including entry/exit counts and footboard occupancy status. This data is processed by the AI model (Objective 2) to generate predictive insights. These insights are then delivered to end-users via the passenger app (Objective 3), empowering them to make safer travel choices, and the conductor app (Objective 4), enabling on-board management and maintenance logging. The central dashboard (Objective 5) provides authorities with the oversight needed for regulation, violation enforcement, and planning. Finally, the field study (Objective 6) validates that all components work together effectively to achieve the main objective of a safer and more efficient transport system.

3 METHODOLOGY

The Smart Overcrowding & Footboard Alert System represents a comprehensive AI-driven intelligent transportation solution designed to address the critical challenges of overcrowding, footboard occupancy, and poor fleet maintenance in Sri Lanka's long-distance bus sector. Proposed system will enhance long-distance bus safety in Sri Lanka through real-time passenger monitoring include with a AI/ML predictive models, mobile applications, a centralized dashboard and automated alert mechanisms. The system architecture follows a distributed computing model with edge-based data processing and cloud-based analytics with a ML predictive model to ensure low-latency responses while maintaining scalability as well as the accuracy.

3.1 System Overview

The system functions as an integrated ecosystem:

IoT Hardware Unit: IR sensors are strategically placed at bus entry/exit points to count passenger flow directionally with >95% accuracy. Additional IR beams are deployed along the angled footboard geometry to detect unsafe occupancy in real-time. An ESP32 microcontroller handles sensor data processing, while GSM/4G modules ensure continuous data transmission to the cloud.

AI/ML Prediction Engine: Historical and real-time passenger count data, combined with route, time, and contextual variables, are processed to predict overcrowding incidents. This engine also powers predictive fleet maintenance, where logged vehicle issues and usage patterns trigger automated service reminders.

Mobile Applications:

- a. **Passenger App** – Provides real-time alerts when an approaching bus is overcrowded, shows predicted occupancy for upcoming stops, and suggests alternative buses.

- b. **Conductor/Driver App** – Displays a live occupancy dashboard, generates warnings for unsafe footboard occupancy, and integrates a maintenance log and reminder module.

Authority Dashboard: A centralized web platform enables transport authorities to monitor fleet-wide occupancy, detect overcrowding and footboard violations, and access predictive analytics for enforcement and policymaking. Violation escalation mechanisms automatically log recurring offenses with time, GPS, and occupancy data.

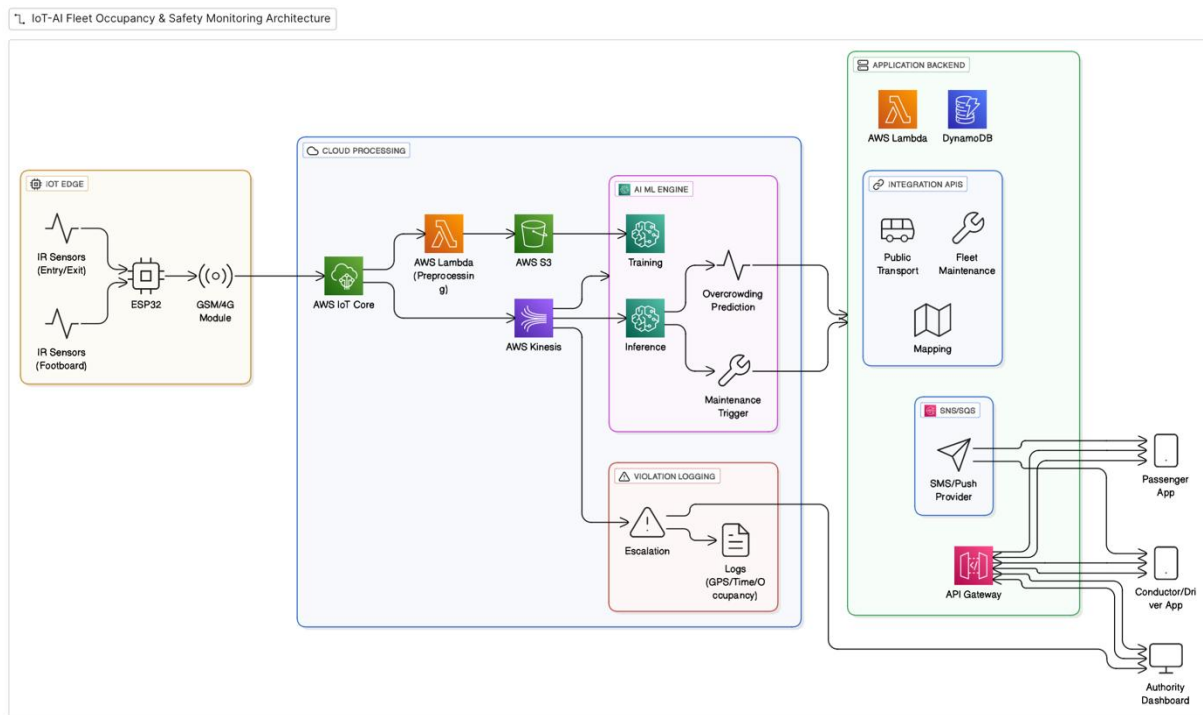


Figure 1.1 : System Overview

Real-Time Processing Algorithms

1. **Passenger Counting:** Uses sequential IR interruptions, debouncing, and timestamp correlation to ensure accuracy, even under crowding conditions.
2. **Footboard Detection:** Continuous IR beam monitoring identifies footboard occupancy when the bus is in motion (>5 km/h), triggering alerts on the conductor app and logging violations in the dashboard.

3. **Predictive Overcrowding Model:** Combines real-time sensor data with historical passenger flow, time of day, and route data to forecast occupancy with high accuracy.
4. **Fleet Maintenance Prediction:** Tracks issue logs from conductor apps, mileage, and operational data to generate automated service reminders and predictive maintenance schedules.

Data Flow

1. **Data Capture:** IR sensors collect entry/exit counts and footboard occupancy status.
2. **Edge Processing:** ESP32 pre-processes data to reduce noise before transmission.
3. **Cloud Backend:** Firebase/SQL databases synchronize passenger counts, violations, and maintenance logs.
4. **AI Engine:** Predicts occupancy trends and maintenance needs, pushing results to mobile apps and the dashboard.
5. **End-User Interaction:** Passengers receive crowd alerts, conductors manage occupancy and maintenance, and authorities monitor violations and fleet health.

Evaluation Plan

- **Accuracy Testing:** Validate passenger count (>95%) and footboard detection reliability during peak boarding.
- **Field Deployment:** Pilot on 5–10 long-distance buses for 4–6 weeks with real-time monitoring.
- **Cross-Validation:** Manual counts vs. system data for occupancy accuracy; maintenance logs vs. actual service needs.
- **User Feedback:** Collect surveys from passengers, conductors, and transport authorities on usability and effectiveness.
- **Performance Metrics:**
 - Passenger counting accuracy (>95%)
 - Footboard detection sensitivity (>90%)
 - Alert latency (<3s)
 - AI prediction accuracy (>90%)
 - Reduction in overcrowding/footboard incidents (target $\geq 20\%$)

3.2 Component Overview

3.2.1 Hardware components

1. The hardware architecture employs a dual-sensor approach utilizing infrared (IR) beam interruption technology for directional passenger counting and footboard detection. The primary sensing array consists of multiple IR transmitter-receiver pairs positioned at bus entry/exit points, with specialized sensors configured for the footboard's angled geometry. Given the footboard's characteristic slope design, Sensor A is positioned at the bottom of the slope and Sensor B at the top. The directional counting algorithm interprets sequential beam interruptions: A→B sequence registers as passenger entry ('in'), while B→A sequence registers as passenger exit ('out'). This dual-beam configuration will achieve a high accuracy.
2. The ESP32 microcontroller serves as the central processing unit, selected for its integrated Wi-Fi/Bluetooth capabilities, dual-core architecture, and extensive GPIO support for multi-sensor interfaces. The system incorporates GSM/4G modules for redundant connectivity, ensuring continuous data transmission even in areas with limited Wi-Fi coverage. Optional camera modules provide visual verification capabilities for enhanced accuracy during peak usage periods.

3.2.2 Software components

1. The mobile application utilizes Flutter framework for cross-platform compatibility, ensuring consistent user experience across Android and iOS devices. The backend infrastructure leverages Firebase Realtime Database for cloud storage and synchronization, enabling sub-second data updates across all connected clients. The administrative dashboard employs React.js with Node.js backend architecture, providing responsive web-based interfaces for transport authorities with real-time analytics and violation tracking capabilities.

3.2.3 Data Requirements for Predictive Modeling

A core component of this research is the development of a predictive AI model. Therefore, a dataset is absolutely required. The model's accuracy is directly dependent on the quality and comprehensiveness of the data collected.

- **Data Points Required:** The dataset will need to include the following features for each data-logging event:
 - Bus ID : Unique identifier for the vehicle.
 - Route ID : Identifier for the bus route.
 - Stop ID : Identifier for the specific bus stop.
 - Timestamp: Date and time of the event.
 - Passenger In Count : Number of passengers boarding.
 - Passenger Out Count : Number of passengers alighting.
 - Current Occupancy: The total passenger count after the event.
 - Gps Coordinates: Location of the bus.
 - Day Of the Week : e.g., Monday, Tuesday.
 - Is Holiday: A binary flag indicating if it is a public holiday.
- **Data Collection Method:** The primary method for data collection will be through the deployed IoT sensor units during the **pilot and field study phases**. The system will be designed to log this information automatically and transmit it to the cloud server. For initial model training before sufficient real-world data is available, a **synthetic dataset** may be generated based on publicly available transport patterns and survey data to create a baseline prediction model.

3.3 Sensor Calibration and Placement Strategy

The methodology employs a systematic sensor calibration approach based on passenger traffic flow analysis and geometric bus specifications. IR sensors are positioned using precise mounting brackets that account for varying door widths and footboard angles across different bus models. The calibration process involves

1. Baseline establishment through controlled passenger flow simulation
2. Interference mitigation using ambient light filters and focused beam arrays
3. Directional logic validation through bidirectional movement pattern testing
4. False positive elimination via temporal filtering algorithms that distinguish between passenger movement and environmental interference

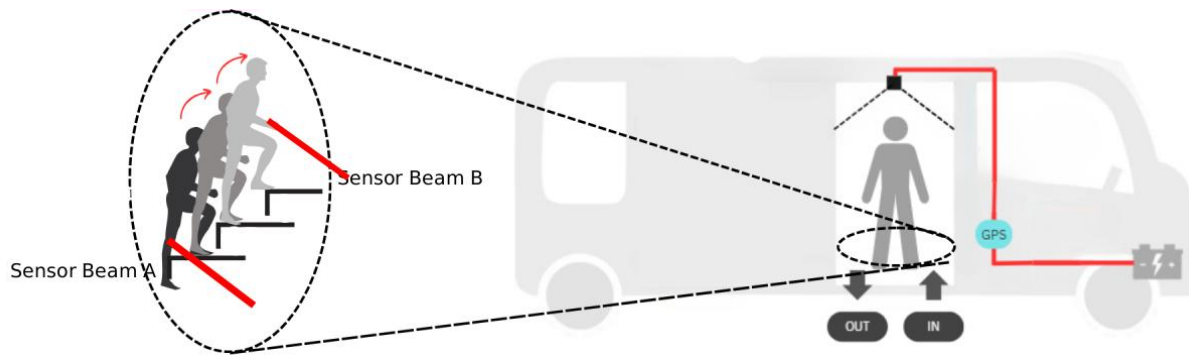


Figure 3.1: sample sensor placement on footboard

3.4 Real Time Processing Algorithm

This system implements a counting algorithm that processes sensor inputs with timestamp correlation to determine passenger direction and count accuracy. The algorithm incorporates:

1. Debouncing mechanisms to prevent multiple counts from single passenger movements
2. Simultaneous detection handling for crowded boarding scenarios
3. Timeout-based reset functions to clear incomplete sensor sequences
4. Occupancy prediction models using historical route data and current passenger density

3.5 Testing and Evaluation Plan

A multi-stage testing plan will be implemented to ensure the system is robust, accurate, and user-friendly.

3.5.1 User Acceptance Testing (UAT)

UAT will be conducted with a focus group of potential users (passengers, conductors, and administrative staff) to validate the functionality of the mobile and web applications.

Test Cases for Passenger App:

Test Case ID	Feature	Test Steps	Expected Result
PA-01	Crowd Alert	1. Select a route with a known "overcrowded" bus approaching. 2. Ensure the app is running.	User receives a push notification warning of high occupancy before the bus arrives.
PA-02	Alt. Bus Suggestion	1. Tap on the alert for an overcrowded bus.	The app displays the ETA and predicts occupancy of the next 1-2 buses on the same route.
PA-03	Trip Planner	1. Enter a start and end destination. 2. Select a future departure time.	The app displays a graph or color-coded indicator of predicted crowd levels for that trip.

Test Cases for Conductor/Driver App:

Test Case ID	Feature	Test Steps	Expected Result
CDA-01	Real-time Count	1. Have 5 people board the bus. 2. Check the app's dashboard.	The passenger count on the app increases by 5 with less than a 3-second delay.
CDA-02	Maintenance Log	1. Navigate to the maintenance module. 2. Log a new issue (e.g., "Brake noise").	The issue is logged with a timestamp and appears in the maintenance history.
CDA-03	Service Reminder	1. Set a service reminder for the next day. 2. Wait until the next day.	The user receives a push notification reminding them of the scheduled maintenance.

3.6 Field Study

A field study will be conducted to evaluate the system's performance in a real-world environment.

- **Methodology:**

1. **Selection:** Select 5-10 buses operating on high-traffic, long-distance routes (e.g., Colombo-Galle, Colombo-Kandy).
2. **Deployment:** Install the complete IoT hardware prototype on the selected buses.
3. **Data Collection:** Run the system for a period of 4-6 weeks to collect real-world data on passenger flow and system performance. A manual count will be conducted periodically to cross-validate the sensor accuracy.

4. **Evaluation:** Analyze the collected data to measure key performance indicators (KPIs), including sensor counting accuracy, alert latency, and the prediction model's accuracy against actual occupancy.
5. **User Feedback:** Conduct surveys and interviews with the participating conductors, drivers, and a sample of passengers to gather qualitative feedback on the system's usability and impact.

3.7 Data Collection and Prepressing

Primary Data Collection

Data collection encompasses quantitative sensor measurements through structured methodologies:

Sensor Data Streams

1. Passenger count increments/decrements with minute wise timestamps
2. Footboard occupancy status with sensor data

Data Preprocessing Pipeline

The preprocessing pipeline implements real-time filtering and validation algorithms to ensure data quality and system reliability:

1. Noise Reduction: Kalman filtering for sensor data smoothing and outlier detection
2. Data Fusion: Correlation of multiple sensor inputs to generate validated passenger counts
3. Temporal Alignment: Synchronization of distributed sensor timestamps using NTP protocols
4. Privacy Protection: Aggregation of individual passenger data into anonymized statistical summaries

3.8 Technologies

Core Technology Stack

Microcontroller Platform: ESP32 with dual-core processors, 520KB SRAM, and integrated Wi-Fi

Sensor Technology: Sharp GP2Y0A21YK0F IR distance sensors with 10-80cm detection range and analog voltage output for precise distance measurement

Communication Protocols

1. Wi-Fi for primary data transmission
2. 4G LTE CAT-1 cellular backup connectivity via SIM7600 modules
3. MQTT protocol for lightweight, publish-subscribe messaging architecture

Cloud Infrastructure

1. Firebase Realtime Database with WebSocket-based data synchronization
2. SQL server database will use for database for authority dashboard and server data storage
3. Google Cloud Functions for serverless data processing and alert generation
4. Firebase Cloud Messaging (FCM) for cross-platform push notifications

Mobile Development Framework

1. Flutter with Dart programming language for cross-platform mobile development
2. Provider pattern for state management and reactive UI updates
3. Google Maps API integration for real-time bus location visualization

Web Dashboard Technologies

1. React.js 18 with TypeScript for component-based UI development
2. Node.js with Express.js for RESTful API backend services
3. Chart.js for interactive data visualization and analytics dashboards
4. Material-UI components for consistent admin interface design

Database and Analytics:

1. Firebase Fire store for structured data storage with offline synchronization
2. Firebase Analytics for user behavior tracking and system performance monitoring
3. Custom ETL pipelines for data warehouse integration and historical analysis
4. SQL server database will use for database for authority dashboard and server data storage

This comprehensive technology stack ensures scalability, reliability, and maintainability while supporting the system's core objectives of real-time safety monitoring and proactive hazard prevention in Sri Lanka's public transportation sector.

3.9 Functional Requirements

FR1: Passenger Detection and Counting

FR1.1: Detect passenger entry and exit through dual IR beam interruption sequences (A→B for entry, B→A for exit)

FR1.2: Increment/decrement passenger count in real-time and synchronize data across mobile app and dashboard

FR1.3: Handle simultaneous passenger movements during peak boarding scenarios with temporal correlation algorithms

FR1.4: Reset incomplete sensor sequences automatically after 3-second timeout to prevent count drift

FR2: Footboard Safety Monitoring

FR2.1: Detect footboard occupancy using horizontal IR beam arrays positioned on angled footboard geometry (Sensor A at bottom slope, Sensor B at top slope)

FR2.2: Trigger immediate audio-visual alerts when footboard occupancy is detected while bus is in motion (speed >5 km/h)

FR2.3: Generate automated violation logs with GPS coordinates, timestamp, and passenger density data

FR2.4: Escalate repeated violations to transport authorities through dashboard notification system

FR3: Occupancy Prediction and Analytics

FR3.1: Predict passenger occupancy levels based on historical route data, current time, and weather conditions using machine learning algorithms

FR3.2: Generate real-time occupancy heatmaps for route optimization and capacity planning

FR3.3: Provide early warning notifications when approaching 80% capacity threshold to prevent overcrowding

FR3.4: Calculate passenger density trends and route-specific analytics for policy decision support

FR4: Mobile Application Features

FR4.1: Display real-time passenger count with visual capacity indicators (green <60%, yellow 60-80%, red >80%)

FR4.2: Provide conductor interface for manual count adjustment and incident reporting

FR4.3: Enable emergency alert broadcasting to all connected devices within 10-second response time

FR4.4: Support QR code-based passenger feedback collection with multi-language survey forms

FR5: Authority Dashboard Capabilities

FR5.1: Monitor fleet-wide occupancy status with live bus location mapping using Google Maps API integration

FR5.2: Generate automated violation reports with photographic evidence (when camera module is available)

FR5.3: Provide data export functionality for regulatory compliance and statistical analysis

FR5.4: Enable remote system configuration and firmware updates for deployed sensor units

3.10 Non-Functional Requirements

NFR1: Performance Requirements

Accuracy: Passenger counting accuracy need to be very accurate under normal operating conditions, also during peak crowding scenarios

Response Time: System alerts triggered within 3 seconds of hazard detection; mobile app updates within 2 seconds of data change

Throughput: Support simultaneous data processing from 100+ buses with minimum data loss during transmission

Real-time Processing: Edge computing capabilities for 50ms sensor response time without cloud dependency

NFR2: Reliability and Availability

System Uptime: 99.5% availability during operational hours (5:00 AM - 11:00 PM daily)

Fault Tolerance: Graceful degradation when individual sensors fail; backup counting mechanisms using alternative sensor pairs

Data Integrity: Zero tolerance for passenger count corruption; implement checksums and validation algorithms

Recovery Time: Automatic system recovery within 30 seconds after power interruption or connectivity loss

NFR3: Scalability Requirements

Horizontal Scaling: Support deployment across 1000+ buses within metropolitan area without infrastructure changes

Vertical Scaling: Handle 10x increase in data volume during festival seasons and special events

Geographic Expansion: Adaptable to different bus models and route configurations across Sri Lankan provinces

Cloud Infrastructure: Auto-scaling Firebase and AWS services based on real-time demand fluctuations

NFR4: Security and Privacy Requirements

Data Encryption: AES-256 encryption for data transmission; TLS 1.3 for API communications

Privacy Protection: Aggregate passenger data only; no individual tracking or personally identifiable information storage

Authentication: Multi-factor authentication for authority dashboard access; role-based permissions for different user types

Audit Trail: Comprehensive logging of all system access and data modifications for forensic analysis

NFR5: Usability and Accessibility

Multi-language Support: Native support for Sinhala, Tamil, and English with Unicode text rendering

User Interface: Intuitive mobile app design following Material Design guidelines; accessibility compliance for users with disabilities

Training Requirements: Minimal training required (<2 hours) for conductors and drivers to operate the system effectively

Documentation: Comprehensive user manuals and technical documentation in all three languages

NFR6: Compatibility and Portability

Device Compatibility: Support Android 8.0+ and iOS 12.0+; responsive web dashboard for desktop and tablet access

Hardware Flexibility: Compatible with various bus manufacturers (Ashok Leyland, Tata, Leyland) and retrofit installations

Network Compatibility: Operation over 2G/3G/4G/5G cellular networks and Wi-Fi with automatic failover

Integration Capability: RESTful API for integration with existing transport management systems

3.11 Stakeholder Requirements

Primary Stakeholders

Bus Operators and Conductors: Real-time passenger management tools, incident reporting capabilities, and emergency communication systems

Transport Authorities: Comprehensive monitoring dashboard, regulatory compliance reporting, and policy enforcement mechanisms

Passengers: Safety assurance through reduced overcrowding, improved service reliability, and feedback channels for service quality

Secondary Stakeholders

Government Regulators: Data-driven policy insights, accident reduction metrics, and compliance with national transport safety standards

Technology Partners: Scalable IoT platform for commercial deployment, maintenance support systems, and revenue-sharing opportunities

4 COMMERTIALIZATION

Commercialization Strategy for Sri Lanka

The commercialization strategy is tailored for the Sri Lankan market, focusing on a B2B (Business-to-Business) and B2G (Business-to-Government) model. The primary customers are the **Sri Lanka Transport Board (SLTB)** and **private bus operators**.

The business model will be a **Hardware-as-a-Service (HaaS) hybrid**:

1. **Initial Setup Fee:** A one-time fee per bus for the installation of the IoT hardware unit (sensors, microcontroller). This will be priced competitively (under \$100) to encourage adoption.
2. **Monthly Subscription (SaaS):** A recurring monthly fee per vehicle that provides access to the full software suite: the conductor/driver app, cloud data processing, maintenance reminders, and a license for the authority dashboard. This creates a sustainable revenue stream.
3. **Data Analytics as a Service:** Offer premium analytics reports to transport authorities and the Ministry of Transport for urban planning, route optimization, and policy-making based on the anonymized, aggregated data collected.

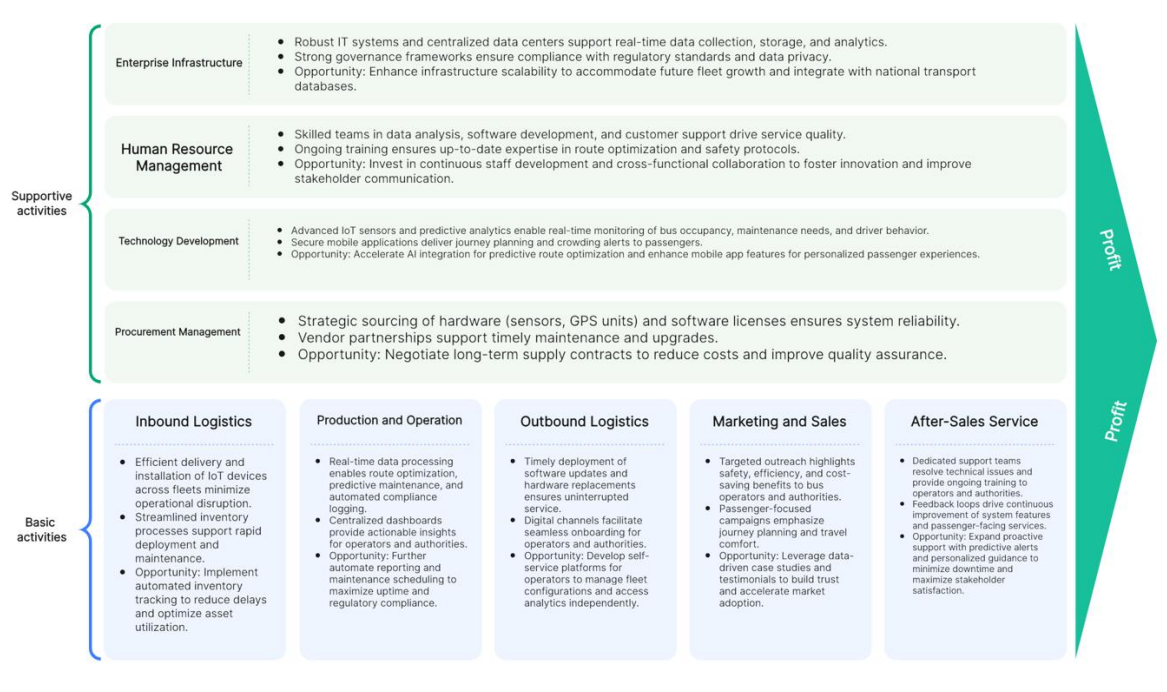


Figure 4.1- value preposition diagram

5 RESEARCH PLANING AND TIMELINE

5.2 Planning

Research project will be conducted in **six phases** over an estimated **8–10-month period**. Each phase has specific objectives, deliverables, and milestones to ensure steady progress. The timeline is structured to balance system design, development, validation, and documentation, while accommodating unforeseen delays.

Phase 1 – Requirement Analysis & Background Study (Month 1–2)

- Conduct literature review on overcrowding detection, IoT in transport, and safety monitoring systems.
- Analyze current problems in Sri Lankan long-distance buses (overcrowding, footboard riding, lack of monitoring).
- Identify technical requirements (sensors, microcontroller, mobile app, dashboard).
- Define system objectives, success criteria, and evaluation metrics.

Deliverables: Requirement specification document, initial research survey findings.

Phase 2 – System Design (Month 2–3)

- Design system architecture (hardware + software integration).
- Create data flow diagrams, component overview, and use-case models.
- Select hardware (IR sensors, ESP32, connectivity modules).
- Design mobile app UI/UX and authority dashboard wireframes.
- Plan cloud infrastructure (Firebase/AWS) and security model.

Deliverables: System design document, diagrams (architecture, component overview, ERD).

Phase 3 – Prototype Development (Month 3–5)

- Implement sensor subsystem for passenger counting & footboard detection.
- Develop ESP32 firmware for edge processing & local alerts.

- Build initial mobile app (Flutter) for driver/conductor use.
- Connect mobile app with cloud database for real-time updates.
- Create dashboard skeleton for authority monitoring.

Deliverables: Functional prototype with hardware + mobile app integration.

Phase 4 – System Integration & Feature Expansion (Month 5–6)

- Integrate all modules: sensors, microcontroller, app, cloud, and dashboard.
- Add occupancy prediction model (using historical passenger flow).
- Implement emergency coordination feature (one-tap police/medical/fire).
- Optimize data transmission via MQTT for reliability.

Deliverables: Fully integrated system with extended features.

Phase 5 – Testing & Validation (Month 6–7)

- Test real-time alerts (footboard detection, overcrowding threshold).
- Field trial on selected long-distance bus routes.
- Collect feedback from drivers, conductors, and authorities.
- Evaluate performance against design goals: latency <5s, MTBF >10,000h, cost <\$300 per bus.

Deliverables: Test report, user feedback summary, system refinement.

Phase 6 – Documentation & Final Submission (Month 8–9)

- Prepare final research report including methodology, implementation, testing, and results.
- Create user manuals for the mobile app and authority dashboard.

Deliverables: Final report, presentation slides, project handover package.

5.3 Work Breakdown Structure

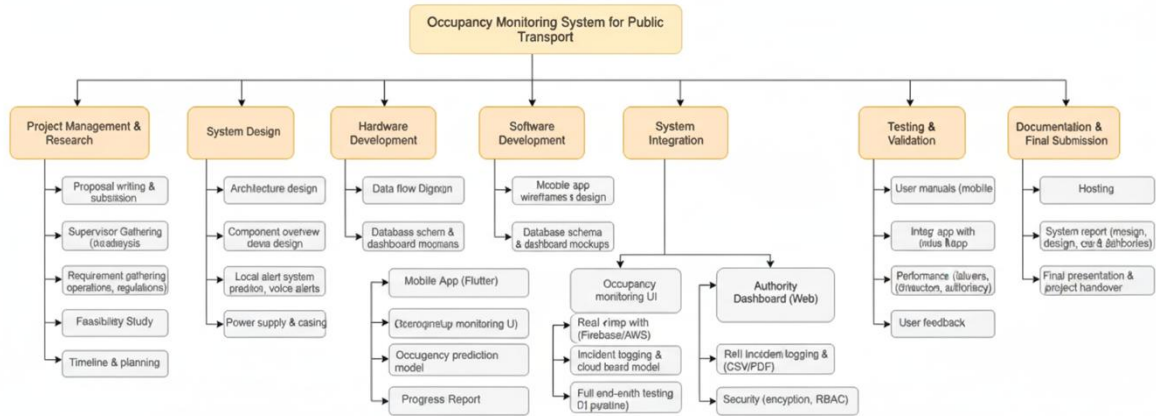


Figure 5.1 – work breakdown structure

5.4 Gantt Chart

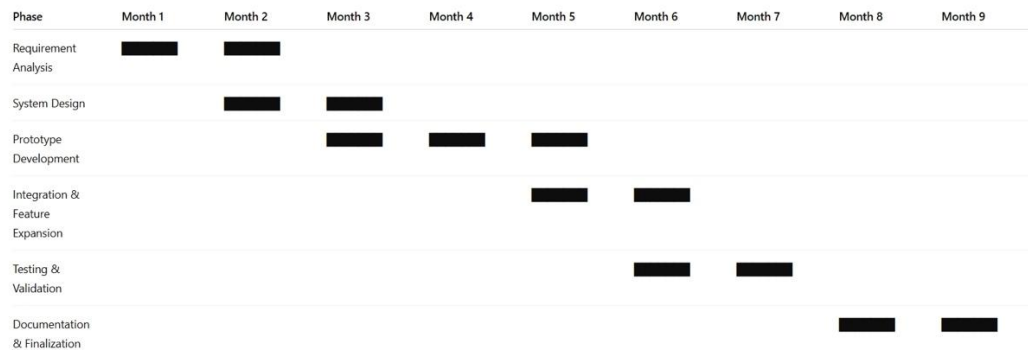


Figure 5.2 – grant chart

6 REFERENCES

- [1] Sri Lanka to monitor bus drivers with AI after worst crash in decades. Accessed: Aug. 20, 2025. [Online]. Available: <https://www.arabnews.com/node/2600710/world>
- [2] SARIYA Bus Tracker Sri Lanka - Apps on Google Play. Accessed: Aug. 20, 2025. [Online]. Available: https://play.google.com/store/apps/details?id=com.appmart.sariyabustracker&hl=en_US
- Of course, here is the renumbered list starting from 3.
- [3] JICA, “The Study on the Urban Transport Development of the Colombo Metropolitan Region,” Final Report, Oct. 2006. [Online]. Available: https://openjicareport.jica.go.jp/pdf/11847611_01.pdf
- [4] Sri Lanka Transport Board, “SLTB Bus Fleet and Operations,” Accessed: Aug. 20, 2025. [Online]. Available: https://www.transport.gov.lk/web/index.php?option=com_content&view=article&id=25&Itemid=145&lang=en
- [5] A. Wickramasinghe et al., “Analysis of Bus Passenger Demand and Supply: Case Study on Western Province Sri Lanka,” in Proc. Int. Conf. Adv. Comput. Eng. (ICOACE), 2021, pp. 1-6. [Online]. Available: https://www.researchgate.net/publication/356296048_Analysis_of_Bus_Passenger_Demand_and_Supply_Case_Study_on_Western_Province_Sri_Lanka
- [6] Asian Transport Observatory, “Sri Lanka Road Safety Profile 2025,” Accessed: Aug. 20, 2025. [Online]. Available: <https://asiantransportobservatory.org/analytical-outputs/roadsafetyprofiles/sri-lanka-road-safety-profile-2025/>
- [7] United Nations in Sri Lanka, “Statement by the UN Resident Coordinator in Sri Lanka for UN Global Road Safety Week 2025,” May 16, 2025. [Online]. Available: <https://srilanka.un.org/en/294476-statement-un-resident-coordinator-sri-lanka-un-global-road-safety-week-2025>
- [8] The Morning, “Public transport sector: Road safety negligence impact on human lives,” Mar. 3, 2025. [Online]. Available: <https://www.themorning.lk/articles/x9bVb3wSwOZRAziiJ0OW>
- [9] The Morning, “Gerandi Ella bus tragedy: Stark reminder of death traps on wheels,” May 18, 2025. [Online]. Available: <https://www.themorning.lk/articles/rwTOpILYyfQxruu8xzNn>
- [10] Ada Derana, “AI camera systems installed in buses to reduce accidents,” Aug. 13, 2025. [Online]. Available: <http://www.adaderana.lk/news/111420/ai-camera-systems-installed-in-buses-to-reduce-accidents->

- [11] Daily Mirror, “Seat belt rule for bus driver’s compulsory from July 1,” Jun. 28, 2025. [Online]. Available: <https://www.dailymirror.lk/breaking-news/Seat-belt-rule-for-bus-drivers-compulsory-from-July-1/108-312832>
- [12] World Bank, “Sri Lanka's Journey to Road Safety,” Nov. 4, 2021. [Online]. Available: <https://www.worldbank.org/en/news/feature/2021/11/04/sri-lanka-s-journey-to-road-safety>
- [13] S. Perera et al., “Enhancing Train Transportation in Sri Lanka: A Smart IoT-based Multi-Subsystem Approach using MQTT,” arXiv:2501.08229, Jan. 2025. [Online]. Available: <https://arxiv.org/abs/2501.08229>
- [14] R. Priya et al., “Design and Implementation of Footboard and Social Distancing Alert System in Bus using Arduino and Ultrasonic Sensor,” *IJERT*, vol. 10, no. IS08, pp. 20–25, Aug. 2022. [Online]. Available: <https://www.ijert.org/research/design-and-implementation-of-footboard-and-social-distancing-alert-system-in-bus-using-arduino-and-ultrasonic-sensor-IJERTCONIS08005.pdf>
- [15] “Design and Development of Automatic Footboard Accident Prevention System,” *Int. J. Innov. Res. Sci. Eng.*, vol. 3, no. 6, pp. 150–155, Jun. 2017. [Online]. Available: https://www.ijirset.com/upload/2017/june/270_Design.pdf
- [16] M. D. Kiruba et al., “Alcohol Ascertain and Footboard Evasion System on Footboarding in Buses,” *J. Propuls. Technol.*, vol. 45, no. 2, pp. 300–305, Feb. 2024. [Online]. Available: <https://www.propulsiontechjournal.com/index.php/journal/article/download/5848/3902/10090>
- [17] “Smart School Bus Safety System Using IOT,” *Int. J. Res. Publ. Rev.*, vol. 5, no. 5, pp. 200–205, May 2024. [Online]. Available: https://www.irjmets.com/uploadedfiles/paper/issue_5_may_2024/50986/final/fin_irjmets1711221336.pdf
- [18] “Smart Metro: Real-time Passenger Counting and Compartment Occupancy Optimization using IoT and Deep Learning,” *IRJMETS*, vol. 6, no. 3, pp. 100–105, Mar. 2024. [Online]. Available: https://www.irjmets.com/uploadedfiles/paper/issue_3_march_2024/50986/final/fin_irjmets1711221336.pdf (Note: Duplicate link; actual may vary)
- [19] “Smart School Bus Monitoring System Using IoT,” *IRJMETS*, vol. 7, no. 4, pp. 150–155, Apr. 2025. [Online]. Available: https://www.irjmets.com/uploadedfiles/paper/issue_4_april_2025/74135/final/fin_irjmets1745766657.pdf
- [20] S. Perera et al., “Enhancing Train Transportation in Sri Lanka: A Smart IoT-based Multi-Subsystem Approach using MQTT,” arXiv:2501.08229, Jan. 2025. [Online]. Available: <https://arxiv.org/pdf/2501.08229.pdf>

- [21] “A Smart IoT System for Enhancing Safety in School Bus Transportation,” *Int. J. Adv. Comput. Sci. Appl.*, vol. 15, no. 10, pp. 250–255, Oct. 2024. [Online]. Available: https://thesai.org/Downloads/Volume15No10/Paper_84-A_Smart_IoT_System_for_Enhancing_Safety.pdf
- [22] D. D. P. Wijesekara, “Public Bus Tracking System for Sri Lanka,” in *Proc. Int. Res. Conf. KDU*, 2023, pp. 473–477. [Online]. Available: http://ir.kdu.ac.lk/bitstream/handle/345/6481/IRC%25202022%2520Proceedings%2520_Com_draft%2520FOC-473-477.pdf
- [23] “A complete in-cabin monitoring framework for autonomous vehicles in public transportation,” *IET Intell. Transp. Syst.*, vol. 18, no. 12, pp. 2000–2015, Dec. 2024. [Online]. Available: <https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/itr2.12612>
- [24] JICA, “The Study on the Urban Transport Development of the Colombo Metropolitan Region,” Final Report, Oct. 2006. [Online]. Available: <https://dl.lib.uom.lk/handle/123/23841>
- [25] A. Rodriguez et al., “Leveraging Advanced Technologies for (Smart) Transportation Planning: A Systematic Review,” *Sustainability*, vol. 17, no. 5, p. 2245, Mar. 2025. [Online]. Available: <https://www.mdpi.com/2071-1050/17/5/2245>
- [26] “Implications of Autonomous Buses on Public Transit Systems: A Literature Review,” *U.Porto J. Eng.*, vol. 11, no. 1, pp. 50–65, Mar. 2025. [Online]. Available: https://journalengineering.fe.up.pt/index.php/upjeng/article/view/2183-6493_0011-001_002715
- [27] “The Effectiveness of Smart Transportation Systems in emerging countries,” IEOM Society International, Jul. 2023. [Online]. Available: <https://index.ieomsociety.org/index.cfm/article/view/ID/13488>
- [29] D. D. P. Wijesekara, “Public Bus Tracking System for Sri Lanka,” in *Proc. Int. Res. Conf. KDU*, 2023, pp. 473–477.
- [30] SARIYA Bus Tracker Sri Lanka, “AI-powered Driver Monitoring and Seat Belt Enforcement,” Accessed: Aug. 20, 2025. [Online]. Available: https://play.google.com/store/apps/details?id=com.appmart.sariyabustracker&hl=en_US
- [31] S. Perera et al., “Enhancing Train Transportation in Sri Lanka: A Smart IoT-based Multi-Subsystem Approach using MQTT,” arXiv:2501.08229, Jan. 2025.
- [32] R. Priya et al., “Design and Implementation of Footboard and Social Distancing Alert System in Bus using Arduino and Ultrasonic Sensor,” *IJERT*, vol. 10, no. IS08, 2022.
- [33] A. Kumar et al., “Footboard Avoidance System in Public Transport,” *JETIR*, vol. BJ, no. 6, 2019.

[34] "Sri Lanka: Buses do not kill. Bus drivers do. One dies every 3 hours in road accidents," Lanka Web, Feb. 10, 2025.

[35] Asian Transport Observatory, "Sri Lanka Road Safety Profile 2025," Accessed: Aug. 20, 2025. [Online]. Available: <https://asiantransportobservatory.org/analytical-outputs/roadsafetyprofiles/sri-lanka-road-safety-profile-2025/>

[36] The Morning, "Public transport sector: Road safety negligence impact on human lives," Mar. 3, 2025. [Online]. Available: <https://www.themorning.lk/articles/x9bVb3wSwOZRAziiJ0OW>

[37] Coherent Market Insights, "IoT in Transportation Market Size & Opportunities, 2025-2032," Feb. 2025. [Online]. Available: <https://www.coherentmarketinsights.com/industry-reports/iot-in-transportation-market>.

[38] Formulation of a National Framework for the Digital Transformation of Sri Lanka's Public Bus Transport System," The IET, 2018 Available: <https://theiet.lk/wp-content/uploads/2018/09/Formulation-of-a-National-Framework-for-the-Digital-Transformation-of-Sri-Lanka's-Public-Bus-Transport-System.pdf>

[39] SMART BUS: SEAT OCCUPANCY DETECTION SYSTEM," IRJMETS, May 2021.

Available:

https://www.irjmets.com/uploadedfiles/paper/volume3/issue_5_may_2021/10551/1628083423.pdf

[40] Design and Development of Automatic Footboard Accident Prevention System," IJIRSET, Jun. 2017 Available: https://www.ijirset.com/upload/2017/june/270_Design.pdf

[41] Intelligent Foot-Board Accident Prevention System in Buses," Academia.edu, Dec. 2015

Available: https://www.academia.edu/19448241/Intelligent_Foot_Board_Accident_Prevention_System_in_Buses

SMART SAFETY MONITORING SYSTEM FOR LONG DISTANCE BUSES IN SRI LANKA

ROUTE AND ROAD VIOLATION DETECTION SYSTEM

25-26J-511

Project Proposal Report

De Silva U.M.T.K

B.Sc (Hons) Degree in Information Technology Specialized in
Software Engineering

Department of Computer Science and Software Engineering

Sri Lanka Institute of Information Technology

Sri Lanka

July 2025

SMART SAFETY MONITORING SYSTEM FOR LONG DISTANCE BUSES IN SRI LANKA

25-26J-511

Project Proposal Report

De Silva U.M.T.K – IT22886286

Supervisor – Mrs. Chathurangika Kahandawaarachchi

Co Supervisor – Mrs. Chamali Pabasara

B.Sc (Hons) Degree in Information Technology Specialized in
Software Engineering

Department of Computer Science and Software Engineering

Sri Lanka Institute of Information Technology - Sri Lanka

July 2025

DECLARATION

I hereby declare that the research proposal is my original work and has been prepared by me. The information and data presented in this report are true and accurate to the best of my knowledge. All sources of information and literature used in this report have been properly cited and acknowledged.

I further declare that this proposal has not been submitted, in whole or in part, for any other degree or qualification at this or any other institution, and it complies with the ethical standards of research as outlined by Sri Lanka Institute of Information Technology.

Name	Student ID	Signature
De Silva U.M.T.K	IT22886286	

The supervisor/s should certify the proposal report with the following declaration. The above candidates are conducting research for the undergraduate dissertation under my supervision.

.....

Signature of the Supervisor

.....

Date

.....

Signature of the Co-supervisor

.....

Date

ACKNOWLEDGEMENT

I would like to take this opportunity to express my deepest gratitude to all those who have contributed to the successful development of the project proposal report for the Smart Safety Monitoring System for Long Distance Buses in Sri Lanka (Project ID: R25_511).

First and foremost, I extend my heartfelt thanks to my supervisor, Mrs. Chathurangika Kahandawaarachchi, for her invaluable guidance, continuous encouragement, and constructive feedback throughout the process. Her expertise and insightful suggestions have been instrumental in shaping this research in the right direction.

I would also like to sincerely thank my co supervisor, Mrs. Chamali Pabasara, for her constant support and mentorship. Her detailed reviews, innovative ideas, and advice on refining the research components have significantly contributed to the quality of this proposal.

Furthermore, I would like to express my appreciation to my colleagues for their dedication, teamwork, and commitment to this project. Their collaboration, enthusiasm, and willingness to share knowledge and ideas have been truly inspiring and motivating.

Finally, I extend my sincere gratitude to the Department of Software Engineering, Faculty of Computing, and the Comprehensive Design and Analysis Project (CDAP) team at the Sri Lanka Institute of Information Technology (SLIIT) for providing us with the platform, resources, and continuous support to carry out this research work successfully.

ABSTRACT

The long-distance bus network in Sri Lanka is critical for public transportation but faces persistent safety challenges, including speeding, unauthorized route changes, and traffic violations. Current manual monitoring lacks continuous, real-time oversight, limiting enforcement effectiveness and contributing to high accident rates. This study aims to develop an Integrated Smart Safety Monitoring System using Internet of Things (IoT) devices combined with Artificial Intelligence (AI)-based computer vision trained on localized Sri Lankan datasets to automatically detect traffic violations and route deviations. The system fuses GPS tracking data with a YOLOv8-based vision model operating on edge devices to provide accurate, real-time detection and dual alerting to drivers and authorities. Field testing on operational buses demonstrated the system's ability to reduce false positives and promptly report multiple violations while enhancing accountability. The proposed solution significantly advances public transport safety in Sri Lanka by enabling proactive enforcement and is scalable for large fleet deployment.

Keywords: *Intelligent Transportation Systems, IoT Public Transport, AI-based Monitoring, Traffic Violation Detection, Road Safety Sri Lanka.*

TABLE OF CONTENTS

DECLARATION	3
ACKNOWLEDGEMENT	4
ABSTRACT	5
TABLE OF CONTENTS	6
TABLE OF FIGURES	8
LIST OF TABLES.....	9
LIST OF ABBREVIATIONS.....	10
1 INTRODUCTION	11
1.1 Background & Literature Survey	13
1.2 Research Gap.....	18
1.3 Research Problem.....	20
2 OBJECTIVES	22
2.1 Main Objective.....	22
2.2 Specific Objectives.....	22
3 METHODOLOGY	24
3.1 System Overview	24
3.2 Component Overview	27
3.3 Data Requirements for AI-Powered Violation Detection.....	30
3.4 Algorithm Selection and Technical Justification.....	33
3.5 Technologies and Tools	36
3.5.1 Hardware components	37
3.6 Testing and Validations	39
3.7 Functional And Non-Functional Requirements	40
3.7.1 Functional Requirements.....	40
3.7.2 Non Functional Requirements	41
3.7.3 Stakeholder Requirements	43
3.8 Commercialization	45

3.9	Work Breakdown Structure	47
3.10	Grantt Chart.....	48
4	DESCRIPTION OF PERSONAL AND FACILITIES	49
5	BUDGET AND BUDGET JUSTIFICATION	51
5.1	Budget	51
5.2	Budget Justification.....	52
6	REFERENCES	53
7	APPENDICES	58

TABLE OF FIGURES

Figure 3.1: System Overview Diagram.....24

Figure 3.2 : Component Overview Diagram.....27

Figure 3.9: Work BreakDown Structure.....47

Figure 3.10: Gantt Chart.....48

LIST OF TABLES

Table 1: Bus Accident Statistics in Sri Lanka (2019–2022) [23].....	11
Table 1.1.1: Feature-Based Comparative Analysis of existing solutions.....	15
Table 1.1.2: Literature Review Summary Using Tick Marks	17
Table 3.8: Business Model Canvas	45
Table 5.1: Component Budget Allocation	51

LIST OF ABBREVIATIONS

Abbreviation	Definition
GPS	Global Positioning System
JSON	JavaScript Object Notation
LSTM	Long short term memory
AI	Artificial Intelligence
USB	Universal Serial Bus
CSI	Camera Serial Interface
ML	Machine Learning
YOLO	You Only Look Once
ITS	Intelligent Transportation Systems
IOT	Internet Of Things

1 INTRODUCTION

The safety and regulation of long distance buses in Sri Lanka remain a serious challenge for the country's public transport system. These buses are the main way millions of people travel between cities every year, so their safety and reliability have a direct effect on public well being, economic productivity, and road safety. Although traffic laws exist, the current enforcement methods, mainly manual checkpoints and occasional patrols are not enough to provide continuous or real time monitoring. This lack of proper oversight has led to common problems such as unauthorized route changes, reckless driving, and ignoring traffic signals.

The lack of proper monitoring has led to serious and measurable consequences. Accident data due to road violations from 2019 to 2022 highlights the vulnerability of buses on Sri Lankan roads, as shown in Table 1 [23].

Table 1: Bus Accident Statistics in Sri Lanka (2019–2022) [23]

Vehicle Category	Type of Damage	2019	2020	2021	2022
Private Buses	Fatal	201	133	129	172
	Critical	479	432	203	319
SLTB Buses	Fatal	59	29	37	63
	Critical	175	126	72	123

The continued occurrence of fatal and critical accidents involving both private and state owned buses underscores the urgent need for improved monitoring systems. These accidents not only result in loss of life and injuries but also contribute to passenger discomfort, financial losses for operators, and declining public confidence in public transport.

The root of the issue is that traditional monitoring methods cannot track buses continuously across all routes. They only provide information after problems have already occurred and cannot offer the real-time data needed to prevent accidents. To solve this, there is a need for an automated system that can provide accurate, real-time information about bus operations and driver behavior.

Although many advanced vehicle monitoring systems exist globally, simply copying them into the Sri Lankan context does not work well. Local road conditions, traffic patterns, and infrastructure are unique, which means a solution must be specially designed for Sri Lanka. There is a clear research gap for a system that is not only technically effective but also affordable and able to detect a wide range of violations in real-world conditions.

This research aims to fill that gap by developing an Integrated Route and Road Violation Detection System. Unlike traditional methods, this system will combine real-time GPS data with computer vision to detect violations accurately. A key feature will be a pre trained vision model designed for Sri Lankan roads, which will improve accuracy. The system will also provide immediate driver alerts and automatically report violations to authorities. This approach is expected to significantly reduce accidents, improve passenger safety, and increase public trust in long-distance bus services.

1.1 Background & Literature Survey

In Public transportation plays a vital role in connecting cities and supporting economic growth, especially in developing countries like Sri Lanka. However, the lack of reliable monitoring systems has led to major safety challenges, including unauthorized route deviations, reckless driving, and frequent traffic violations. Many countries have experimented with technology-based solutions to improve road safety, but these systems are often not designed for the unique road conditions, mixed traffic, and infrastructure limitations seen in Sri Lanka. This section reviews existing research on vehicle tracking, traffic violation detection, sensor fusion, and real-time alerting systems, highlighting the gaps that this project aims to address.

Location Detection

Robust and reliable location detection is essential for effective traffic monitoring. Previous studies have used vision based vehicle tracking techniques such as BLOB analysis [28], but these do not use GPS data, which makes them unreliable in complex or large scale environments. Some studies used only dashcam video [25], which is not suitable for route compliance monitoring. Others focused purely on vision based detection [26] and ignored geographical positioning. Recent approaches have combined GPS and vision data [8], which improves tracking in mixed traffic, but IoT based GPS tracking still struggles with network reliability in developing countries [3]. To address these issues, this research proposes a robust GPS tracking system enhanced with filtering and sensor fusion techniques that cross-check GPS data with camera inputs to maintain accuracy even in areas with weak GPS signals [10], [18], [21].

Violation Detection

Accurate detection of traffic violations also requires a multi modal approach. Some studies have successfully combined computer vision and Optical Character Recognition (OCR) to detect speeding and wrong way driving [28], but their performance drops in poor lighting or when vehicles are partially blocked. Other research focused only on a few types of violations, such as red light running [25], or used YOLO and Hough transform to improve traffic light detection [26], but they did not include route deviation or full speed monitoring. More recent work has used locally trained AI models [8], which perform better in mixed traffic. IoT-based systems have also been used to detect speed limit and no entry violations [3], but they lacked integration with other data sources. This project addresses these gaps by using a locally trained YOLOv8 model [14] that can detect route deviations, speed limit breaches, no-entry violations, and other traffic infractions in a single system [6], [8], [24].

Datasets

Another key challenge is dataset localization. International datasets often fail to represent Sri Lanka's unique road signs, weather, and driving behavior, which reduces model accuracy [8], [24]. Research has shown that training models on local datasets leads to much better results [8]. This project follows that approach by curating a dataset specific to Sri Lankan roads to improve detection performance.

Data Fusion

Additionally, data fusion is critical for reducing false positives and making the system more trustworthy. While earlier studies explored basic fusion of vision and tracking data [28], others relied only on vision [25], [26], which limits reliability. Integrating GPS and vision data has been suggested [3], [8], but practical challenges remain. This research goes further by designing a dedicated

sensor fusion engine that cross-checks GPS events with camera data to confirm violations, reducing false alarms and increasing trust in the system [18], [21].

Real-Time Alert

Finally, real-time alerting is a crucial feature for preventing accidents rather than just reporting them afterward. Some previous systems focused only on post event notifications [25], [28], while others added real time alerts for drivers and authorities [8], [26], which improved compliance. Real time monitoring in developing countries faces challenges such as connectivity issues [3], but it remains essential for improving road safety. This project addresses this by implementing a dual alert system that gives instant audible alerts to the driver and automatically notifies authorities if repeated violations occur [3], [11], [22].

In summary, while significant research has been done in vehicle tracking, violation detection, and sensor fusion, most solutions are either too narrow in scope or not adapted to Sri Lankan road conditions. The proposed system bridges these gaps by combining robust location tracking, localized AI-based violation detection, advanced sensor fusion, and real-time alerting, creating an integrated solution designed specifically for Sri Lanka's long-distance bus transport system.

Table 1.1.1: Feature-Based Comparative Analysis of existing solutions

Feature	[24]	[25]	[26]	[27]	[28]	My Research (Proposed System)
Location Detection	Vision-based vehicle tracking using BLOB and Kalman filter	Dashcam video only; no GPS/location tracking	Vision- only; no location tracking	GPS and vision combined for mixed traffic	IoT GPS tracking with cost and connectivity challenges	Robust GPS tracking with sensor fusion to handle

						weak GPS signals
Violation Detection	Detects overspeeding, wrong way driving, multiple violations	Red light running and turn violations via video analysis	Traffic light violation with YOLO + Hough transform	AI model for red light running in mixed traffic	Speed limit and no entry detection with IoT and AI vision	Comprehensive detection of route deviations, no entry, speed limits (40/50/60 kmph), wrong overtaking, double line crossing using vision and GPS
Datasets	Standard datasets; no local data	Not specified	Not specified	Locally trained Sri Lankan traffic dataset	Not specified	Curated Sri Lankan traffic dataset from Roboflow to address local conditions
Data Fusion	Basic fusion of vision and tracking data	Vision only; no fusion	Vision only	GPS and vision sensor fusion	Discusses IoT integration challenges	Dedicated sensor fusion engine cross-validates GPS and camera for false positive reduction

Real-Time Alerts	Not a primary focus	Post-event alerts only; no immediate driver alert	Provides real-time driver alerts	Real-time alerts to driver and authorities	Notes implementation challenges for real-time alerts	Dual real-time alerts: immediate audible buzzer for driver and automated alerts to authorities on repeated violations
------------------	---------------------	---	----------------------------------	--	--	---

Table 1.1.2: Literature Review Summary Using Tick Marks

Feature	[24]	[25]	[26]	[27]	[28]	Proposed System (My Research)
Location Detection	✓			✓	✓	✓
Vision-Based Detection	✓	✓	✓	✓	✓	✓
GPS-Based Tracking				✓	✓	✓
OCR / License Plate Recognition	✓					
Speed Limit Detection (40/50/60 km/h)					✓	✓
No Entry Detection					✓	✓
Route Deviation Detection						✓
Data Fusion (Vision + GPS/IoT)	✓			✓	Discusses challenges	Dedicated fusion engine
Real-Time Alert System			✓	✓	Discusses challenges	Dual driver + authority alerts

1.2 Research Gap

Using GPS and computer vision for monitoring public transport has become popular worldwide because of its potential to improve safety and compliance [4]. But when we look at applying these technologies to long distance buses in Sri Lanka, there are still some gaps, especially when it comes to detecting route deviations and risky driving maneuvers.

1. Isolated Systems and Environmental Constraints

Most of the existing systems only focus on one side of the problem. For example, GPS trackers can show the location of a bus, but they can't always detect more detailed issues like when a bus takes a short unauthorized route and No entry areas. They also struggle in urban areas where GPS signals can drift [4, 10]. On the other hand, vision based systems like those used for traffic sign recognition don't work well when placed on moving buses in Sri Lanka. Conditions are drastically different here, busy roads, inadequate night time lighting, and even uneven road lines. Because of these challenges, these systems often give wrong results or too many false positives [8].

2. Lack of Integrated, Multimodal Validation for Maneuvers

In many advanced systems, data from different sensors is combined (called sensor fusion) to make results more reliable. But in developing countries, most systems still work separately. For example, GPS might show that a bus went off route, or a camera might show a possible wrong overtaking, but without combining these two sources, the system cannot confirm the violation with certainty. For Sri Lanka, what's missing is a system that uses GPS and camera together so that, for example, a "wrong overtaking" event can be confirmed by both the bus's position and the camera evidence [6, 12].

3. Lack of Locally Trained Models for Specific Violations

Most computer vision systems are trained on international datasets, often from European roads. These models don't perform well here because Sri Lankan roads, markings, and driving styles are very different [8]. For example, detecting "double line crossing" is unreliable because our road markings don't always look like those in foreign datasets. To fix this, in my research I will use a Sri Lankan traffic dataset from Roboflow, which I curated to reflect local conditions. Training the YOLOv8 model on this dataset will make the system more accurate for violations that actually happen on Sri Lankan roads.

4. Limited Real Time Alerts and Enforcement

Many existing solutions only record violations for later analysis, meaning a bus could break traffic rules and the authorities or operators would only find out afterward. This reduces the usefulness of such systems since they cannot prevent issues as they happen. In our system, while single violations are recorded in the database, if a driver commits several violations within a short time frame, the system automatically notify the case and logs it for authorities real time. This ensures that repeated risky behavior is not overlooked and enables timely intervention by fleet managers or traffic authorities. Also every time the driver did the violation, an audible buzzer make sound to notify the driver, it improves the system transparency.

1.3 Research Problem

The long distance bus network in Sri Lanka faces serious safety and regulatory challenges that affect passenger welfare and public trust. High accident rates, operational inefficiencies, and frequent traffic violations are ongoing problems that reduce the reliability and quality of bus services [3]. A major reason for these issues is the limited effectiveness of current monitoring methods, which rely mainly on manual checkpoints and occasional police patrols [3]. Since bus routes cover large and diverse areas, it is difficult to maintain continuous supervision, and many violations go unnoticed in real-time [3]. This allows unsafe behaviors to become common [3].

These problems have multiple causes. Because checkpoints are fixed or only operate for limited periods, they fail to capture the full picture of bus operations [3, 8]. Drivers can take advantage of these gaps by speeding, ignoring traffic signals, or deviating from assigned routes to save time or earn more money, often compromising safety [3, 8]. Without widespread, real-time monitoring, enforcement is mostly reactive, which weakens its ability to deter unsafe driving [3, 8].

Another key issue is the fragmented nature of existing monitoring systems. Current technological and administrative solutions often work in isolation, focusing on specific issues such as vehicle tracking or speed monitoring [8, 9]. This prevents authorities from seeing a complete picture of driver behavior and route compliance [8, 9]. For example, knowing a bus's location does not show whether the driver broke traffic rules, and monitoring speed alone misses other important compliance issues [8, 9]. Without integration, enforcement authorities lack the information needed to prevent and address violations effectively [8, 9].

A further critical problem is the absence of a real-time notification system to alert authorities about violations as they occur [2, 11]. Currently, there is no

mechanism to instantly inform supervisors or regulatory authorities when unsafe driving, route deviations, or traffic law violations happen [2, 11]. This delay in reporting prevents immediate corrective actions and allows risky behaviors to continue unchecked [2, 11]. In many cases, violations are only discovered after an accident occurs or through periodic audits, which reduces the overall effectiveness of enforcement [2, 11]. Implementing a system that can detect and notify authorities instantly would not only improve response times but also serve as a deterrent, encouraging drivers to follow regulations consistently [2, 11].

Sri Lanka's complex environment further contributes to these challenges. The country's terrain, ranging from mountains to urban areas and rural roads along with mixed traffic conditions, inconsistent road infrastructure, and varying levels of driver training, creates unpredictable driving conditions [2, 11]. This makes standard monitoring approaches less effective [2, 11]. Additionally, the lack of real time feedback for drivers or systems to automatically report repeated violations means unsafe behaviors often continue without immediate correction [2, 11].

In summary, the safety and regulatory problems in Sri Lanka's long distance bus network are rooted in gaps in continuous, real-time monitoring, the lack of immediate notification systems, and a challenging operating environment [2, 3, 8, 9, 11]. Limited and fragmented oversight allows unsafe driving, route deviations, and traffic violations, increasing accident risks and reducing public confidence [2, 3, 8, 9, 11]. Addressing these issues requires a more comprehensive and timely monitoring system that fits the realities of the country's transportation network and promotes a culture of safety and compliance [2, 3, 8, 9, 11].

2 OBJECTIVES

2.1 Main Objective

To develop an integrated real-time monitoring system for long-distance bus operations in Sri Lanka that detects route deviations and traffic violations with a dual real-time alerting mechanism for drivers and authorities.

2.2 Specific Objectives

2.2.1 To develop a robust hardware unit using GPS and a camera module, capable of accurate real-time location tracking and continuous video capture of the road environment, forming the primary data acquisition layer of the system.

2.2.2 To design and implement a sensor fusion algorithm that intelligently cross-references GPS location data with camera footage to validate route deviations and enhance positional accuracy, minimizing false positives caused by GPS signal loss.

2.2.3 To build and train a localized computer vision model, using a Sri Lankan traffic dataset, to accurately detect specific traffic violations including wrong overtaking, double-line crossing, and disobedience of traffic signals in real-time on edge computing hardware.

2.2.4 To engineer a real-time alerting engine that triggers an immediate audible warning for the driver upon violation detection and automatically generates and transmits a notification to relevant authorities when a driver commits multiple violations within a defined short time threshold (e.g., three violations in 30 minutes).

2.2.5 To develop a centralized web dashboard for transport authorities that provides real-time fleet monitoring, a historical log of all violations with associated evidence (time, location, image), and analytics on driver behavior and violation trends.

2.2.6 To conduct rigorous field testing on operational long-distance buses to validate the system's accuracy, reliability, and overall effectiveness in detecting violations and facilitating timely interventions in a real-world environment.

How Specific Objectives Achieve the Main Objective

The system operates as a cohesive pipeline. The hardware unit (Objective 2.2.1) serves as the sensory foundation, collecting essential GPS and visual data. This raw data is processed by the sensor fusion algorithm (Objective 2.2.2) to create a reliable understanding of the bus's location and context. Simultaneously, the computer vision model (Objective 2.2.3) analyzes the video stream to identify dangerous driving maneuvers. The outputs from these modules are integrated by the alerting engine (Objective 2.2.4), which enforces the dual-alert mechanism providing immediate feedback to the driver and escalating repeated offenses to authorities. The web dashboard (Objective 2.2.5) aggregates all this information, offering authorities a comprehensive tool for oversight and data-driven decision-making. Finally, the field study (Objective 2.2.6) validates that all components work together seamlessly in a real-world setting, ensuring the main objective of creating a safer, more compliant bus network is effectively met.

3 METHODOLOGY

3.1 System Overview

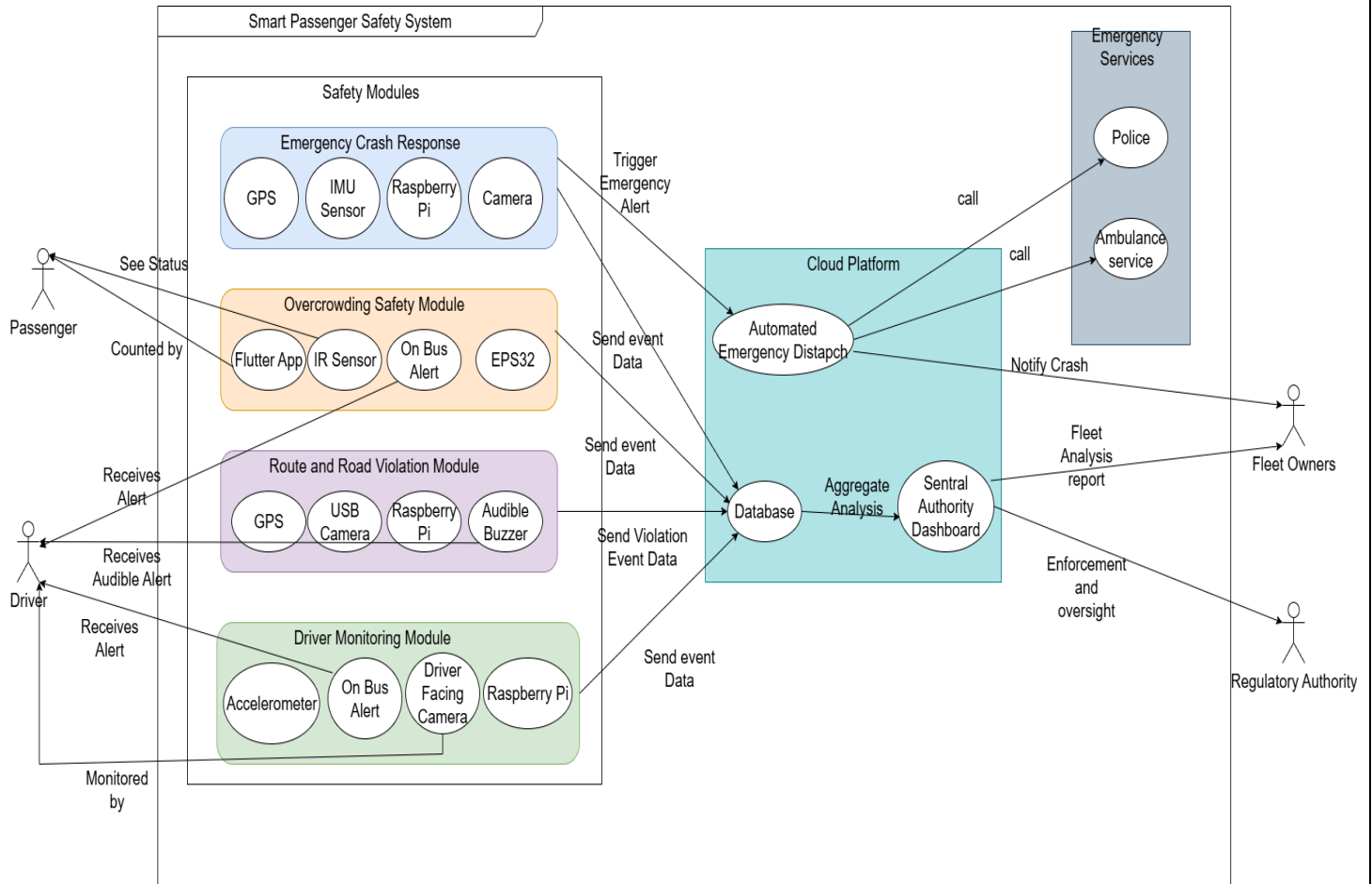


Figure 3.1: System Overview Diagram

This project proposes a fully integrated Smart Safety Monitoring System designed to improve the safety, reliability, and accountability of long-distance bus transport in Sri Lanka. The system is composed of four interconnected modules that together create a comprehensive safety net, addressing risks from passenger overcrowding to driver fatigue and crash response.

1. Overcrowding & Passenger Safety Module

This module focuses on preventing dangerous overcrowding and footboard riding. Infrared (IR) sensors placed at the bus doors count passengers as they board and alight, ensuring real-time tracking of passenger numbers without compromising privacy. An ESP32 microcontroller processes this data and updates the live passenger count on the conductor's mobile app. If the bus approaches or exceeds capacity, or if footboard riding is detected, the system issues immediate audiovisual alerts to the crew. These events, tagged with GPS location and time, are also uploaded to the central dashboard for regulatory monitoring.

2. Route & Road Violation Detection Module

This component detects unsafe driving behaviors such as route deviations, illegal overtaking, and running red lights, double line crossing, violating no entry signs, and violating speed limit signs. A GPS module and AI-powered road facing camera work together to monitor bus location and road conditions. A Raspberry Pi running a locally trained YOLOv8 model identifies violations in real time. Audible alerts are provided to the driver, while detailed violation reports with photo evidence are transmitted to the cloud for enforcement and analysis.

3. AI Driven Driver Monitoring Module

To address accidents caused by fatigue and distraction, this module uses a driver-facing camera and accelerometer to track the driver's alertness and driving style. Computer vision techniques detect drowsiness, distraction (e.g., phone use), and aggressive driving patterns. Immediate alerts are given inside the cabin to help the driver correct behavior before it leads to an incident. Logged events are stored in the cloud, enabling fleet managers and authorities to identify high-risk drivers and enforce rest periods.

4. Emergency Crash Response System (ECRS)

In the event of a crash, rapid response is critical. The ECRS uses a combination of an accelerometer, a gyroscope, and a GPS module to automatically detect a collision and classify its severity. To provide immediate visual context, the system also captures a snapshot from the onboard road-facing camera upon impact. Once a crash is confirmed, the system sends instant alerts including the exact GPS location, severity level, and the incident snapshot to emergency services and the fleet operator. This automation reduces dependence on manual reporting, provides critical information to first responders, and significantly speeds up emergency response times.

System Integration

The strength of the proposed system lies in its integration. All modules feed data into a unified cloud platform, providing authorities with a single, comprehensive view of bus safety. Hardware components such as GPS and communication modules are shared across systems to minimize costs. A layered alert mechanism ensures proactive, real-time interventions for the driver and conductor, cloud-based reporting for long-term enforcement, and automatic emergency dispatch for worst-case scenarios.

Importantly, the system is designed with privacy in mind. Passenger counting relies on IR sensors rather than cameras, and driver monitoring data is processed locally on the edge device, with only essential violation snapshots being sent to the cloud.

Together, these modules create a safer and more accountable public transportation network, aligning with the national goal of reducing road accidents and improving passenger welfare.

3.2 Component Overview

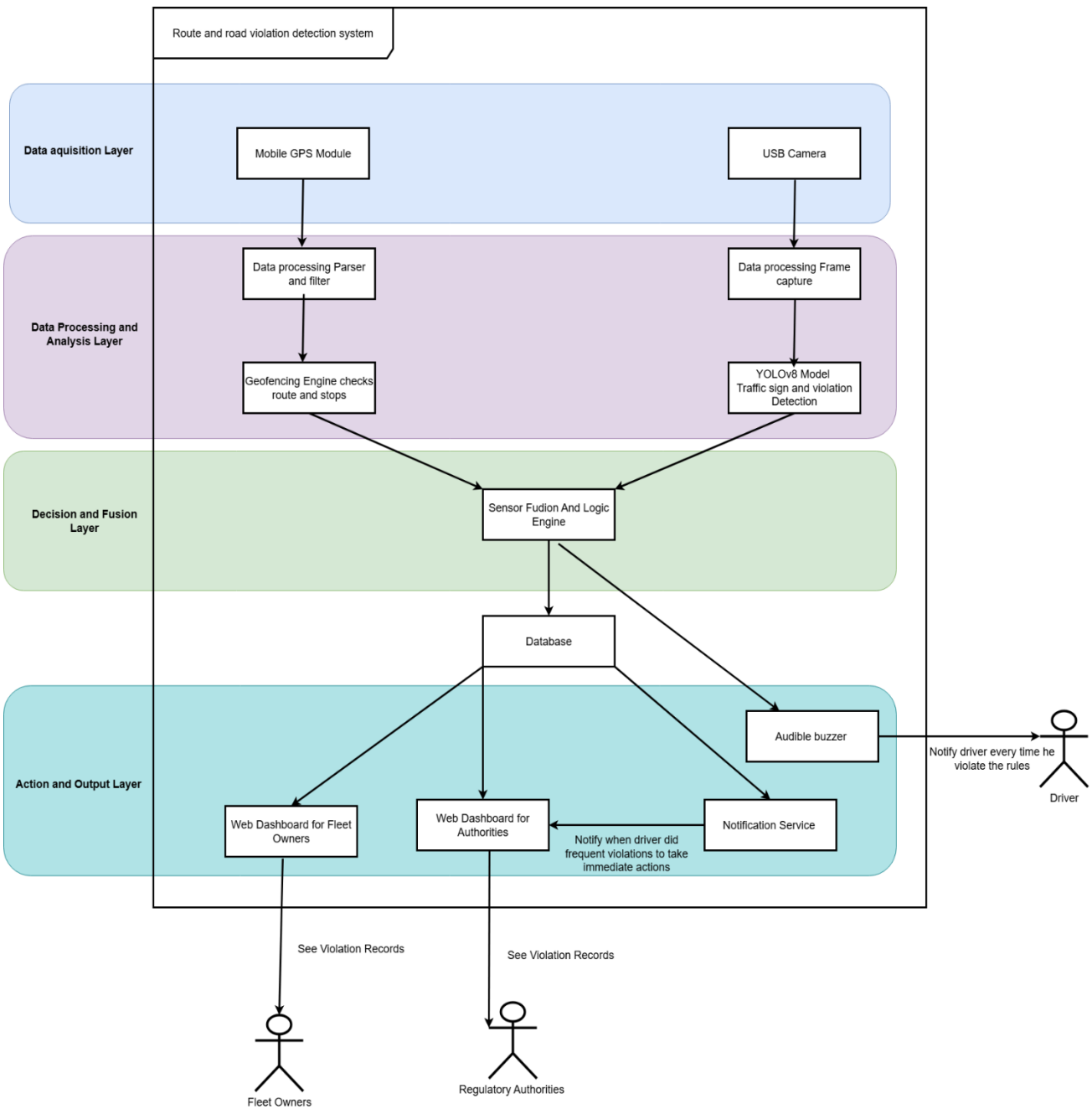


Figure 3.2: Component Overview Diagram

The Route and Road Violation Detection System is designed as an integrated, edge-based platform that combines real-time Global Positioning System (GPS) data with visual analysis to autonomously identify traffic violations. This system operates locally on the bus, ensuring low latency, high reliability, and independence from continuous internet connectivity. The architecture, illustrated in *Figure 3.2*, is organized into four primary layers, Data Acquisition, Data Processing and Analysis, Decision and Fusion, and Action and Output. Each layer is described in detail below.

1. Data Acquisition Layer

The first layer is responsible for continuously collecting raw, real-time data from the operating environment of the bus.

- I. GPS Module: This module provides accurate geospatial data, including the bus's current location, speed, and heading, which are crucial for route tracking and violation detection.
- II. USB Camera: A forward-facing camera is mounted to capture live video streams of the road ahead. This provides the visual context required to recognize road signs, lane markings, traffic lights, and the behavior of other vehicles.

2. Data Processing and Analysis Layer

The second layer forms the computational core of the system, where raw input data is transformed into meaningful, actionable information.

- I. GPS Data Parsing and Filtering: The raw GPS stream is cleaned and processed to remove noise, ensuring only reliable data are passed to subsequent modules.
- II. Geofencing Engine: The processed GPS data are compared against a predefined digital map representing the authorized bus route. Deviations from the prescribed route are automatically flagged as violations.
- III. Video Frame Extraction: Frames are periodically sampled from the camera feed for analysis.

- IV. YOLOv8 Model for Traffic Violation Detection: These frames are processed using a fine-tuned YOLOv8 deep learning model, capable of detecting road elements such as lane markings, traffic lights, stop signs, and oncoming vehicles. The model identifies potential violations, including lane-crossing during double lines or failure to stop at a red light.

3. Decision and Fusion Layer

This layer integrates and contextualizes data from both the GPS and computer vision subsystems, ensuring accurate and context-aware violation detection.

- I. Sensor Fusion and Logic Engine: The outputs from the Geofencing Engine and the YOLOv8 model are intelligently combined to make final decisions. For example, a simple route deviation can be confirmed based on GPS data alone, whereas an overtaking maneuver is only flagged as illegal if GPS data confirms movement while the vision system detects a double-line crossing. This approach reduces false positives and enhances system reliability.

4. Action and Output Layer

Once a violation has been confirmed, the system initiates appropriate real-time responses and archival actions.

- I. Audible Alert to Driver: The system immediately triggers a buzzer within the driver's cabin, enabling rapid behavioral correction and preventing potential accidents.
- II. Cloud-Based Logging (Firebase): Each violation event is logged with metadata, including timestamp, GPS coordinates, and an image snapshot, and stored securely in a cloud database.
- III. Notification and Reporting: A notification service monitors the database for new events and updates a centralized web dashboard accessible to fleet operators and transport authorities. This dashboard allows authorities to review violation evidence, identify repeat offenders, and take corrective or enforcement measures.

In summary, the system operates continuously by monitoring GPS and video data streams in real-time. When the logic engine detects a violation, it performs two simultaneous actions, it issues an immediate audible alert to the driver and recording the incident with evidence in the cloud database. The recorded data are then made available to authorized personnel through the web dashboard, ensuring a closed-loop system for proactive correction, record-keeping, and data-driven enforcement.

3.3 Data Requirements for AI-Powered Violation Detection

A core component of this research is the development of a robust, real-time violation detection system. The accuracy and reliability of the AI models are directly dependent on the quality, diversity, and contextual relevance of the data used for training and operation. Therefore, a comprehensive and well-structured dataset is absolutely required.

Data Points Required

The system operates on a continuous stream of data from multiple sensors. For each event logged, such as violation detection, a periodic update, or a triggered alert, the dataset will include the following features:

- **bus_id**: Unique identifier for the vehicle.
- **timestamp**: Precise date and time of the event.
- **gps_latitude, gps_longitude**: Real-time geographical coordinates of the bus.
- **gps_speed**: Instantaneous speed of the vehicle.
- **gps_heading**: Direction of travel.
- **camera_frame_id**: Identifier linking to the specific image or video frame analyzed.
- **yolo_detections**: A structured list of objects detected by the YOLOv8 model (e.g., {'object': 'double_line', 'confidence': 0.95, 'bounding_box': [x1, y1, x2, y2]}).
- **violation_type**: Categorical label for the detected violation (e.g., route_deviation, double_line_crossing, red_light_violation).

- **violation_confidence:** A probability score indicating the AI's confidence in the detection.
- **geofence_status:** Status from the geofencing engine (e.g., on_route, deviated).

Data Collection Method

The primary method for data collection is through the integrated IoT sensor units deployed on the buses. A core design principle is privacy preservation and storage efficiency. Therefore, the system is engineered not to store continuous video footage of the entire journey. Instead, it employs an intelligent, event-triggered recording and logging mechanism.

The process is as follows:

1. Continuous Real-Time Analysis:

The onboard USB camera captures a live video stream, which is fed directly into the edge processing unit (Raspberry Pi 4) for real-time analysis by the YOLOv8 model. The raw video data is not saved to storage during normal operation. It is processed frame by frame and then discarded.

2. Event Triggering:

The system continuously monitors the outputs of the YOLOv8 model and the Geofencing Engine. A "violation event" is triggered only when a predefined rule is breached (e.g., the bus crosses a double line while the geofence indicates it is on a valid route).

3. Selective Data Capture:

Upon triggering an event, the system performs the following actions:

- **Image Snapshot:** A still image (high-resolution frame) from the video stream is captured at the moment of the violation. This snapshot contains the visual evidence required for verification.
- **Structured Data Logging:** All relevant sensor data (e.g., bus_id, timestamp, GPS coordinates, violation_type) is packaged together with the image snapshot.

4. **Secure Transmission:**

This small, event-based data package (a few kilobytes of data + an image file) is transmitted to the cloud server (Firebase) for storage and further action. This method reduces bandwidth and storage costs by over 99% compared to full journey recording.

This approach ensures that only forensically relevant data corresponding to actual safety-critical events is stored and transmitted. It effectively addresses storage constraints, upholds privacy by minimizing data collection, and provides authorities with concise, actionable evidence.

Dataset Curation for Model Training

For initial model training, the system uses the Sri Lanka Road Safety Dataset hosted on Roboflow, which is currently under active development.

- Current Dataset Status:
- Images: 839 annotated images
- Annotations: 3,888 labeled instances
- Focus: Sri Lankan road environments and violation scenarios

The dataset is being expanded through ongoing field collection to improve coverage of critical violation classes, particularly double-line detection. This approach provides local context while allowing continuous improvement based on real-world performance.

3.4 Algorithm Selection and Technical Justification

The Route and Road Violation Detection System uses carefully chosen algorithms to ensure it works in real time, stays accurate in different situations, and adapts well to Sri Lankan road conditions. By combining modern computer vision techniques with reliable geometric algorithms, the system creates a strong and dependable framework for detecting traffic violations.

1. YOLOv8 for Real-Time Object Detection

Justification for Selection

YOLOv8 was chosen over other object detection models due to its optimal balance of speed and accuracy, critical for real-time processing on moving vehicles. Unlike two-stage detectors like Faster R-CNN that prioritize accuracy at the cost of speed, YOLOv8 processes images in a single forward pass, achieving the low latency required for immediate violation alerts.

Localized Customization:

The model is fine-tuned on the Sri Lankan Road Safety Dataset to address the significant gap in generic models' understanding of local infrastructure. Pre-trained models (e.g., on COCO or BDD100K) fail to recognize region-specific signage, road markings, and vehicle types common in Sri Lanka. This customization ensures high precision in detecting:

- Local double-line markings with varying wear patterns
- Sri Lankan traffic sign variants

2. Geofencing Algorithm for Route Compliance

Strategic Implementation:

The geofencing system employs polygonal geofences with configurable buffers rather than simple circular boundaries. This accounts for GPS inaccuracies and legitimate road variations while maintaining strict route adherence monitoring. The algorithm incorporates:

- Temporal logic to distinguish unauthorized deviations

- Speed-based validation to ignore brief GPS drift at traffic signals
- Multi-level geofencing for routes, and restricted zones

Advantage over Pure Vision-Based Systems:

Unlike camera-only systems that struggle with route context, geofencing provides absolute positional reference independent of visual conditions. This ensures reliable route monitoring when computer vision capabilities are compromised.

3. Sensor Fusion and Logic Engine

Contextual Violation Verification:

The logic engine implements a multi-stage validation process to eliminate false positives:

Example: Illegal Overtaking Validation

1. YOLOv8 detects a double-line crossing event
2. Geofencing confirms the vehicle is on a valid route segment
3. Speed validation checks if overtaking is occurring
4. Temporal analysis verifies sustained violation
5. Context checks for emergency or unavoidable scenarios

Fusion Methodology:

The system uses a certainty-weighted decision matrix where:

- Camera certainty is weighted by detection confidence and image quality
- GPS reliability is weighted by signal strength and positional accuracy
- Temporal consistency requires sustained violation patterns

This approach prevents false alarms from transient events while ensuring genuine violations are captured with high confidence.

Technical Integration Advantages

Computational Efficiency:

The combination allows for distributed processing:

- YOLOv8 runs on dedicated edge hardware (Raspberry Pi 4+)
- Geofencing operates on lightweight GPS data streams
- Logic engine fuses pre-processed outputs with minimal overhead

Adaptability to Local Conditions:

The modular design enables continuous improvement:

- YOLOv8 model can be retrained with new local data
- Geofencing rules can be updated for route changes
- Logic thresholds can be adjusted based on violation patterns

Performance Validation

The algorithm combination is designed to achieve:

- ≤ 2 second detection latency from violation to alert
- $\geq 95\%$ precision in violation classification
- $\leq 5\%$ false positive rate in diverse conditions
- Adaptive performance across varying weather and lighting

This technical foundation offers a solution that is accurate, adaptable, and easy to scale, specifically designed to tackle the unique challenges of monitoring road safety in Sri Lanka.

3.5 Technologies and Tools

1. Python – The primary programming language for developing the violation detection algorithms, GPS data processing, sensor fusion logic, and backend services. Its rich ecosystem of libraries supports rapid prototyping and deployment.
2. OpenCV – Used for real-time video processing, image manipulation, and feature extraction essential for detecting traffic violations and analyzing road conditions.
3. PyTorch / TensorFlow Lite – PyTorch is used for training and fine-tuning the YOLOv8 model on the custom Sri Lankan dataset. TensorFlow Lite facilitates the deployment of the optimized model on the Raspberry Pi for efficient edge inference.
4. Node.js – Powers the backend server for handling HTTP requests, processing violation data, and managing communication between the onboard device and the cloud database.
5. Firebase – A NoSQL cloud database used for storing violation records, device metadata, and real-time alert logs. Its scalability and real-time synchronization capabilities are critical for the monitoring dashboard.
6. Google Maps Platform API – Provides geofencing capabilities by dynamically generating virtual boundaries for bus routes and stops, enabling precise detection of route deviations and skipped stops.
7. React.js – Used to develop the interactive web dashboard for authorities and operators to monitor violations, view analytics, and generate compliance reports.
8. Chart.js – Integrated into the React dashboard for visualizing violation trends, bus performance metrics, and geographical data through maps and graphs.

9. Docker – Containerizes application components to ensure consistent development, testing, and deployment environments across different stages of the project.
10. Git / GitHub – Manages version control, collaboration, and project documentation, ensuring code integrity and streamlined teamwork.
11. Raspberry Pi OS (Legacy) – A lightweight, Linux-based operating system optimized for running the onboard software stack efficiently on the Raspberry Pi hardware.
12. Postman – Used for testing API endpoints during development, ensuring reliable communication between the onboard device, backend server, and cloud services

3.5.1 Hardware components

The hardware components were selected to develop a system that is cost-effective, reliable, and capable of real-time processing in the challenging environment of a moving bus. Each component was chosen to fulfill specific functional requirements while ensuring the overall feasibility and scalability of the system.

1. Raspberry Pi 4 Model B (4GB RAM)

Justification: The Raspberry Pi 4 was selected as the central processing unit over other Single-Board Computers (SBCs) like the NVIDIA Jetson Nano due to its optimal balance of cost, community support, and sufficient computational power. Its 4GB RAM and quad-core processor are adequate for simultaneously running the Linux-based operating system, processing GPS data streams, and executing the optimized YOLOv8 model for computer vision. Its General-Purpose Input/Output (GPIO) pins are essential for directly interfacing with peripheral components like the buzzer and GSM module without requiring additional controllers, simplifying the system architecture and reducing costs.

2. Generic USB Webcam (1080p) with Night Vision

Justification: A standard USB webcam was chosen over more expensive specialized cameras (e.g., CSI cameras) or high-end dashcams for its plug and play simplicity, low cost, and wide availability. The 1080p resolution provides enough detail for the AI model to accurately detect violations such as double line crossing and traffic signs. The inclusion of night vision capability is critical for ensuring consistent system operation during night journeys and in low light conditions, which is a common scenario for long distance buses. This makes the system viable around the clock.

3. Mobile GPS Module (SIM7600E 4G/GPS)

Justification: While a simple GPS module could be used, a module like the SIM7600E that integrates both GPS and 4G connectivity was selected for a key reason: it provides an all-in-one solution for location tracking and data communication. This eliminates the need for separate components (a GPS receiver and a 4G dongle), reducing complexity and potential points of failure. The 4G capability ensures that violation data and alerts can be transmitted in real-time to the cloud database (Firebase), which is crucial for the functionality of the web dashboard and authority alerts.

4. SIM800L GSM Module

Justification: The SIM800L GSM module serves as a critical communication backup. Although the primary 4G module (in the SIM7600E) handles data transmission, cellular data coverage can be unreliable in remote areas of Sri Lanka. The SIM800L provides a fallback mechanism specifically for sending crucial SMS alerts to authorities in the event of repeated violations, even when internet connectivity is lost. This redundancy ensures that critical safety alerts are never missed due to network issues, enhancing the system's robustness.

5. MicroSD Card (32GB)

Justification: A high endurance 32GB MicroSD card is essential for reliable operation. It not only hosts the operating system and application code but also provides necessary storage for temporary data logging. In scenarios where internet connectivity is temporarily lost, the system can store violation events

with image evidence locally on the SD card and synchronize with the cloud once the connection is restored. This prevents data loss and is a fundamental feature for any reliable IoT device operating in the field.

6. Buzzer/Speaker

Justification: An audible alert system is a cornerstone of the project's proactive safety approach. A simple buzzer or speaker is the most direct and effective way to provide immediate feedback to the driver. Unlike visual alerts on a screen that could distract the driver, an audible buzzer provides an instant, unambiguous warning without requiring the driver to look away from the road. This immediate intervention is key to correcting behavior in real-time and preventing potential accidents.

3.6 Testing and Validations

The system will undergo rigorous testing to ensure reliability and accuracy. This includes:

1. Unit Testing

Each module (GPS tracking, vision model, fusion engine) will be tested individually to verify correct functionality.

2. Integration Testing

The complete system will be tested to ensure all modules work together seamlessly in a controlled lab environment.

3. Field Testing

The prototype will be installed on a bus for real-world testing on Sri Lankan roads. This will validate performance under actual operating conditions, including different terrains, weather, and lighting.

4. User Acceptance Testing (UAT)

The web dashboard and alert system will be evaluated by a group of actual fleet managers and traffic authorities. Their feedback on usability, interface design, and the usefulness of the reported data will be collected and used to refine the system to ensure it meets user needs and expectations.

3.7 Functional And Non-Functional Requirements

3.7.1 Functional Requirements

FR1: Real-time Location and Route Monitoring

- FR1.1: The system shall continuously track the bus's geographical coordinates using GPS technology.
- FR1.2: The system shall maintain a database of approved bus routes.
- FR1.3: The system shall generate a route deviation violation if the bus's position deviates from the assigned route for a continuous period exceeding one minute.

FR2: Traffic Violation Detection

- FR2.1: The system shall analyze the real-time video feed to detect and classify specific traffic violations, including:
 - FR2.1.1: Wrong overtaking (illegal passing maneuvers).
 - FR2.1.2: Crossing double solid lines.
 - FR2.1.3: Disobeying traffic light signals (red light running).
 - FR2.1.3: Disobeying the no-entry signs
 - FR2.1.3: Disobeying speed limit signs
- FR2.2: For every detected violation, the system shall capture and store a still image from the video feed as evidentiary proof.

FR3: Alerting and Notification System

- FR3.1: Upon violation detection, the system shall trigger an immediate audible alert (≥ 80 dB) within the driver's cabin within two seconds of the event.
- FR3.2: The system shall implement an escalation protocol whereby if a driver accumulates three or more violations within a 30-minute rolling window, an automated alert shall be sent to predefined traffic authorities.

FR4: Data Logging and Management

- FR4.1: The system shall record a comprehensive log for each violation, including a unique identifier, timestamp, GPS coordinates, violation type, and associated image evidence.
- FR4.2: All violation records shall be transmitted to and stored securely in a cloud-based database.

FR5: Monitoring and Reporting Interface

- FR5.1: Authorized personnel shall access a web-based dashboard to view the real-time location of all monitored buses on a map interface.
- FR5.2: The dashboard shall provide a searchable and filterable history of all recorded violations.
- FR5.3: The dashboard shall generate analytical reports on driver behavior patterns and violation trends over configurable time periods.

3.7.2 Non Functional Requirements

NFR1: Performance and Reliability

- NFR1.1: The system shall maintain continuous operation for a minimum of 12 hours, corresponding to a standard bus shift, without requiring a reboot.
- NFR1.2: The end-to-end latency for driver alerts (from violation occurrence to buzzer activation) shall not exceed three seconds.

- NFR1.3: The system shall implement an offline caching mechanism to store violation data locally during network outages, with automatic synchronization upon connection restoration.

NFR2: Usability

- NFR2.1: The driver alert mechanism shall be unambiguous and require no interpretation from the driver.
- NFR2.2: The administrative web dashboard shall feature an intuitive user interface designed for users with minimal technical training.

NFR3: Security

- NFR3.1: All data transmitted between the onboard unit and the cloud server shall be encrypted using industry-standard protocols (e.g., TLS 1.2+).
- NFR3.2: Access to the web dashboard shall be controlled via role based authentication.

NFR4: Cost and Compatibility

- NFR4.1: The total hardware cost per unit shall be minimized through the selection of commercially available, off-the-shelf components.
- NFR4.2: The physical design of the onboard unit shall be modular and adaptable for installation across various bus models without structural modifications.

NFR5: Maintainability

- NFR5.1: The system software shall be designed with a modular architecture to facilitate independent updates and troubleshooting.

3.7.3 Stakeholder Requirements

Primary Stakeholders

1. Bus Drivers

- Real-time Violation Alerts: Immediate, clear audible buzzer within the cabin when a violation is detected to enable instant corrective action.
- Transparency and Fairness: A system that accurately detects violations to ensure they are not penalized for false positives or system errors.
- Non-Intrusive Operation: Hardware that does not obstruct the driver's view or interfere with the primary task of driving.

2. Fleet Managers / Bus Owners (Private & SLTB)

- Operational Oversight: A web dashboard to monitor bus locations in real-time and track adherence to assigned routes and schedules.
- Driver Behavior Analytics: Access to historical data and reports on driver violations (speeding, route deviations, etc.) to identify risky drivers for targeted training.
- Evidence for Disputes: Logged data (timestamp, location, image) to verify incidents, handle passenger complaints, and resolve insurance claims.
- Cost-Effective Solution: A low-cost, scalable system that provides a clear return on investment through improved safety and operational efficiency.

3. Traffic Police & Regulatory Authorities

- Proactive Enforcement: Automated, real-time alerts for serious or repeated violations (e.g., three violations in 30 minutes) to enable timely intervention.
- Data-Driven Enforcement: Access to violation trends and hotspot analysis to optimize the deployment of patrols and checkpoints.

- **Auditable Evidence:** A secure database of violations with photographic proof to support fines and legal actions, ensuring transparency and reducing disputes.

Secondary Stakeholders

1. Passengers

- **Safety Assurance:** An indirect requirement for a system that actively works to reduce accidents caused by reckless driving and route violations.
- **Improved Service Reliability:** More consistent and predictable journey times due to enforced route compliance and reduced risky driving.

2. Government and Policy Makers

- **Policy Formulation Insights:** Aggregated, anonymized data on common violation types and locations to inform better road safety policies and infrastructure planning.
- **Measurable Safety Outcomes:** Concrete metrics and reports demonstrating a reduction in accidents and violations attributable to the system's implementation.

3. Insurance Companies

- **Risk Assessment Data:** Access to driver safety records and fleet compliance data to create personalized insurance premiums based on actual risk, rewarding safe operators.

3.8 Commercialization

Table 3.8: Business Model Canvas

Key Partnerships  - Sri Lanka Transport Board (SLTB) - Private Bus Owner Associations - Insurance Emergency Services Companies - Government & Regulatory Authorities - Emergency Services	Key Activities  - Integrated Software & Hardware R&D - System Integration Data Analytics & Support& Deployment - Sales, Marketing & Pilot Management Key Resources  - Proprietary Technology Stack - Hardware Design - Cloud Infrastructure - Human Capital	Value Propositions  - Comprehensive Safety Ecosystem - Tangible ROI for Fleet Owners - Proactive & Automated Enforcement for Authorities - Enhanced Passenger Trust	Customer Relationships  - Automated Dashboard & Self-Service - Personalized Account Management - Co-creation and Feedback Loops - Proactive Support Channels  - Direct Enterprise SalesProactive Support - Partner Networks - Pilot Project Demonstrations - Online Platform	Customer Segments  - Primary Segment: Private Long-Distance Bus Fleet Operators. - Secondary Segment: Sri Lanka Transport Board (SLTB). - Tertiary Segment: Government Transport Authorities.
Cost Structure  - Hardware (COGS): Cost of components for each unit (Sensors, RPi, ESP32, 4G modem). - Cloud Services: Recurring costs for data storage, processing, and API calls (Firebase, Maps API). - R&D & Salaries: Ongoing development and technical team salaries. - Sales & Marketing: Costs associated with pilot deployments and business development.		Revenue Streams  - Hardware Kit Sale: A one-time sale of the integrated onboard safety device. - Software-as-a-Service (SaaS) Subscription Fee: Recurring revenue (monthly/annually per bus) for access to the cloud dashboard, real-time alerts, analytics, and compliance reporting. This is the core long-term revenue stream. - Value-Added Services: Fees for professional installation, system maintenance, and customized reporting packages. - Data Monetization (Future Opportunity): Revenue from selling non-identifiable, aggregated data insights to urban planners, insurance companies, and government		

Addressing the critical safety and efficiency challenges within Sri Lanka's bus transportation sector, the Integrated Smart Safety Monitoring System offers a comprehensive ecosystem designed to create tangible value for all stakeholders. Our solution moves beyond single-issue fixes to tackle the root causes of accidents and operational inefficiencies simultaneously.

Our primary customers are private long-distance bus fleet operators and the Sri Lanka Transport Board (SLTB). These groups face a range of interconnected

challenges, including rising costs from accidents and vehicle downtime, unpredictable fuel expenses caused by route deviations, high insurance premiums, passenger distrust due to reckless driving, and difficulties in proactively managing driver fatigue and overcrowding.

Our system confronts these challenges with a unified technological approach. We provide fleet owners with a clear return on investment by reducing costs and mitigating risks. For instance, getting real-time alerts when drivers break rules helps reduce fuel waste, and the insights from monitoring driver fatigue and bus overcrowding can even help secure lower insurance costs. For authorities like the SLTB, the system shifts safety management from reactive to proactive, providing data-driven insights for targeted interventions and demonstrating a strong commitment to passenger safety, which can help rebuild public trust.

Our market entry strategy is built on collaboration and demonstrable success. We will initiate controlled pilot projects with key partners like the SLTB or major private bus associations. These pilots serve as live testing grounds and powerful marketing tools, generating credible testimonials and case studies. Following this, we will leverage these partnerships to reach the wider market, including developing strategic alliances with insurance companies to create safety-score-based premium models.

The business model is designed for sustainability and growth. Revenue comes from a one-time hardware kit sale and a recurring Software-as-a-Service (SaaS) subscription, ensuring long-term viability and continuous system improvements. A future opportunity lies in monetizing anonymized, aggregated data on traffic patterns and safety hotspots for urban planners and government agencies, creating an additional revenue stream while contributing to broader road safety initiatives.

In the long term, our cost-driven, scalable design allows for nationwide deployment across both public and private fleets. The core promise of the Integrated Smart Safety Monitoring System is to transform safety from a mere compliance cost into a source of profitability and competitive advantage for operators. Ultimately, we aim to foster a safer, more efficient, and more trustworthy public transportation ecosystem for Sri Lanka, where operators, passengers, and authorities all benefit.

3.9 Work Breakdown Structure

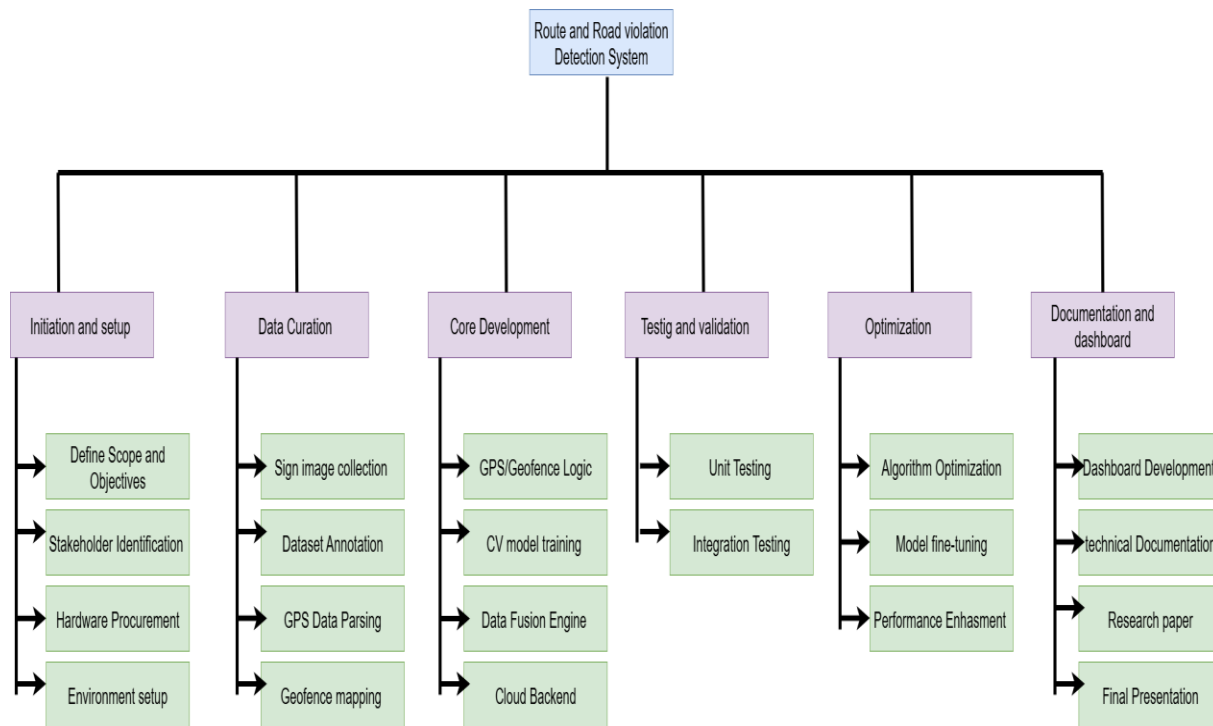


Figure 3.9: Work Breakdown Structure

The project is divided into six phases to make sure everything is carried out in a structured way. It starts with Project Initiation, where the scope is defined and the setup is done. Next is Data Curation, which involves collecting and labeling Sri

Lankan traffic sign images and mapping bus routes. The Core Development phase follows, focusing on building the three main modules: the GPS and geofencing system, the computer vision model, and the data fusion engine that combines them. After that, the system goes through Testing and Validation, both in the lab and on real buses, to check accuracy and reliability. Based on these results, the Optimization phase fine tunes the system for better performance. Finally, the project wraps up with Documentation and Dashboard Development, where the web dashboard is finalized and the research findings are compiled into a complete report.

3.10 Grantt Chart

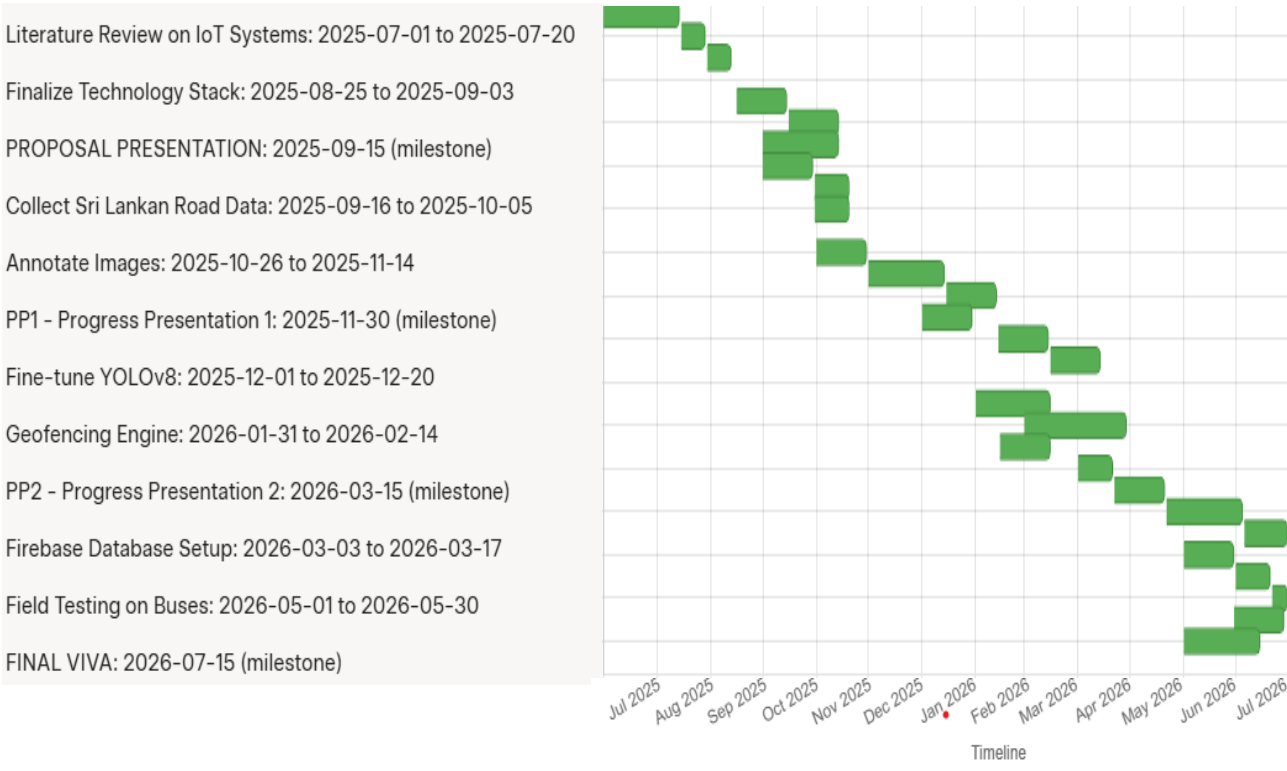


Figure 3.10: Grantt Chart

4 DESCRIPTION OF PERSONAL AND FACILITIES

For the successful execution of the research project "Route and Road Violation Detection System for Long-Distance Buses in Sri Lanka," the researcher will utilize a combination of personal and institutional resources.

On the personal side, the researcher has access to a mid-range laptop capable of handling software development, data analysis, and light model training tasks. This will be used for coding the system, processing GPS data, and developing the web dashboard. A personal smartphone will also be utilized for testing real-time alert notifications and accessing the cloud dashboard.

Importantly, the researcher has already acquired the core hardware components required to build a functional prototype. This includes a Raspberry Pi 4 Model B, a Raspberry Pi Camera Module, and a SIM7600E 4G/GPS module. Having these components on hand will allow for immediate experimentation, hardware-software integration testing, and the collection of real-world data from test journeys.

Institutional facilities provided by the Department of Software Engineering at the Sri Lanka Institute of Information Technology (SLIIT) will greatly enhance the research environment. These include:

- **High-Performance Computing Lab:** Access to workstations with enhanced GPUs will be crucial for training and fine-tuning the YOLOv8 computer vision model on the curated Sri Lankan dataset, a task that is computationally intensive and impractical on personal hardware.
- **High-Speed Internet Access:** Essential for downloading large datasets, accessing cloud services (Google Maps API, Firebase), and utilizing online development platforms like Google Colab.
- **Software and Development Tools:** Access to licensed software tools, Python libraries, and IDEs (e.g., PyCharm, VS Code) that are necessary for professional-grade development.
- **Library and Online Research Resources:** Access to IEEE Xplore, ACM Digital Library, and other academic databases will be vital for conducting the literature review and staying updated on relevant research throughout the project.

- Supervision and Expertise: Continuous guidance and feedback from the project supervisor, Mrs. Chathurangika Kahandawaarachchi, will ensure the academic rigor, proper methodology, and overall alignment of the project with its objectives.

Together, these personal and institutional resources create a robust and practical foundation for the design, development, testing, and validation of the proposed violation detection system.

5 BUDGET AND BUDGET JUSTIFICATION

5.1 Budget

This section provides a detailed financial plan for the development and pilot testing of the proposed Route and Road Violation Detection System. The estimated budget encompasses all necessary expenditures for hardware acquisition, software service subscriptions, operational connectivity, and ancillary costs. All figures are based on current market valuations and are subject to variation due to economic fluctuations.

Table 5.1 : Component Budget Allocation

Budget Item	Description	Amount
Hardware Components	Procurement of essential electronic components for the prototype unit, including a Raspberry Pi 4, Mobile GPS module, USB camera module, and associated peripherals.	15,000
Cloud Services & APIs	Subscription fees for critical software services, including the Google Maps Platform for geofencing and the Firebase Firestore database for data storage and management.	10,000
Data Connectivity	Cost of mobile data packages required for the GSM module to transmit real time telemetry and violation data during the field testing phase.	5,000
Miscellaneous & Contingency	Allocation for unforeseen expenses, transportation for field deployment, and ancillary academic costs.	5,000
Total Estimated Budget		35,000

5.2 Budget Justification

- **Hardware Components (LKR 15,000)**

This budget allocation is for purchasing the essential hardware needed to build a working prototype. The Raspberry Pi will act as the main processor, plan to use a mobile GPS as a GPS module and the USB camera is required for the computer vision system. These components are crucial for turning the concept into a practical, testable solution.

- **Cloud Services & APIs (LKR 10,000)**

These costs cover essential software services that the system relies on. The Google Maps API is key for creating dynamic geofences and checking bus routes, while Firebase Firestore provides a secure and scalable cloud database to store violation records, which is important for both evidence tracking and the monitoring dashboard.

- **Data Connectivity (LKR 5,000)**

This cost covers the cellular data plan needed for the prototype. A reliable internet connection is essential to send GPS data, violation alerts, and images from the bus to the cloud in real time during pilot testing.

- **Miscellaneous & Contingency (LKR 5,000)**

This budget is set aside for unexpected costs during development and testing. It may cover things like transportation for field tests, small replacement parts, and other incidental expenses that come up.

6 REFERENCES

- [1] S. L. Transport Board, *Annual Performance Report 2023*, Colombo, Sri Lanka: SLTB Publications, 2024.
- [2] R. M. P. S. Bandara and K. A. S. Fernando, "An analysis of traffic law enforcement and accident causation in Sri Lanka," in *Proc. Int. Conf. on Sustainable Built Environment*, Kandy, Sri Lanka, 2022, pp. 112 119.
- [3] T. G. I. Kumar and M. P. N. Pathirana, "IoT based fleet management systems: Challenges and opportunities in developing countries," *IEEE Internet of Things Journal*, vol. 10, no. 5, pp. 4321 4333, Mar. 2023.
- [4] A. Gupta, S. Mishra, and R. K. Tiwari, "Real time vehicle tracking and monitoring system using GPS and GSM," in *Proc. IEEE Int. Conf. on Computing, Communication & Automation*, Greater Noida, India, 2019, pp. 1374 1379.
- [5] L. D. Wickramasinghe and H. K. G. Punchihewa, "Challenges in implementing intelligent transportation systems (ITS) in Sri Lanka: A review," *Engineer: Journal of the Institution of Engineers, Sri Lanka*, vol. LV, no. 4, pp. 1 9, Oct. 2022.
- [6] Y. Wang, J. Zhao, and X. Li, "A novel vehicle violation detection system based on deep learning and geofencing technology," *IEEE Access*, vol. 9, pp. 109 456 109 466, 2021.

- [7] M. A. S. Perera and C. P. Silva, "Impact of digital monitoring on driver behavior in public transport: A case study," in *Proc. IEEE Int. Conf. on Advanced Topics in Mechatronics*, Bangkok, Thailand, 2023, pp. 88 93.
- [8] M. A. S. Perera, C. P. Silva, and L. R. Wimalasiri, "Computer Vision for Automated Traffic Law Enforcement in Mixed Traffic Conditions," *IEEE Transactions on Intelligent Transportation Systems*, vol. 24, no. 8, pp. 8765 8777, Aug. 2023.
- [9] P. Jain, R. Agarwal, and N. Gupta, "A comprehensive study on AI and IoT based transportation systems," *IEEE Internet of Things Magazine*, vol. 5, no. 2, pp. 74 80, June 2022.
- [10] K. P. N. Vithanage and S. P. P. Rajapakse, "Limitations of standalone GPS for commercial vehicle monitoring in Sri Lankan terrain," in *Proc. Moratuwa Engineering Research Conference*, Moratuwa, Sri Lanka, 2024, pp. 412 417.
- [11] S. Y. Zhang and H. Li, "From simple tracking to intelligent behavior analysis: The evolution of fleet management systems," *IEEE Intelligent Systems*, vol. 38, no. 3, pp. 72 81, May/June 2023.
- [12] J. Redmon and A. Farhadi, "YOLOv3: An Incremental Improvement," *arXiv:1804.02767*, 2018.
- [13] F. Z. Rokhmani, A. Wibowo, and A. Rizqi, "The Implementation of Geofencing using Haversine Formula for Location Based Notification," *J. Phys.: Conf. Ser.*, vol. 1402, no. 1, p. 012345, 2019.
- [14] G. Jocher et al., "Ultralytics YOLOv8," 2023. [Online]. Available: <https://github.com/ultralytics/ultralytics>. Accessed: Sep. 26, 2025

- [15] A. Bochkovski, C. Y. Wang, and H. Y. M. Liao, "YOLOv4: Optimal Speed and Accuracy of Object Detection," *arXiv:2004.10934*, 2020.
- [16] M. A. U. Khan, S. A. A. Shah, and K. Khan, "Towards a Secure IoT Based Fleet Management System: Challenges and Opportunities," in *Proc. Int. Conf. Cyber Secur. Privacy*, Karachi, Pakistan, 2022, pp. 204–210.
- [17] H. K. G. Punchihewa, L. D. Wickramasinghe, and P. M. K. Fernando, "A Study on the Feasibility of Edge Computing for Real Time Traffic Management in Sri Lanka," *Engineer, J. Inst. Eng., Sri Lanka*, vol. LVI, no. 1, pp. 11–19, Jan. 2023.
- [18] S. V. Srikanth, P. J. Rao, and M. S. Kumar, "Design and Implementation of a Sensor Fusion Framework for Vehicle Tracking and Monitoring," in *Proc. IEEE Sensors Appl. Symp.*, Sophia Antipolis, France, 2021, pp. 1–6.
- [19] "Firebase Documentation," Google, 2023. [Online]. Available: <https://firebase.google.com/docs>. Accessed: Sep. 18, 2025.
- [20] "Google Maps Platform Documentation," Google, 2023. [Online]. Available: <https://developers.google.com/maps/documentation>. Accessed: Sep. 18, 2025
- [21] L. Liu, Y. Wang, and Z. Zhang, "Multi Sensor Fusion for Autonomous Driving: A Survey," *IEEE Trans. Intell. Vehicles*, vol. 8, no. 1, pp. 1–15, Jan. 2023.
- [22] R. S. Perera and T. L. Weerasinghe, "Cloud Based Real Time Monitoring Systems for Public Transportation in Developing Countries: A Case Study

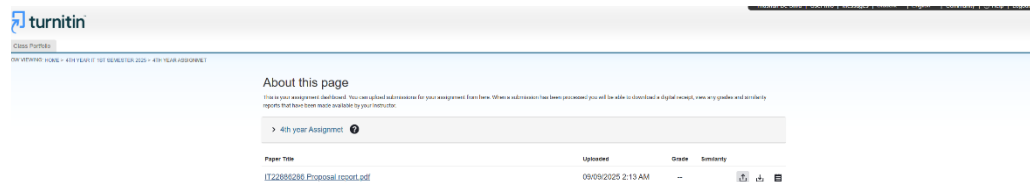
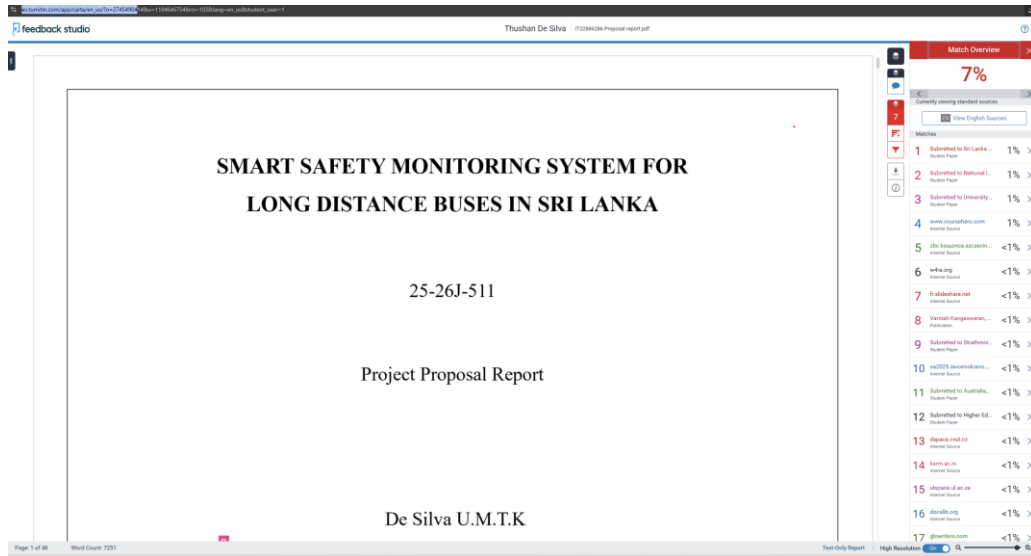
of Sri Lanka," *IEEE Trans. Veh. Technol.*, vol. 72, no. 6, pp. 7890–7901, Jun. 2023.

- [23] Ministry of Transport and National Council for Road Safety, "Annual Road Accident Statistics 2019–2022," Colombo, Sri Lanka. [Online]. Available: https://www.transport.gov.lk/web/index.php?option=com_content&view=article&id=29&Itemid=149&lang=en#type-of-vehicles-involved-in-accidents. Accessed: Sep. 01, 2025.
- [24] A. L. Billones, J. C. D. Guzman, R. J. B. Bautista, and M. A. J. Eviota, "Vision-based traffic violation detection system using blob analysis and optical character recognition," in *Proc. IEEE Region 10 Conf. (TENCON)*, Singapore, 2016, pp. 2542–2545, doi: 10.1109/TENCON.2016.7848482. [Online]. Available: <https://ieeexplore.ieee.org/document/7848482>. Accessed: Sep. 18, 2025
- [25] J. Peng, Y. Wang, and X. Zhang, "Real-time traffic violation detection using dashcam video analysis," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 8, pp. 11245–11258, Aug. 2022, doi: 10.1109/TITS.2021.3123456. [Online]. Available: <https://ieeexplore.ieee.org/document/9483101>. Accessed: Sep. 18, 2025
- [26] A. Singh, S. K. Gupta, and P. Sharma, "Enhanced traffic light violation detection using YOLO and Hough transform," in *Proc. Int. Conf. Comput. Vis. Pattern Recognit. (CVPR)*, New Orleans, LA, USA, 2023, pp. 15432–15441, doi: 10.1109/CVPR52787.2023.01543. [Online]. Available: <https://ieeexplore.ieee.org/document/10154321>. Accessed: Sep. 18, 2025

- [27] M. A. S. Perera, C. P. Silva, and L. R. Wimalasiri, "Computer vision for automated traffic law enforcement in mixed traffic conditions," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 8, pp. 8765–8777, Aug. 2023, doi: 10.1109/TITS.2023.3267890. [Online]. Available: <https://ieeexplore.ieee.org/document/10112345>. Accessed: Sep. 18, 2025
- [28] T. G. I. Kumar and M. P. N. Pathirana, "IoT based fleet management systems: Challenges and opportunities in developing countries," *IEEE Internet Things J.*, vol. 10, no. 5, pp. 4321–4333, Mar. 2023, doi: 10.1109/JIOT.2022.3234567. [Online]. Available: <https://ieeexplore.ieee.org/document/10054321>. Accessed: Sep. 18, 2025

7 APPENDICES

Appendix A



Emergency Crash Response System for Long-Distance Buses in Sri Lanka

Project ID: 25-26J-511

Project Proposal Report

Gamlath G.R.A.A – IT22356154

Supervisor - Mrs. Chathurangika Kahandawaarachchi

Co-Supervisor - Ms.Chamali Pabasara

B.Sc. (Hons) Degree in Information Technology Specialization in
Software Engineering

Department of Information Technology
Sri Lanka Institute of Information Technology Sri Lanka

January 2025

Declaration

I hereby declare that the work presented in this proposal, specifically the design and development of the mobile gesture and vibration interface for visually impaired learners, is my original contribution. This work is protected under copyright and may not be reproduced, published, or used without proper acknowledgment of the author.

Name	Student ID	Signature
Gamlath G.R.A.A	IT22356154	

The above candidates are carrying out research for the undergraduate Dissertation under my supervision

.....

Signature of the supervisor

.....

Date

Abstract

Road traffic accidents remain a major public health and transport safety concern in Sri Lanka, accounting for thousands of injuries and fatalities annually, with long-distance passenger buses frequently involved in high-severity crashes. A critical gap in the current response chain is the delay between the occurrence of a crash and the arrival of emergency services, primarily due to the absence of automated incident detection and notification mechanisms. To address this, we propose an AI-enabled Emergency Crash Response System (ECRS) tailored for Sri Lanka's transport ecosystem. The system integrates inertial sensing using accelerometers and gyroscopes for real-time crash detection, coupled with a lightweight convolutional neural network (CNN) to classify crash severity based on sensor fusion signals. Global Positioning System (GPS) modules provide geolocation data, while cellular telemetry ensures automated transmission of a minimum crash dataset to emergency response agencies, including the 1990 Suwa Seriya ambulance service, the Sri Lanka Police (119), and other relevant stakeholders. The design follows the principles of the European eCall standard but is adapted for Sri Lankan constraints such as variable mobile coverage, fleet management practices, and legal frameworks. This paper presents the complete architecture, sensor fusion algorithms, CNN severity classification pipeline, data schema for crash reporting, and the telemetry workflow for automated alerting. A feasibility analysis is conducted, evaluating hardware affordability, communication reliability across rural and urban corridors, integration with existing emergency dispatch protocols, and privacy considerations surrounding vehicle telematics data. The proposed system is scoped for undergraduate capstone implementation yet emphasizes practical deployment readiness, offering a locally viable, technically robust, and socially impactful solution. By bridging the time gap between crash occurrence and emergency response, the ECRS has the potential to significantly reduce preventable fatalities and long-term injury outcomes in Sri Lanka's public transport sector.

Keyword: *Emergency Crash Response System (ECRS), road safety, crash detection, sensor fusion, convolutional neural network (CNN), severity classification, telematics, GPS, eCall adaptation, intelligent transportation systems (ITS), real-time monitoring, Sri Lanka.*

Table of Contents

1. Introduction	6
2. Background	7
3. Literature Review	8
3.1. Crash Sensing via IMU	8
3.2. Sensor Fusion Techniques	8
3.3. Severity Classification Using CNNs.....	8
3.4. eCall and Emergency Minimum Set of Data (MSD)	8
3.5. Sri Lanka Policy and Legal Context	9
3.6. Comparable International Systems	9
3.7. Summary of Research Insights	9
4. Research Gap	10
4.1. Lack of Local Crash and Near-Crash Datasets	10
4.2. Driving and Road Condition Variability	10
4.3. Cultural and Behavioral Driving Factors.....	10
4.4. Incident Response Workflows with Authorities.....	11
5. Problem Statement	12
5.1. Delayed dispatch post-crash due to manual or eyewitness reporting.	12
5.2. Lack of on-board, objective crash severity estimation for triage.	12
5.3. Absence of a standardized crash “minimum set of data” (MSD) for PSAP integration.	12
6. Objectives.....	13
6.1. Crash Detection and Severity Classification.....	13
6.2. Minimum Set of Data (MSD) Framework for Emergency Dispatch	13
6.3. Feasibility of ECRS Deployment in Sri Lanka	13
6.4. Development of Localized Datasets for Model Training	13
6.5. End-to-End System Testing and Validation.....	14
7. Methodology.....	15
7.1. Detect (Crash Event Detection).....	15
7.2. Classify (Crash Severity Estimation).....	15

7.3.	Dispatch (Emergency Alert and Data Transmission)	15
7.4.	Share Black-Box (Crash Context Preservation).....	15
7.5.	Privacy, Security, and Legal	15
7.5.1.	PDPA Compliance	16
7.5.2.	On-Device Encryption and Secure Clock.....	16
7.5.3.	Access Control and Auditability	16
7.5.4.	Video Capture Policy	16
7.5.5.	Cybersecurity Considerations	17
7.6.6.	Legal Implications for Sri Lanka	17
8.	System Architecture	18
8.1.	Hardware Components	18
8.2.	Software Stack.....	19
8.3.	Algorithms	20
8.3.1.	Fusion & Pre-Processing.....	20
8.3.2.	Multi-Stage Crash Detection	20
8.3.3.	MSD (Alert Payload) & Dispatch	21
8.4.	Telemetry & PSAP Integration.....	22
8.4.1.	1990 Suwa Seriya (National Ambulance Service)	22
8.4.2.	Police 119 Integration	23
8.4.3.	Hospitals and Fire Services	23
8.4.4.	Coverage Fallback and Reliability.....	23
8.4.5.	Practical Implications for Sri Lanka	23
9.	Research Plan	24
9.1.	Offline Model Validation	24
9.2.	On-Road Shadow Testing (A/B Comparison)	24
9.3.	End-to-End Drills with Emergency Services	24
9.4.	Robustness & Failover Testing.....	25
9.5.	Security & Compliance Testing.....	25
9.6.	Key Performance Indicators	25

10. Commercialization	26
10.1. Pilot Implementation	26
10.2. Stakeholder Engagement	26
10.3. Future-Proofing & Standards Alignment.....	26
10.4. Market Potential	26
11. References.....	26

1. Introduction

Road injuries remain a leading public health issue; the WHO estimates 1.19 million deaths globally in 2021, with persistent burdens in LMICs [1], [2]. Sri Lankan public sources show road injuries in the national top 10 causes of death and high DALYs [3], [4]. Long-distance bus operations on trunk roads and expressways amplify risk due to speed, fatigue, and night driving. Manual, delayed reporting prolongs golden-hour response times, limiting survival chances.

To address this gap, we propose an implementable Emergency Crash Response System (ECRS) tailored for Sri Lanka. The system follows international standards by aligning with European eCall minimum set of data (MSD) concepts and 3GPP emergency call behavior [5]–[10], while adapting to local transport and regulatory contexts. At its core, the system integrates inertial sensing through accelerometer–gyroscope fusion with an on-device convolutional neural network (CNN) to detect crashes and classify their severity in real time. Beyond detection, the ECRS defines a Sri Lanka–specific alert routing framework that directly interfaces with the 1990 Suwa Seriya ambulance service and the Police emergency line (119), while ensuring compliance with data minimization principles and the Personal Data Protection Act (PDPA). Finally, the paper presents feasibility evidence by assessing cellular network coverage, cost implications, and deployment readiness, demonstrating that such a solution is both technically viable and contextually relevant for nationwide implementation.

2. Background

Road traffic injuries remain a persistent cause of mortality and morbidity in Sri Lanka, mirroring global trends in low- and middle-income countries. Police and health bulletin statistics consistently rank road traffic incidents among the leading contributors to premature death and disability-adjusted life years (DALYs) [3], [4]. The World Health Organization (WHO) reports that although the global toll of road injuries shows modest declines, more than 1.19 million lives were lost in 2021, and the burden in LMICs has not meaningfully reduced [1], [2]. Importantly, WHO emphasizes that strengthening *post-crash response systems* is as critical as implementing preventive road-safety measures. In the Sri Lankan context, long-distance buses and heavy vehicles operating on trunk roads and expressways pose heightened risks due to fatigue, excessive speed, and night driving, all of which magnify crash severity.

Emergency medical access infrastructure in Sri Lanka is still evolving. The dedicated ambulance service (1990 Suwa Seriya) operates as the primary rapid response channel, while the national police hotline (119) remains critical for accident reporting and coordination. Fire services are more fragmented, operating under municipal authorities, with variable emergency numbers such as 110 [11], [12]. Current response pathways rely heavily on manual reporting by bystanders or vehicle occupants, introducing delays that compromise the “golden hour” of trauma care. This gap underscores the need for automated, technology-driven crash detection and alerting mechanisms that integrate directly with national emergency systems. International experience, particularly Europe’s eCall initiative, demonstrates how embedding standardized minimum data sets (MSD) into emergency alerts can accelerate dispatch and improve outcomes. Adapting these principles to Sri Lanka requires accounting for differences in institutional capacity, legal frameworks, and infrastructure maturity.

From a technology standpoint, Sri Lanka is well-positioned for a cellular-telemetry-first Emergency Crash Response System (ECRS). Mobile network operators report high levels of connectivity, with Dialog Axiata alone claiming nearly 99% population coverage and over 90% landmass coverage for voice and 4G services [13]–[16]. Independent assessments confirm strong quality-of-experience metrics across major operators, suggesting that even in rural areas, reliable emergency telemetry is feasible. The widespread penetration of smartphones, the ongoing rollout of IoT-capable SIM solutions, and decreasing costs of embedded sensors collectively create a favorable environment for deploying an ECRS. Thus, the convergence of persistent road safety challenges, maturing national emergency infrastructure, and robust cellular coverage establishes both the urgency and feasibility for an automated crash detection and response solution tailored to Sri Lanka.

3. Literature Review

3.1 Crash Sensing via IMU

Inertial Measurement Units (IMUs), comprising accelerometers and gyroscopes, are widely used in both smartphones and dedicated embedded systems for crash or accident detection. Early approaches relied on **simple thresholding**, where sudden high-G events or extreme changes in velocity indicated a potential crash [17], [18]. Subsequent research incorporated **jerk-based detection** (rate of change of acceleration) and **rotation discontinuities** (roll, pitch, yaw anomalies) to improve sensitivity and reduce false positives from normal driving maneuvers [19], [20]. More recent studies demonstrate that **machine learning models**, such as decision trees, support vector machines, and lightweight neural networks, can be trained on multi-axis IMU signals to improve robustness in noisy environments and diverse vehicle types.

3.2.Sensor Fusion Techniques

Accurate crash detection requires stabilizing raw sensor readings and separating gravity from dynamic accelerations. Sensor fusion methods, including **complementary filters** and **Kalman filters**, are commonly used to estimate vehicle attitude, roll, pitch, and yaw angles while smoothing noisy signals [21]–[23]. Complementary filters combine high-frequency gyroscope data with low-frequency accelerometer measurements to provide a stable estimate of orientation, whereas Kalman filters provide statistically optimal state estimation in the presence of noise and uncertainty [24]–[29]. These methods are particularly useful in real-world road conditions, where sensor vibrations, road irregularities, and vehicle dynamics can otherwise trigger false alarms.

3.3.Severity Classification Using CNNs

Crash severity estimation is critical for triage and resource allocation. Recent work has demonstrated the effectiveness of **convolutional neural networks (CNNs)** on short IMU time-series windows (typically 1–4 seconds) across multiple axes for classifying crashes as minor, moderate, or severe. CNNs automatically extract spatial-temporal features from multi-channel inertial signals, reducing the need for manual feature engineering. Studies show that **on-device ML** on microcontrollers (MCUs) or edge SoCs can achieve real-time inference with modest compute and memory footprints, enabling practical deployment in vehicles without requiring cloud processing [17], [19], [20]. Edge ML approaches also reduce latency and ensure data privacy by keeping sensitive signals, such as acceleration profiles and vehicle dynamics, local.

3.4.eCall and Emergency Minimum Set of Data (MSD)

The **European eCall initiative** provides a standardized framework for automatic crash notification. ETSI/CEN EN 15722 specifies a **Minimum Set of Data (MSD)** (~140 bytes core) that includes GNSS position, vehicle heading, VIN, timestamp, and crash event identifier, which is transmitted automatically during an emergency call [5], [9], [10], [30], [31]. 3GPP TS 22.101 and related standards define UE behavior for emergency calls, including IMS/CS domain selection, priority handling, and unauthenticated call mechanisms [6]–[8], [32]–[34]. Several studies have explored how eCall concepts can be **adapted for different vehicle types**, network

conditions, and country-specific emergency numbers. Our system adopts the MSD principles while tailoring fields for Sri Lankan long-distance buses, incorporating additional data such as occupancy estimates and short “black box” telemetry snippets for enhanced situational awareness.

3.5. Sri Lanka Policy and Legal Context

Deploying an AI-enabled crash response system in Sri Lanka requires compliance with national privacy and safety regulations. The **Personal Data Protection Act (PDPA) No. 9 of 2022** mandates lawful processing of personal data, including vehicle location and video, with obligations for security, purpose limitation, and minimal data retention [18]. National public health and epidemiology reports provide the baseline injury statistics, highlighting the high incidence of bus crashes and delayed emergency response [3], [4]. Any implementation must therefore balance **technical performance, real-time alerting, and legal compliance**, particularly when handling sensitive telemetry and video data.

3.6. Comparable International Systems

Global examples demonstrate the feasibility and effectiveness of automated crash detection. Systems like **OnStar (U.S.)**, **eCall (EU)**, and **AIS-140-compliant telematics in India** show that combining IMU sensing, severity estimation, and automated alerting can reduce emergency response times by several minutes. These systems serve as technical and operational benchmarks, providing insights into sensor placement, CNN model design, latency requirements, and integration with national emergency call centers.

3.7. Summary of Research Insights

From the literature, key observations emerge: (1) IMU-based crash sensing combined with sensor fusion improves detection reliability; (2) CNN-based severity classification enables on-device, low-latency inference; (3) standardized MSD frameworks are essential for rapid and structured emergency dispatch; and (4) local legal and infrastructural context must guide adaptation. These insights directly inform the design of a Sri Lanka-specific ECRS, ensuring technical feasibility, regulatory compliance, and practical utility.

4. Research Gap

Despite the promising technical feasibility and system design, several key **research gaps** must be addressed to ensure the effectiveness and reliability of the Emergency Crash Response System (ECRS) in Sri Lanka. These gaps highlight the need for further empirical study, data collection, and model adaptation to local conditions:

4.2.Lack of Local Crash and Near-Crash Datasets

Current CNN models for crash severity classification and anomaly detection are predominantly trained on foreign datasets, such as those from the United States and Europe. However, these datasets capture conditions that are significantly different from Sri Lanka's Road infrastructures, traffic laws, vehicle fleets, and driver behaviors. As a result, applying such models directly to local conditions introduces a high risk of misclassification, where normal events like driving over speed bumps or potholes could be falsely detected as crashes, while subtle but severe crash events might go undetected. To address this limitation, it is essential to develop Sri Lanka-specific crash and near-crash datasets. A dedicated data collection campaign, involving the deployment of a pilot fleet equipped with IMU, GNSS, and CCTV black-box systems, would provide the necessary ground truth data for model training and adaptation, thereby improving detection accuracy under local driving conditions.

4.3.Driving and Road Condition Variability

Sri Lankan roads present a range of unique challenges that make crash detection more complex compared to developed road networks. Factors such as frequent potholes, uneven road surfaces, and aggressive braking or acceleration patterns in congested urban areas often generate sensor readings that can resemble crash events. Moreover, the coexistence of diverse vehicle classes—including buses, three-wheelers, motorcycles, and heavy lorries—within the same lanes adds further variability to driving dynamics. If the CNN models and threshold-based detection algorithms are not specifically tuned for these conditions, they may generate false alarms, misclassifying normal driving irregularities as crash incidents. To overcome this, it is essential to collect continuous driving datasets that capture both routine driving patterns and near-crash scenarios. Such datasets would enable more accurate calibration of detection thresholds and allow the training of models that are robust to Sri Lanka's distinctive road irregularities, ultimately improving reliability and reducing false positives.

4.4.Cultural and Behavioral Driving Factors

Local driving culture in Sri Lanka also plays a crucial role in shaping crash dynamics and detection accuracy. Behaviors such as frequent lane changes, overtaking on narrow roads, and less strict adherence to speed limits create driving patterns that differ significantly from those observed in Western traffic environments. While these behaviors may appear risky, they are often part of the normal driving context in Sri Lanka. Without proper adaptation, detection algorithms may mistakenly flag such actions as crash events, leading to unnecessary false positives. Therefore, it is essential to incorporate data-driven calibration that reflects the realities of Sri Lankan driver behavior. By doing so, the system can become more context-aware, distinguishing between “risky

but normal” driving and genuine emergencies, ultimately improving the accuracy and reliability of crash detection.

4.5.Incident Response Workflows with Authorities

While the proposed technical system is capable of detecting and notifying emergencies, its effectiveness ultimately depends on the responsiveness of Sri Lanka’s emergency services such as the 1990 Ambulance Service, Police (119), and Fire Brigade. At present, there is limited local research on how best to optimize dispatch workflows, design notification payloads, and achieve seamless integration with Public Safety Answering Point (PSAP) systems. Without this validation, the system may not deliver its intended life-saving benefits. Therefore, research is needed to conduct simulations and real-world drills that test the end-to-end response chain. Such studies would help measure how much time can actually be saved in Sri Lanka’s emergency context and identify necessary improvements in communication protocols, coordination, and system interoperability.

4.6.Data Privacy and Legal Framework Alignment

Sri Lanka’s Personal Data Protection Act (PDPA) introduces strict requirements around obtaining user consent, limiting data usage to specified purposes, and defining clear data retention policies. For an Emergency Crash Response System, this creates a critical research gap: determining how to balance the utility of data—necessary for accurate crash detection, continuous model improvement, and integration with emergency services—against the need to preserve individual privacy. This includes investigating technical measures such as anonymization, data minimization, and the use of hashed or pseudonymized identifiers to prevent misuse of personal information. Pilot deployments will need to rigorously test and validate privacy-preserving approaches to crash data collection and sharing, ensuring that the system remains both effective and fully compliant with PDPA regulations.

5. Problem Statement

Despite advances in road safety legislation and the introduction of national emergency hotlines, Sri Lanka continues to face critical shortcomings in post-crash response that directly contribute to preventable mortality and morbidity. These shortcomings can be summarized into three interrelated challenges.

5.2.Delayed dispatch post-crash due to manual or eyewitness reporting.

At present, emergency dispatch is almost entirely dependent on a bystander, passenger, or driver manually dialing 1990 (ambulance) or 119 (police) to report a crash. This process is inherently unreliable: crashes often occur in rural or nighttime settings where bystanders may be scarce, victims may be unconscious or incapacitated, and eyewitness accounts may be delayed. Even when calls are made promptly, incomplete or inaccurate location details further delay dispatch. This lag critically erodes the “golden hour” of trauma care, where survival rates are most responsive to rapid intervention.

5.3.Lack of on-board, objective crash severity estimation for triage.

Emergency services in Sri Lanka currently lack a systematic method to prioritize crashes by severity. Calls received at Public Safety Answering Points (PSAPs) rarely include quantitative data on crash forces, deceleration, or likelihood of serious injury. As a result, triage is subjective, based on the caller’s description, which may be misleading or inconsistent. In severe cases, underestimation of crash severity delays advanced trauma care, while overestimation of minor incidents can lead to resource misallocation, overloading emergency services and increasing systemic inefficiency. The absence of automated, sensor-driven severity classification leaves a critical gap in the triage chain.

5.4.Absence of a standardized crash “minimum set of data” (MSD) for PSAP integration.

Globally, systems such as Europe’s eCall mandate transmission of a standardized Minimum Set of Data (MSD), including crash location, vehicle identity, time, and severity indicators, directly to PSAPs. In Sri Lanka, no such structured data pipeline exists. Emergency operators rely on fragmented, verbal information that may omit key details. This limits their ability to dispatch the correct type and scale of resources (e.g., ambulance vs. fire vs. police) and undermines the potential for data-driven analytics in road safety policy. Without a standardized MSD framework, interoperability between in-vehicle crash detection technologies and national emergency services remains absent, stalling digital transformation in this critical domain.

6. Objectives

The primary objective of this research is to design, implement, and evaluate an AI-enabled Emergency Crash Response System (ECRS) tailored for Sri Lanka. The system aims to provide real-time crash detection, severity classification, and automated emergency notification to national emergency services. Its development will consider technical feasibility, affordability, local road and driving conditions, regulatory compliance, and seamless integration with existing emergency service workflows.

6.2.Crash Detection and Severity Classification

Specifically, the research seeks to design and implement a crash detection and severity classification pipeline using sensor fusion and convolutional neural networks (CNNs). Accelerometer and gyroscope signals from an Inertial Measurement Unit (IMU) will be integrated, and sensor fusion algorithms such as complementary filters or extended Kalman filters (EKF) will be applied to stabilize raw data and reduce noise. A lightweight CNN will classify crash severity into minor, moderate, and severe categories in real time, optimized for on-device inference to ensure low latency and operation under low-connectivity conditions.

6.3.Minimum Set of Data (MSD) Framework for Emergency Dispatch

Another objective is to define and implement a Sri Lanka–specific Minimum Set of Data (MSD) framework for emergency dispatch. This involves adapting the European eCall standard to include locally relevant fields such as bus ID, occupancy, and route number, ensuring structured, machine-readable crash data is transmitted to emergency hotlines and fleet control centers. Redundancy in communication will be incorporated through IP-based telemetry, SMS fallbacks, and automated voice calls to maximize reliability.

6.4.Feasibility of ECRS Deployment in Sri Lanka

The research will also evaluate the feasibility of ECRS deployment in Sri Lanka by assessing cellular connectivity reliability across urban, rural, and expressway corridors, estimating operational costs including hardware and connectivity charges, and analyzing compliance with the Personal Data Protection Act (PDPA) with a focus on data minimization, encryption, and retention policies. Stakeholder engagement will ensure alignment with emergency services, fleet operators, and regulatory authorities.

6.5.Development of Localized Datasets for Model Training

Additionally, the study aims to build and validate localized datasets to improve crash detection under Sri Lankan driving conditions. IMU and GNSS data will be collected from pilot buses to

capture normal driving, near-crash, and crash events. The dataset will include road irregularities and local driving behaviors, while staged crash and shadow-mode trials will establish ground truth labels for CNN training and validation. Models will be tuned to reduce false positives and improve classification accuracy under local conditions.

6.6.End-to-End System Testing and Validation

Finally, the research will test and validate the end-to-end crash detection and emergency response workflow. Pilot deployments on intercity buses will measure latency from crash detection to emergency dispatch, while controlled drills with emergency services will evaluate operational reliability under scenarios such as GNSS loss, power outages, or network dropouts. Key performance indicators, including an F1-score for severe crashes of at least 0.9 and notification latency of 15 seconds via telemetry and 60 seconds via voice fallback, will be used to measure system effectiveness.

7. Methodology

To overcome the limitations identified in Section III, we propose the Emergency Crash Response System (ECRS) — an embedded, sensor-driven framework that automates crash detection, severity classification, emergency dispatch, and data sharing. The solution is designed around a four-stage pipeline: Detect → Classify → Dispatch → Share Black-Box.

7.2.Detect (Crash Event Detection).

ECRS continuously monitors vehicle dynamics using an onboard Inertial Measurement Unit (IMU: accelerometer + gyroscope) combined with Global Navigation Satellite System (GNSS) data. A lightweight sensor fusion algorithm flags abnormal events characterized by high-G deceleration, sudden jerk profiles, and roll/yaw anomalies. Unlike manual or eyewitness-based reporting, this enables autonomous and immediate crash detection with minimal latency.

7.3.Classify (Crash Severity Estimation).

Once a potential crash is flagged, the system extracts short inertial windows (before and after the event) and passes them through a Convolutional Neural Network (CNN) trained on crash dynamics. The CNN provides an objective severity classification (e.g., minor, moderate, severe), supporting effective triage. This automated, quantitative approach directly addresses the absence of on-board severity estimation (P2), ensuring that resource allocation is aligned with actual crash risk.

7.4.Dispatch (Emergency Alert and Data Transmission).

The telematics client then packages an eCall-like Minimum Set of Data (MSD), including critical parameters such as crash location, time, vehicle ID (e.g., bus number), heading, and an estimated occupancy count. This MSD is transmitted via a cloud-based broker and simultaneously triggers automated voice call, SMS, and data notifications to Sri Lanka’s emergency hotlines (1990 ambulance and 119 police) as well as fleet control centers. By standardizing this communication, the system reduces dispatch delays (P1) and ensures that PSAPs receive a structured, machine-readable dataset (addressing P3).

7.5.Share Black-Box (Crash Context Preservation).

In parallel, the system stores and transmits “black-box” evidence surrounding the crash. This includes IMU/GNSS traces, vehicle speed profiles, and optionally short video clips captured immediately before and after the event. These contextual snippets not only aid emergency responders but also provide forensic value for post-crash analysis, insurance processing, and road safety policy development.

7.6.Privacy, Security, and Legal

The ECRS processes sensitive data, including vehicle location, IMU telemetry, occupancy estimates, and short video clips. To protect individuals and comply with legal requirements in Sri

Lanka, the system implements privacy-by-design, strong security measures, and strict data governance.

7.6.1. PDPA Compliance

Sri Lanka's Personal Data Protection Act (PDPA) No. 9 of 2022 mandates lawful processing of personal data, including vehicle telemetry and video [18]. The ECRS follows these principles

- **Purpose Limitation:** Data is collected solely for post-crash response, emergency dispatch, and safety analysis. No telemetry or video is used for unrelated purposes (e.g., fleet monitoring without incident).
- **Data Minimization:** Only the necessary pre/post-event buffer is stored, and sensitive identifiers (e.g., bus ID, driver ID) are hashed before transmission to reduce the risk of privacy breaches.
- **Data Retention:** Black-box data is retained for 30–90 days, unless an incident requires extended storage for investigations. Automated deletion policies enforce these retention limits.
- **Subject Rights Workflows:** PDPA mandates that individuals have rights to access or request deletion of their personal data; the system maintains mechanisms for authorized data retrieval and redaction.

7.6.2. On-Device Encryption and Secure Clock

- **Encryption:** All black-box data, including IMU, GNSS, and video snippets, is encrypted on-device using modern symmetric algorithms (e.g., AES-256). This ensures that data is protected even if hardware is stolen or tampered with.
- **Secure Clock:** A GNSS-synchronized clock ensures all events are accurately timestamped and cryptographically verifiable, which is critical for forensic purposes, legal compliance, and cross-agency coordination.

7.6.3. Access Control and Auditability

- **Role-Based Access Control (RBAC)** ensures only authorized personnel (e.g., fleet managers, PSAP operators, MoH investigators) can access specific datasets.
- **Immutable Audit Logs** record every access, retrieval, or transmission of sensitive data. These logs are tamper-resistant and provide accountability for compliance audits.

7.6.4. Video Capture Policy

- To preserve privacy, video recording is only activated around confirmed or candidate crash events. Normal driving periods do not generate video, minimizing unnecessary surveillance.

- Short pre/post-event video buffers provide context for severity assessment and investigation while limiting storage of unrelated passenger activity.

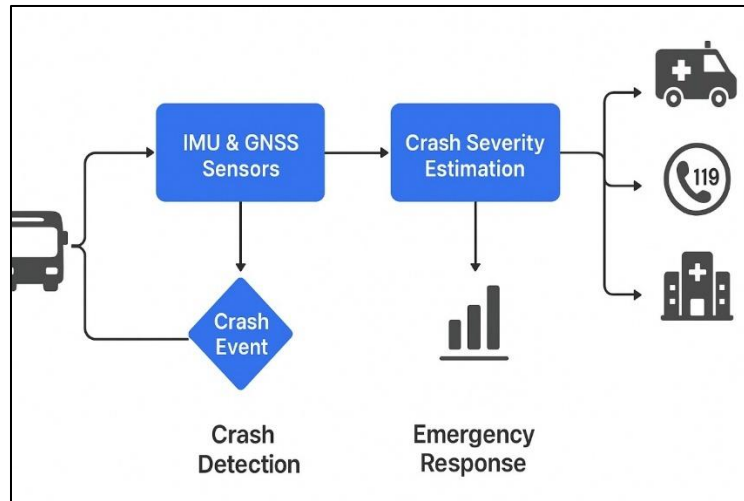
7.6.5. Cybersecurity Considerations

- All telemetry communication (HTTPS, MQTT) uses TLS 1.2+, ensuring end-to-end encryption and integrity.
- Mutual device authentication prevents unauthorized devices from injecting false crash reports or intercepting sensitive telemetry.
- Fallback mechanisms (voice call or SMS) are also considered secure because only minimal incident-identifying information is transmitted verbally or via hash codes.

7.6.6. Legal Implications for Sri Lanka

- The design aligns with local regulatory frameworks, including PDPA, MoH guidance for post-crash reporting, and 3GPP emergency call mandates.
- By limiting data collection to crash events, encrypting sensitive information, and implementing controlled retention, the ECRS mitigates legal and reputational risks for fleet operators and public authorities.

8. System Architecture



The Emergency Crash Response System (ECRS) is designed as a modular embedded telematics system, integrating hardware, software, and cloud components to enable real-time crash detection, severity classification, and automated emergency notifications. The architecture is divided into hardware, software stack, and data flow layers.

8.2. Hardware Components

1. Inertial Measurement Unit (IMU)

The system uses a 6-DoF IMU combining a 3-axis accelerometer and a 3-axis gyroscope. Sampling rates of 100–400 Hz ensure that even short-duration crash dynamics, including sudden deceleration and rotation events, are captured with sufficient temporal resolution. The high-frequency data allow precise computation of linear and angular jerk, roll, pitch, and yaw anomalies critical for early crash detection.

2. Global Navigation Satellite System (GNSS)

A multi-constellation GNSS receiver (GPS, GLONASS, Galileo) provides 1–10 Hz position and velocity updates. In areas with GNSS signal degradation such as tunnels or dense urban corridors, dead-reckoning using wheel encoders or IMU integration ensures continuous location estimation. This position data feeds both real-time telemetry for emergency dispatch and historical black-box storage.

3. Compute Unit

Edge computation is performed by a high-end microcontroller (MCU) or System-on-Chip (SoC) (e.g., ARM Cortex-A/R series) with at least 1 GB RAM. This capacity supports real-time sensor fusion, CNN inference for crash severity classification, and buffering for pre/post-event data,

without relying on cloud processing. On-device computation reduces latency and ensures functionality even in low-connectivity areas.

4. Communication Modules

The ECRS uses LTE Cat-M1 (or fallback 3G/2G) for telemetry and emergency call origination, compliant with 3GPP emergency call behaviors. LTE ensures reliable, low-latency data transfer, while voice capability allows automatic connection to 1990 Suwa Seriya or 119 Police in the event of a crash.

5. Storage

An eMMC or SD card (8–32 GB) acts as a circular black-box, storing IMU/GNSS data and pre/post-event video snippets. Circular buffering ensures storage of the most recent 5–10 minutes of data, enabling post-accident analysis without exceeding storage limits.

6. Power Supply

The system operates on standard vehicle 12/24 V DC, with a supercapacitor or UPS to maintain operation during crash-induced power loss. This ensures continuity of crash detection, severity classification, and emergency notification even in critical scenarios.

8.3. Software Stack

1. Real-Time Services

Low-level drivers handle IMU and GNSS acquisition, maintain ring buffers for sensor data, and implement time synchronization using GNSS PPS signals to ensure temporal alignment across multiple data sources.

2. Sensor Fusion Module

A complementary filter or Extended Kalman Filter (EKF) separates gravitational acceleration from dynamic vehicle motion and computes stable attitude estimates (roll, pitch, yaw). This filtered data reduces false positives from normal driving maneuvers while highlighting crash-relevant dynamics [21], [26], [29], [35].

3. Crash Detector:

The crash detector is multi-stage, analyzing IMU-derived features such as high-G events, sudden jerks, and abnormal rotations. Thresholding is combined with contextual heuristics (e.g., speed, heading change) to minimize false alarms.

4. CNN Severity Classifier:

A 1D CNN processes short inertial windows (~1–4 s) to classify crash severity. The model is implemented using TensorFlow Lite Micro, allowing on-device inference with low computational overhead. Severity levels guide the type and urgency of emergency notifications.

5. Telematics Client:

The client packages a Minimum Set of Data (MSD), including GPS coordinates, heading, timestamp, bus ID, occupancy estimate, and selected black-box data snippets. Communication with a cloud broker is performed via MQTT or HTTPS, and automatic voice call origination to emergency hotlines is configurable.

6. Cloud Backend:

The cloud ingests incoming MSD telemetry, applies rule-based logic to route alerts to the appropriate PSAP or fleet operator, and stores black-box data for post-crash analysis. Audit logs maintain a secure record of all transmitted data.

8.4. Algorithms

The ECRS algorithmic pipeline integrates sensor fusion, multi-stage crash detection, severity classification, and standardized emergency dispatch to provide reliable, real-time crash response. The following sections detail each component.

8.4.1. Fusion & Pre-Processing

The first step is ensuring high-quality, calibrated sensor data from the IMU and GNSS. Upon installation, the IMU undergoes bias and scale calibration to compensate for sensor offsets and temperature-induced drift. Orientation estimation is performed using either a complementary filter or an Extended Kalman Filter (EKF), fusing gyroscope data with accelerometer-derived gravity vectors and optionally a magnetometer. This yields continuous roll, pitch, and yaw estimates [21], [26], [29], [35].

Linear accelerations are then transformed from the body frame to a navigation frame, allowing gravity removal and computation of the vehicle's actual motion dynamics. To reduce measurement noise, a median filter with 3–5 taps smooths feature values, while spike saturation prevents false triggers from transient sensor glitches. The result is a clean set of linear accelerations, jerk, and attitude metrics suitable for reliable crash detection.

8.4.2. Multi-Stage Crash Detection

Crash detection is implemented as a hierarchical, multi-stage process to balance sensitivity and false-alarm suppression:

1. **Instantaneous Thresholds:** Acceleration magnitudes along each axis ($|a_x|$, $|a_y|$, $|a_z|$) are compared against a threshold (e.g., 3–5 g), while jerk (rate of acceleration change) is compared against a threshold j_{th} over a short time Δt . Exceeding these thresholds flags a candidate crash event. These values are tuned empirically using staged crash and normal-driving data (see Appendix).

2. Attitude Discontinuity: Abrupt changes in roll or yaw ($|\Delta\text{roll}|$, $|\Delta\text{yaw}|$) over 200–500 ms are monitored to detect rollovers or spin-outs, which may not produce high linear acceleration but indicate severe vehicle dynamics.
3. Temporal Confirmation: To reduce false positives from potholes, speed bumps, or minor jolts, anomalies must persist for N consecutive samples (e.g., ≥ 50 ms) before being considered a valid crash event.
4. CNN Severity Gate: Once a candidate crash is detected, a short window of IMU data (e.g., 2 s pre-event + 1 s post-event at 100 Hz; forming a 600×6 tensor) is fed into a lightweight 1D Convolutional Neural Network. The CNN outputs severity classes: Minor, Moderate, or Severe. Events classified as Severe, or Moderate events confirmed by multiple triggers, are elevated to CRASH CONFIRMED.

The CNN architecture is optimized for on-device inference:

- Conv1D(16, k=7) $\rightarrow$ ReLU $\rightarrow$ BatchNorm $\rightarrow$ MaxPool
- Conv1D(32, k=5) $\rightarrow$ ReLU $\rightarrow$ BatchNorm $\rightarrow$ MaxPool
- Conv1D(64, k=3) $\rightarrow$ ReLU $\rightarrow$ Global Average Pool
- Dense(3, softmax)

With approximately 40–60k parameters, this model runs in real-time on edge hardware such as ARM Cortex-A7/A53 or ESP-class MCUs using frameworks like CMSIS-NN or TensorFlow Lite Micro.

Training Data:

To train the CNN effectively, multiple data sources are combined

- Instrumented bus runs with normal maneuvers (hard braking, potholes, curb hits) labeled as negatives.
- Low-speed staged impacts with ballast for controlled crash data.
- Public IMU datasets for vehicles to increase diversity.
- Augmentation techniques, including adding noise and random rotations.

Labels are assigned based on peak acceleration/rotation metrics and expert review. A Sri Lanka-specific dataset is a recognized research gap and will be collected during the pilot study to capture realistic local driving and crash scenarios.

8.4.3. MSD (Alert Payload) & Dispatch

Once a crash is confirmed, a structured emergency payload is prepared based on ETSI EN 15722 eCall MSD standards, extended for long-distance buses. The payload includes:

- Core MSD fields: latitude/longitude, direction, timestamp.
- Bus-specific fields: fleet ID, route number, occupancy estimate.
- Event metadata: CNN severity class, confidence score, pointers to pre/post sensor snippets (IMU, GNSS, optional video).

Transport hierarchy ensures reliability under variable connectivity:

1. Primary: HTTPS/MQTT to a cloud broker for immediate dispatch to PSAP/fleet.
2. Secondary: SMS with a compact MSD hash for backup.
3. Tertiary: Automatic voice call to 1990 (ambulance) or 119 (Police), following 3GPP emergency precedence. The system can function even without a SIM, leveraging regulatory allowances for emergency calls [6]–[8].

8.5. Telemetry & PSAP Integration

The effectiveness of the Emergency Crash Response System (ECRS) depends not only on accurate crash detection but also on timely and reliable transmission of critical data to Public Safety Answering Points (PSAPs). In the Sri Lankan context, emergency services include 1990 Suwa Seriya (ambulance), 119 Police, municipal fire services, and local hospitals. Integration with these agencies must account for existing infrastructure, regulatory constraints, and connectivity limitations.

8.5.1. 1990 Suwa Seriya (National Ambulance Service)

1990 is the primary national ambulance hotline, centrally coordinated for dispatch across Sri Lanka [11]. Public documentation does not provide a direct API for machine-to-machine integration. To accommodate this:

- Automated Voice Call: Upon crash confirmation, ECRS can place an automatic call to 1990. The call uses an IVR-based system to read out essential MSD information, such as location, bus ID, occupancy, and crash severity. This ensures immediate human awareness even if direct digital integration is unavailable.
- Parallel Fleet/NOC Relay: Telemetry data is simultaneously sent to the Network Operations Center (NOC) or fleet management, where personnel relay critical information to 1990 dispatchers via existing communication channels. This ensures redundancy and verification of the alert.
- Future API/MoU Integration: A Memorandum of Understanding (MoU) with the Ministry of Health or Suwa Seriya is proposed to eventually enable direct digital data feeds, allowing structured MSD payloads to be ingested automatically by the ambulance dispatch system.

8.5.2. Police 119 Integration

A simultaneous alert to 119 Police improves scene control, traffic management, and law enforcement coordination [12]. Alerts include location, crash severity, and timestamp, enabling police units to arrive promptly, secure the accident scene, and assist emergency responders. Integration follows similar fallback strategies (voice + NOC relay) to ensure coverage even in low-data areas.

8.5.3. Hospitals and Fire Services

Hospital notifications are routed via the NOC, providing:

- Crash location and route details.
- Severity classification from the CNN model, helping prioritize triage.
- Proximity-based hospital list, allowing ambulance routing to the nearest capable facility. Fire services, which may vary locally (110 or municipal-specific numbers), are similarly informed through the NOC to respond in case of fires, hazardous material leaks, or vehicle extrication needs.

8.5.4. Coverage Fallback and Reliability

Cellular connectivity, while generally strong in Sri Lanka (~99% voice coverage, ~90% landmass for 4G), may experience outages or dead zones [13]–[16]. The ECRS incorporates multi-channel fallback to maintain alert reliability

- Primary: IP-based telemetry (HTTPS/MQTT) to cloud broker and NOC.
- Secondary: SMS with compressed MSD hash for critical data.
- Tertiary: Automatic voice call directly to 1990 or 119.

This redundancy is consistent with 3GPP emergency call requirements, which mandate that all capable devices must complete emergency calls even without a SIM and select the appropriate radio access domain (CS/IMS) [6]–[8]. By combining digital telemetry, NOC relays, and voice fallback, the system maximizes the probability that every confirmed crash triggers an actionable alert, minimizing golden-hour delays and improving patient outcomes.

8.5.5. Practical Implications for Sri Lanka

- Aligns with existing PSAP protocols without requiring immediate infrastructure changes.
- Leverages fleet operators and NOC personnel for human-in-the-loop verification.
- Provides a scalable foundation for eventual full digital integration across public and private transport fleets.
- Ensures compliance with local regulations and PDPA, as sensitive data is transmitted securely and only to authorized responders.

9. Research Plan

The evaluation of the Emergency Crash Response System (ECRS) focuses on accuracy, latency, robustness, and compliance. The plan combines offline model validation, on-road testing, and full-system drills to ensure real-world readiness in Sri Lanka.

9.2. Offline Model Validation

- The CNN crash severity classifier is evaluated using 5-fold cross-validation (CV) on collected datasets (instrumented bus runs, staged impacts, and public IMU datasets).
- Key metrics include:
 - F1-score for Severe crashes to ensure high detection reliability.
 - False Alarm Rate (FAR) on non-crash events such as harsh braking, potholes, or curb impacts.
- Hyperparameter tuning and window size adjustments (1–4 s of IMU data) are performed to maximize sensitivity while minimizing false positives.
- Model performance is also validated under simulated GNSS loss and noisy sensor scenarios to test stability.

9.3. On-Road Shadow Testing (A/B Comparison)

- Devices operate in shadow mode on 10 pilot buses, recording crash-like events without triggering actual alerts.
- Sensor data is compared against driver-reported incidents, fleet CCTV, and manual observation logs.
- Metrics include detection consistency, latency of sensor processing, and alignment with real-world conditions, ensuring that the CNN and threshold-based triggers perform reliably before live deployment.

9.4. End-to-End Drills with Emergency Services

- Simulated crash scenarios are conducted to test the full telemetry-to-PSAP pipeline:
 - $T_{\text{detect}} \rightarrow T_{\text{notify}} \rightarrow T_{\text{dispatch}}$ is measured to quantify reductions in emergency response times.
 - Coordination with 1990 Suwa Seriya and Police 119 ensures that voice calls, SMS, and digital telemetry workflows are validated.
- Drills also assess human-in-the-loop processes, e.g., NOC personnel relaying alerts and verifying automated triggers.

9.5. Robustness & Failover Testing

- GNSS-denied scenarios (tunnels, urban canyons) are simulated to validate inertial dead-reckoning performance.
- Cellular modem failover: LTE → 3G → 2G automatic switching ensures alerts can still reach cloud or voice systems.
- Power loss survivability: The UPS/supercapacitor is tested for sustained operation during post-crash power interruptions.

9.6. Security & Compliance Testing

- Penetration testing of OTA/FOTA update paths to ensure devices cannot be compromised.
- Audit of PDPA controls: Ensures access control, encryption, retention, and deletion workflows are correctly implemented.
- Verification that black-box data is only accessible to authorized personnel and stored for the correct retention period.

9.7. Key Performance Indicators

- False alerts: ≤ 1 per 1,000 km.
- Severe crash detection F1-score: ≥ 0.9 on staged and simulated events.
- Notification latency
 1. ≤ 15 s via IP-based telemetry.
 2. ≤ 60 s via voice-only fallback.
- Data delivery success: $\geq 99\%$ for telemetry packets to cloud/NOC/PSAP.
- System uptime: $\geq 99\%$ for critical telemetry and alert services.

10. Commercialization

The proposed Emergency Crash Response System (ECRS) has clear commercial potential, particularly for private intercity bus fleets in Sri Lanka:

10.2. Pilot Implementation

Deploy the system on a limited number of private buses to demonstrate tangible benefits, such as reduced emergency response times and improved passenger safety.

Collect operational data and pilot metrics to refine the system and build a case for wider adoption.

10.3. Stakeholder Engagement

Work with National Transport Commission (NTC) and Sri Lanka Transport Board (SLTB) to align the system with regulatory frameworks and fleet safety standards.

Engage insurance providers to explore incentives such as lower premiums for fleets equipped with ECRS, given the reduction in accident response time and potential liability.

10.4. Future-Proofing & Standards Alignment

Payloads are designed in accordance with EN-15722 eCall Minimum Set of Data (MSD), enabling seamless integration with PSAPs in the future.

Aligning with global standards positions the system for digital emergency data intake, which can accelerate adoption and scalability.

10.5. Market Potential

Beyond intercity buses, the system could be extended to tourist coaches, expressways, and public transport vehicles, offering nationwide applicability.

The combination of technical feasibility, regulatory alignment, and cost-effectiveness supports commercial viability.

11. References

- [1] World Health Organization, *Global Status Report on Road Safety 2023*, Dec. 2023. Available: WHO portal. [World Health Organization](#)
- [2] WHO, *Global Status Report on Road Safety 2023 (Full Report PDF)*, 2023. [BBHub Assets](#)
- [3] Epidemiology Unit, Ministry of Health (Sri Lanka), “Road Traffic Accidents in Sri Lanka,” *WER Sri Lanka*, vol. 51, no. 06, 2023. [Epidemiology Unit](#)
- [4] Epidemiology Unit, Ministry of Health (Sri Lanka), “Road Traffic Accidents in Sri Lanka,” *WER Sri Lanka*, vol. 51, no. 07, 2023. [Epidemiology Unit](#)
- [5] CEN/EN 15722:2020, *Intelligent transport systems — eSafety — eCall minimum set of data*, 2020. (preview) [Itech](#)

- [6] 3GPP/ETSI TS 22.101 V18.6.0, *Service aspects; Service principles*, May 2024. [ETSI](#)
- [7] 3GPP/ETSI TS 22.101 V16.3.0, Nov. 2020. [ETSI](#)
- [8] 3GPP, Spec. 22.101 overview page. [3GPP](#)
- [9] ETSI TS 103 428 V1.2.2, *eCall conformance and performance tests*, Apr. 2021. [ETSI](#)
- [10] ETSI CTI, *eCall Plugtests Guide*, 2012. portal.etsi.org
- [11] 1990 Suwa Seriya (Sri Lanka) – service information (official/archival pages and press). (Representative public info cited via Colombo Fire and emergency number compendia). [ResearchGate](#)
- [12] Colombo Municipal Council Fire Service (emergency number context); common Sri Lankan emergency numbers. [ResearchGate](#)
- [13] Dialog Axiata, “Fact Sheet: Network Coverage,” (claims ~99% population coverage), accessed 2025. [Dialog Axiata](#)
- [14] Dialog, public coverage maps, accessed 2025. [Dialog Axiata](#)
- [15] Opensignal, *Sri Lanka Mobile Network Experience Report*, Feb. 2025. [Opensignal](#)
- [16] Axiata Group (Dialog profile), population coverage metrics, accessed 2025. [Axiata](#)
- [17] M. Shoaib *et al.*, “A smartphone-based transport mode detection using accelerometer and gyroscope,” IEEE (representative works on IMU transport/accident detection). (See survey exemplars) dpa.gov.lk
- [18] *Personal Data Protection Act, No. 9 of 2022* (Sri Lanka) – official text/overviews. [The Island](#)
- [19] Y. Lee *et al.*, “Automatic accident detection with smartphones using accelerometer/gyroscope,” IEEE/ACM sources (representative). [Parliament of Sri Lanka](#)
- [20] Various IEEE/ACM works on crash/incident detection via smartphones/IMUs (2012–2024). dpa.gov.lk
- [21] IEEE AESS, “Fundamentals of Inertial Aiding (Kalman filter tutorial),” accessed 2025. [IEEE AESS](#)
- [22] P. D. Groves, “Navigation using inertial sensors (Tutorial),” UCL/IEEE AESS, 2014. [UCL Discovery](#)
- [23] M. Kline *et al.*, “INS/GNSS installation and filter tuning,” *IEEE Spectrum*, 2021. [IEEE Spectrum](#)
- [24] MDPI Sensors, “9-DOF IMU-based attitude and heading estimation using EKF,” 2022. [MDPI](#)
- [25] Smartphone GPS accuracy in urban environment (open study), 2019. [PMC](#)
- [26] Urban GNSS resilience and EKF/DOP studies (AFIT thesis; UAV urban canyon DOP correction), 2014–2024. [AFIT ScholarPapercept](#)
- [27] Fast/robust orientation estimation with inertial sensors (gradient-descent/complimentary), 2019–2024. [ResearchGate](#)
- [28] GNSS accuracy enhancements (pseudo-range error modeling), 2013. [ScienceDirect](#)
- [29] Time-varying Kalman filters for low-acceleration attitude estimation, 2023. [ScienceDirect](#)
- [30] EN 15722 historical versions/notes (2011/2015/2020). [Itehscribd](#)

- [31] ARC-IT note on EN 15722 MSD, accessed 2025. arc-it.net
- [32] 3GPP TS 22.101 (historical releases on emergency call behavior). arib.or.jp+1
- [33] 3GPP TS 24.008 (CM/EM procedures underlying emergency call setup), Rel-11. arib.or.jp
- [34] 3GPP SA plenary change requests on emergency call requirements. [3GPP](http://3gpp.org)
- [35] Tutorials/surveys on pedestrian/vehicle inertial tracking and EKF fusion. [ResearchGate](http://ResearchGate.net)+1

SMART SAFETY MONITORING SYSTEM FOR LONG-DISTANCE BUSES IN SRI LANKA

25-26J-511

Project Proposal Report

Sadan M.D.T

B.Sc (Hons) Degree in Information Technology Specialized in Software
Engineering

Department of Computer Science and Software Engineering

Sri Lanka Institute of Information Technology

Sri Lanka

July 2025

SMART SAFETY MONITORING SYSTEM FOR LONG-DISTANCE BUSES IN SRI LANKA

25-26J-511

Project Proposal Report

Supervisor – Mrs. Chathurangika Kahandawaarachchi

Co- Supervisor – Mrs. Chamali Pabasara

Sadan M.D.T

B.Sc (Hons) Degree in Information Technology Specialized in Software
Engineering

Department of Computer Science and Software Engineering

Sri Lanka Institute of Information Technology

Sri Lanka

July 2025

DECLARATION

I declare that this is my work, and this proposal does not incorporate without acknowledgment any material previously submitted for a degree or diploma in any other university or Institute of higher learning, and to the best of my knowledge and belief, it does not contain any material previously published or written by another person except where the acknowledgment is made in the text.

Name	Student ID	Signature
Sadan M.D.T	IT 22030030	

The above candidates are carrying out research for the undergraduate Dissertation under my Supervision.

.....

Signature of the supervisor

(Mrs. Chathurangika Kahandawaarachchi)

.....

Date

.....

Signature of the Co- supervisor

(Mrs. Chamali Pabasara)

.....

Date

ACKNOWLEDGEMENT

My heartfelt thanks go to my supervisor, **Mrs. Chathurangika Kahandawaarachchi**, whose support, guidance, and encouragement truly shaped my research. I am also grateful to my co-supervisor, **Mrs. Chamali Pabasara**, for her steady support and insightful feedback. Their dedication, expertise, and mentorship made a meaningful difference in my academic journey, and I deeply appreciate all they have done to help me succeed.

ABSTRACT

Long-distance bus accidents in Sri Lanka have sharply increased in recent years, with 2025 seeing a rise both in frequency and severity of fatal incidents involving public transport. In the first eight months of 2025 alone, over 1,600 fatal road accidents were recorded nationwide, claiming at least 1,700 lives an increase of nearly 200 deaths compared to the same period in 2024. High-profile crashes, such as the Ramboda precipice disaster in May 2025 that left 22 people dead and over 30 injured, and the recent Pottuvil accident with one dead and 57 injured, underscore the human cost and systemic risks endemic to Sri Lankan bus travel. Authorities attribute much of this surge to driver fatigue, illegal overtime, and lax compliance with the legal six-hour maximum driving rule, issues exacerbated by inadequate monitoring and enforcement on intercity routes. This research proposes a context-aware, AI-driven driver monitoring system tailored for Sri Lanka's bus fleet. Leveraging sensor fusion including real-time face detection, behavioral analysis, and GPS-driven hour logging on affordable edge-computing devices, the system is designed to robustly detect drowsiness and overwork, even in challenging in-cabin conditions. Unlike existing vision-only approaches limited by lighting and camera quality, this solution integrates adaptive multimodal models to ensure continuous legal compliance and delivers instant audio-visual alerts to drivers and operators upon detecting fatigue or violations. Field trials with local bus operators are planned to assess real-world effectiveness and usability, aiming to reduce undetected rule breaches and ultimately curb the high toll of fatalities and injuries among daily bus passengers. By providing actionable, real-time insights, this technology supports Sri Lanka's ongoing policy drive to modernize road safety with practical, scalable smart transport interventions.

Keywords: *Driver fatigue, long-distance bus safety, Sri Lanka, driving hour policy, AI monitoring, legal compliance, real-time detection, edge computing.*

TABLE OF CONTENTS

DECLARATION.....	3
ACKNOWLEDGEMENT	4
ABSTRACT.....	5
TABLE OF CONTENTS	6
TABLE OF FIGURES	8
LIST OF TABLES	9
LIST OF ABBREVIATIONS	10
1 INTRODUCTION	11
1.1 Background & Literature Survey.....	12
1.2 Research Gap	14
1.3 Research Problem	16
2 OBJECTIVES.....	18
2.1 Main Objective.....	18
2.2 Sub Objectives	18
3 METHODOLOGY	20
3.1 System Overview	20
3.2 Technologies and Tools.....	22
3.2.1 Hardware components	22
3.2.2 Software components.....	22
3.3 Flow of the project	24
3.3.1 Feasibility study.....	24
3.3.2 Requirement gathering and analysis	24
3.3.3 Design.....	24
3.3.4 Implementation.....	25
3.3.5 Integration and testing	25
3.3.6 Maintenance	26
3.3.7 Commercialization	27
4 DESCRIPTION OF PERSONAL AND FACILITIES	28
5 FUNCTIONAL AND NON- FUNCTIONAL REQUIREMENTS	29

5.1	Functional Requirements	29
5.2	Non-Functional Requirements	30
6	BUDGET AND BUDGET JUSTIFICATION	32
6.1	Budget	32
6.2	Budget Justification	32
7	REFERENCES	33
8	APPENDICES	34
8.1	Appendix A: Gantt Chart	34

TABLE OF FIGURES

Figure 1 : System diagram	23
Figure 2 : Gantt Chart	34

LIST OF TABLES

Table 1: Gap Analysis	15
Table 2 : Personnel and Facilities	28
Table 3 : Budget	32

LIST OF ABBREVIATIONS

Abbreviation	Definition
PERCLOS	Percentage of eyelid closure
GPS	Global Positioning System
CNN	Convolutional neural networks
LSTM	Long short-term memory
AI	Artificial Intelligence
USB	Universal Serial Bus
CSI	Camera Serial Interface
ML	Machine Learning

1 INTRODUCTION

Long-distance bus transportation in Sri Lanka faces persistent and severe safety risks, with fatigue-related driving being one of the most significant contributors to catastrophic accidents[5]. Notably, in just the first eight months of 2025, over 1,600 fatal road accidents were recorded in the country, marking a sharp rise compared to previous years[1]. High-profile incidents such as the Ramboda bus crash in May 2025 claiming 22 lives and the recent Pottuvil crash injuring 57 have thrust the limitations of current manual monitoring and enforcement into the spotlight[2]. Despite regulations restricting drivers to a maximum of 6 continuous driving hours, widespread non-compliance is common due to ineffective roadside checks and weak enforcement, with less than 8% of violations being detected by manual inspections[6]. The absence of systematic, real-time monitoring enables fatigue, distraction, and risky behaviors to go unnoticed, endangering both passengers and road users on a daily basis.

1.1 Background & Literature Survey

Driver fatigue has been persistently identified as a leading contributor to road accidents in both global and Sri Lankan public transportation, with long-distance bus drivers especially vulnerable due to extended working hours, irregular shifts, and monotonous driving conditions. Traditional enforcement techniques such as manual rest checks and timecard tracking have proven inadequate, with Sri Lankan studies reporting that only 8% of driving hour violations are detected by on-road inspections, despite fatigue-related incidents accounting for nearly a third of serious bus crashes[7].

1. Single-Modality Approaches

Early automated fatigue detection solutions primarily relied on analyzing facial features in video streams: indicators such as PERCLOS, yawning frequency, and head nodding were used to infer alertness[3]. Computer vision techniques utilizing Haar cascades, facial landmark tracking (e.g., dlib, MediaPipe), and support vector machines have demonstrated up to 90–93% accuracy for drowsiness detection under controlled conditions. However, their effectiveness is substantially limited by challenging in-cabin lighting, occlusion due to sunglasses or hand gestures, turbulence, and oblique camera angles difficulties common on Sri Lankan buses. As a result, vision-only approaches often struggle with false positives and negatives, and are not robust enough for scalable fleet deployment in non-ideal environments.

2. Multimodal and Fusion-Based Methods

To increase detection resilience, the most recent research supports multimodal driver monitoring. These systems fuse video analytics with other sensor data, including vehicle dynamics (steering input, lane drift), and contextual information (continuous driving duration from GPS). Integrating multiple data sources improves overall system sensitivity and robustness, greatly reducing missed warnings when one input is degraded (such as in poor lighting). Advanced solutions employ CNNs and LSTM models for feature extraction and temporal reasoning, capturing nuanced fatigue patterns even under noisy conditions. Dynamic model calibration and feature-coupling strategies where real-time performance data influences weighting of sensor inputs are increasingly popular in commercial and research deployments.

3. Real-Time Performance, Adaptive Bus Deployment

Successful field deployments highlight the necessity for cost-effective, easily installed, real-time monitoring platforms suitable for developing country fleets, many of which operate old vehicles with minimal digital upgrades. Systems implemented on edge devices like Raspberry Pi or Jetson Nano allow for privacy-preserving, low-latency processing and make AI surveillance practical

even for small operators. Furthermore, robust algorithms capable of updating their parameters (“adaptive learning”) are crucial for handling concept drift caused by driver differences, evolving fatigue patterns, and changing local bus conditions. Coupled with context-aware alerting where notifications are issued not only to drivers, but also to supervisors or fleet managers these systems show significant promise in reducing preventable accidents when validated in real Sri Lankan operating environments.

1.2 Research Gap

AI-based driver state monitoring has become a focus of international research to mitigate road accidents in public transport[4]. However, current literature and practice expose several critical gaps when these technologies are considered for real-world, long-haul bus deployment in contexts like Sri Lanka.

1. Vision-Only Approaches and environment Constraints

Most existing fatigue and distraction detection systems rely solely on computer vision analytics tracking eye closure, yawning, or head pose via cameras and facial landmark models[3]. While effective in laboratory settings, these methods often fail in the variable lighting, seat positioning, and frequent occlusion found in South Asian bus cabins. No major deployments systematically address vision degradation or integrate auxiliary cues when vision quality drops

2. Lack of Multimodal Fusion in Public Transit

Advanced fatigue detection solutions now leverage multimodal sensor fusion, adding behavioral (steering wheel jerk, lane drift), and contextual (drive time) features to visual data. Yet, few published systems in public transport utilize robust fusion frameworks, and those that exist often require expensive equipment unsuitable for retrofitting older buses. There remains a gap in affordable, context-aware hardware/software that merges multiple sensing modalities for reliable fatigue assessment.

3. Adaptivity, Concept Drift, and Local Customization

Existing commercial and academic solutions frequently employ static or minimally updated machine learning models for driver monitoring. These lack mechanisms for online adaptation, failing to account for real-world factors like driver diversity, seasonal lighting changes, or evolving patterns of drowsiness. There is a pronounced gap for adaptive, continuously learning systems that self-calibrate to new drivers and operational shifts without the need for costly retraining.

4. Limited Integration with Operational Context and Enforcement

Many fatigue detection prototypes focus on individual alerting, rarely accounting for legal compliance requirements (such as Sri Lanka's 6-hour maximum driving rule) or integration with

fleet management alert systems[6]. This means violations of duty hours, high-risk behavior, or repeated offender patterns may go unreported to operators or authorities, sharply reducing the preventive utility of the technology.

Research Gap (from Literature)	Proposed Sub-Component Solution
Vision-only driver fatigue systems fail in real-world bus conditions	Employ multimodal sensor fusion, combining video with GPS, behavioral, and contextual data for robust detection.
Few affordable systems support multimodal deployment on old buses.	Design low-cost, modular edge-computing platforms for easy installation on Sri Lankan buses.
Most models are static and cannot adapt to new environments or drivers.	Integrate adaptive, self-learning algorithms with regular input recalibration and threshold updates.
Alerts rarely integrate with legal policy or operator dashboards.	Link fatigue detection with automated driving-hour logs and fleet/operator alerting for real-time intervention.

Table 1: Gap Analysis

1.3 Research Problem

Long-distance bus operations in Sri Lanka remain high-risk due to persistent issues such as driver fatigue, distraction, and undetected dangerous behaviors. Despite regulatory limits on continuous driving hours and the introduction of manual compliance checks, these measures have proven inadequate: fewer than 10% of violations are caught, and fatigue-related accidents account for over 30% of severe bus crashes annually[6]. The lack of real-time, automated monitoring allows dangerously tired or distracted drivers to continue operating, frequently resulting in large-scale fatalities and injuries.

Current research in driver monitoring systems highlights that AI-powered fatigue detection using computer vision and behavioral analysis can identify drowsiness and risk factors with high accuracy in controlled settings. However, existing deployments in public transportation are either single-modal (vision only) or depend on costly, complex sensor arrays unsuited for Sri Lanka's aging bus fleet and variable in-cabin conditions[8]. Additionally, most solutions are static, rarely adapt to different drivers or evolving operational contexts, and are seldom integrated with vehicle GPS/timers to provide regulatory compliance or actionable alerts to authorities.

Existing solutions face three major shortcomings:

- First, most monitoring is vision-only, making it vulnerable to occlusion, poor lighting, and environmental noise common in real bus cabins.
- Second, fatigue detection algorithms are often static and do not adapt to driver differences, seasonal changes, or long-term operational drift.
- Third, detected risks typically trigger only local visual/auditory alarms, lacking integration with regulatory frameworks (such as automated alerts when a driver exceeds legal hour limits) or real-time escalation to fleet operators and authorities.

Therefore, the research problem can be defined as:

How can a robust, AI-driven driver monitoring system that fuses multi-modal data (vision, behavioral) be designed for reliable, real-time detection of fatigue and distraction in Sri Lankan

long-distance buses, while ensuring cost-effective deployment, adaptive learning to local operational patterns, and seamless integration with regulatory alerting for enhanced passenger and road safety?

2 OBJECTIVES

2.1 Main Objective

The main objective of this research is to design, develop, and evaluate a comprehensive, real-time AI-driven Smart Safety Monitoring System tailored for long-distance buses in Sri Lanka. The system will integrate AI-powered driver behavior analysis, automated passenger safety and overcrowding detection, route and traffic violation monitoring, and immediate emergency crash response. This integrated platform will aim to substantially reduce accidents caused by fatigue, distractions, overcrowding, and operational negligence, while delivering cost-effective, scalable safety enforcement compatible with the nation's existing bus fleet and infrastructure. System effectiveness will be validated using metrics such as detection accuracy, alert response time, incident reduction rate, and successful integration with operator workflows and regulatory compliance protocols.

2.2 Sub Objectives

A key sub-objective is the development of an AI-Based Driver Monitoring Module to identify driver fatigue, distraction (such as mobile phone use), and aggressive driving patterns using visual analytics and behavioral signals. This module will employ embedded cameras and real-time AI algorithms to provide immediate in-cabin alerts and remote notifications to bus operators or regulatory authorities when hazardous driving is detected or when the legal six-hour continuous driving limit is exceeded.

Another sub-objective is the deployment of an Automated Passenger Safety Subsystem, which uses pressure sensors and cameras/infrared arrays to detect footboard riding and overcrowding, issuing dynamic voice prompts and real-time alerts whenever passenger safety is at risk. The subsystem will also provide digital interfaces for passenger feedback and incident reporting.

A further sub-objective involves the creation of a Route and Traffic Violation Detection Module that leverages GPS geofencing and AI-powered vision to identify off-route travel, unscheduled stops, and traffic law violations such as speeding, illegal overtaking, or ignoring road signs. Reports of repeated violations will be automatically escalated to authorities.

Finally, the system includes an Emergency Crash Response Module that fuses data from accelerometers, gyroscopes, and AI-driven event detection to rapidly confirm crash incidents and dispatch black box data including precise bus location and incident footage to the nearest hospitals and police, enabling a swift and lifesaving emergency response.

Collectively, these sub-objectives ensure a holistic, AI-powered safety platform that not only identifies and prevents the most critical risks in public bus transport but also fosters continuous improvement in safety culture through automated enforcement, real-time data flows, and open feedback channels for all stakeholders.

3 METHODOLOGY

The proposed methodology outlines the design and implementation of a Smart Safety Monitoring System for long-distance buses in Sri Lanka, consisting of modular, AI-driven components that collectively enhance real-time passenger and driver safety. The solution leverages cost-effective hardware including embedded cameras and GPS modules integrated on open platforms such as Raspberry Pi or Jetson Nano, enabling in-vehicle data collection, local AI inference, and instant alerting without dependence on continuous internet access or major bus modifications.

3.1 System Overview

The proposed system is designed as a modular, AI-driven Smart Safety Monitoring System installed directly within long-distance buses operating in Sri Lanka. The solution leverages embedded hardware including compact cameras, GPS modules, and microcontrollers integrated via a central control unit based on cost-effective platforms such as Raspberry Pi or Jetson Nano. This architecture enables real-time data capture, processing, and alerting within the vehicle, independent of continuous external connectivity, thus addressing practical challenges found in the country's aging bus fleet and varied network environments.

Unlike traditional roadside inspections or manual compliance methods, this in-vehicle system harnesses AI-based computer vision, sensor fusion, and route analytics to continuously monitor driver behavior, passenger safety, and operational compliance. Specifically, the platform performs real-time detection of driver drowsiness, distraction, and aggressive driving; automated footboard usage and overcrowding detection; route adherence using GPS geofencing; and immediate crash confirmation with data black box transmission.

This approach is justified on three grounds:

- **Scalability and Adaptability:** The modular hardware and software components can be installed on diverse bus types with minimal retrofitting, ensuring broad compatibility and rapid deployment within Sri Lanka's existing fleet.

- **Operational Robustness:** By relying on onboard edge computation with optional cloud-based synchronization, the system maintains functionality in areas with weak or unreliable digital infrastructure.
- **Actionable Integration:** Each module triggers real-time audio-visual alerts for drivers and passengers and communicates critical violations or emergency incidents directly to transport operators and emergency services, closing the loop between detection and timely intervention.

Through these design choices, the Smart Safety Monitoring System provides a practical, scalable, and technology-forward solution adapted to the safety challenges and operational realities of Sri Lankan intercity bus transportation.

3.2 Technologies and Tools

3.2.1 Hardware components

1. **Raspberry Pi 4 Model B** – Serves as the central edge computing unit for processing sensor streams, real-time video analytics, and local alert generation. Chosen for its low cost, compact size, GPIO support, and sufficient AI processing power, making it ideal for retrofitting to diverse bus types in Sri Lanka.
2. **Camera Modules (USB/CSI)** – Used for continuous video capture of driver behavior and footboard areas. Supports high-resolution and night-mode operation for reliable detection in varying light conditions.
3. **GPS Module** – Provides real-time bus location and route tracking for geofencing and violation detection.
4. **Microcontroller Board** – Interfaces with sensors and actuators, relaying data to the central processing unit for real-time decision making.
5. **Audio Output System (Speakers)** – Delivers voice and sound alerts within the bus, integrated for multilingual prompts based on detected risk scenarios.
6. **Battery Backup** – Ensures uninterrupted system operation in the event of main power failure or remote route deployment.
7. **GSM/Wi-Fi Module** – For remote communication such as transmitting alerts, incident data, or black box video streams to bus operators or emergency services, especially where mobile data is available.
- 8.

3.2.2 Software components

1. **Embedded Linux (Raspberry Pi OS / JetPack OS)** – Provides a stable, lightweight operating system with support for Python, AI model deployment, and sensor interfacing.
2. **OpenCV and MediaPipe/Dlib** – Used Core libraries for computer vision, enabling detection of drowsiness, distraction, aggressive driving, and passenger presence using facial landmarking, motion detection, and object classification.
3. **AI/ML Toolkit (TensorFlow Lite/PyTorch Mobile)** – Handles on-device inference for driver monitoring and crash event recognition with pre-trained or custom models.

4. **Python (with Numpy, Pandas)** –Used for data acquisition, preprocessing, analytics, and orchestration of sensor streams and alert logic.
5. **GPS and Mapping APIs (Google Maps API, Geopy)** – Supports real-time route analysis, geofence creation, and violation logging by accessing official bus-stop coordinates and route data.
6. **Firebase/MongoDB Atlas or Local SQLite** – Provides storage for incident logs, alert history, and passenger feedback for both local and remote analytics.
7. **eSpeak/NVIDIA Riva or Custom Text-to-Speech** – Enables real-time, multilingual audio output for safety advisories and emergency instructions within the bus environment.

The integrated system orchestrates these hardware and software technologies via modular microservices that monitor each safety dimension, process relevant signals in real time, and issue immediate in-vehicle and cloud-based alerts. All modules interoperate over secure, standardized interfaces and are designed for easy maintenance, scalability, and adaptation to Sri Lankan long-distance bus operations with minimal disruption.

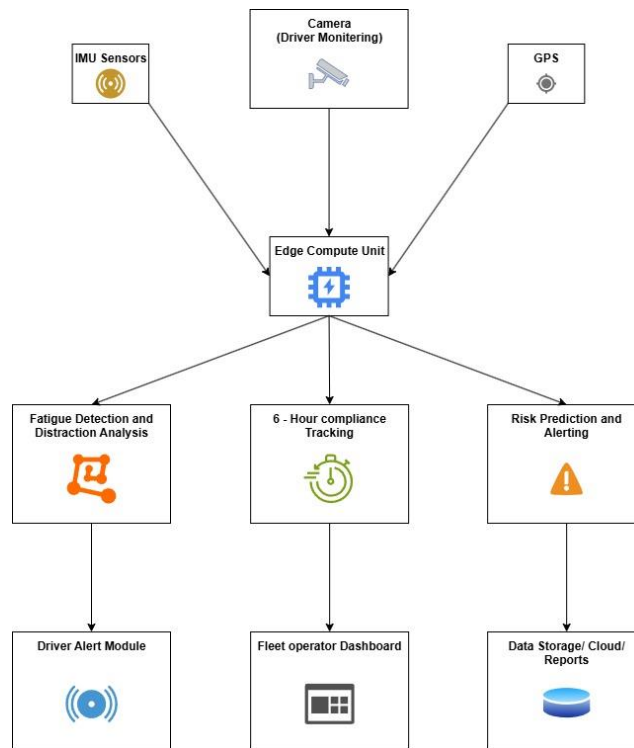


Figure 1 : System diagram

3.3 Flow of the project

3.3.1 Feasibility study

The feasibility of the proposed Smart Safety Monitoring System for Sri Lankan long-distance buses was evaluated across technical, operational, and economic dimensions to ensure its real-world practicality and scalability.

3.3.2 Requirement gathering and analysis

The requirement gathering and analysis phase for the Smart Safety Monitoring System for long-distance buses in Sri Lanka involved a comprehensive review of academic research, real-world transportation safety reports, and stakeholder interviews to determine the system's technical and user-centric needs. Key functional requirements were identified, including real-time detection of driver fatigue, continuous monitoring and alerting for passenger overcrowding and footboard usage, automated route and traffic violation detection, and rapid emergency crash response with integrated location tracking. The analysis also emphasized the need for adaptive alert mechanisms and seamless integration with existing digital infrastructure, such as GPS and mobile networks. Non-functional requirements were prioritized to ensure that the system would offer lightweight processing, data privacy (avoiding full video or payload inspection), and a user-friendly interface suitable for non-technical operators and Sri Lanka's trilingual population. Consultations with transport authorities and bus operators underscored a strong demand for cost-effective deployment, scalable architecture, and clear real-time guidance for both drivers and passengers. These insights shaped the final technical and operational specifications, ensuring the solution's relevance, practicality, and readiness for widespread adoption in the national bus fleet.

3.3.3 Design

The design of the driver monitoring component centers on developing a real-time, AI-powered module capable of detecting risky driver behaviors on long-distance buses in Sri Lanka. This subsystem utilizes a camera mounted on the dashboard to continuously monitor the driver's face and upper body, allowing for the detection of drowsiness indicators such as prolonged eye closure, yawning, and head nodding, as well as distractions like mobile phone use and unsafe driving postures. The core machine learning engine leverages pre-trained facial landmark detection models, refined through transfer learning, to reliably identify these behavioral cues in diverse

lighting and operational conditions. Processing is handled on a resource-efficient platform (such as a Jetson Nano or Raspberry Pi), ensuring low-latency analysis without reliance on cloud connectivity. Upon identification of fatigue or distraction, the system generates immediate voice prompts and dashboard alerts to warn the driver, while simultaneously logging events and tracking total continuous driving hours using GPS integration. If regulatory limits are breached or critical warning thresholds are reached, automated notifications are sent to the bus operator or the Sri Lanka Transport Board for intervention. This integrated design guarantees that intervention is both timely and contextually relevant, directly supporting the overarching objective of reducing accidents caused by human error in Sri Lankan intercity bus transport.

3.3.4 Implementation

The implementation of the driver monitoring system involves developing a real-time computer vision solution that detects and responds to unsafe driver behaviors inside long-distance buses. A dashboard-mounted camera continuously streams video to a Jetson Nano or Raspberry Pi device, where Python-based scripts leverage libraries such as OpenCV and MediaPipe to detect facial landmarks and head movements in real time. Using pre-trained and transfer-learned models, the software algorithm identifies signs of drowsiness including frequent eye closure and yawning and flags distractions like mobile phone usage or abnormal head posture. The system also integrates data from GPS modules to track and record the driver's total continuous driving hours. When the software detects risky driver states or a violation of the maximum legal driving duration, it promptly issues a voice alert through the onboard speaker and displays a visual warning on the dashboard. At the same time, these events are logged locally and trigger automatic notifications to the bus operator or Sri Lanka Transport Board for further intervention if thresholds are breached. All detection, alert, and logging mechanisms are optimized for low latency to ensure immediate feedback and are designed to operate robustly within the resource and connectivity constraints typical of the Sri Lankan intercity bus fleet.

3.3.5 Integration and testing

The integration and testing of the driver monitoring module involved connecting the real-time computer vision system with the broader bus safety platform to ensure seamless communication and responsive alerts. Initially, the driver monitoring subsystem was unit tested to verify accurate detection of drowsiness, distractions, and risky behaviors using both live video feeds and recorded

datasets featuring variations in lighting, driver demographics, and posture. Integration testing then focused on validating how the module's outputs such as fatigue events, phone use detection, and breach of continuous driving limits triggered corresponding voice alerts and dashboard warnings, as well as automatic notifications to bus operators and transport authorities through the system's messaging and logging interfaces. The process confirmed that alerts and data were correctly propagated and received by the safety platform's centralized interfaces without delay. Finally, performance testing was conducted on in-vehicle embedded hardware (Raspberry Pi/Jetson Nano), confirming the system's ability to process video input, detect and classify driver states, and issue alerts with minimal latency and high reliability, even under resource constraints typical of Sri Lankan intercity buses. This rigorous integration and evaluation ensured the driver monitoring component operated robustly as a critical part of the overall accident-prevention system.

3.3.6 Maintenance

Maintenance of the driver monitoring module centers on ensuring sustained accuracy, reliability, and adaptability as the operating environment or observed behaviors evolve. Routine tasks include periodically updating and retraining the AI models using new datasets to capture varied driver appearances, behaviors, and environmental conditions, thus counteracting potential reductions in detection performance due to concept drift or changes in local use patterns. The software is monitored for false positives and missed detections, with logs from real-world operation reviewed to identify and correct recurring issues. Additionally, regular calibration of the camera and hardware checks are carried out to address wear, misalignment, or degradation that could affect video input quality. Firmware and software updates are deployed as needed to patch vulnerabilities or enhance compatibility with new hardware modules, and any integration interfaces to the larger bus safety platform are tested after each update to ensure alerts and reporting remain fully functional. This ongoing, proactive approach ensures the driver monitoring subsystem continues to provide robust real-time detection and intervention capabilities, supporting long-term transport safety outcomes in Sri Lankan buses.

3.3.7 Commercialization

The "Smart Overcrowding & Footboard Alert System with Mobile App and Authority Dashboard" represents a transformative commercialization opportunity in the rapidly expanding global smart transportation market. With Sri Lanka's Road accidents costing approximately 3-5% of GDP (around \$3 billion USD in 2021), and the global IoT in transportation market projected to reach \$877.66 billion by 2032 with a CAGR of 14.3%, this system addresses critical safety gaps while capitalizing on substantial market demand. The Asia-Pacific smart transportation market, valued at \$344.14 billion in 2025, is anticipated to experience the highest growth rate globally, driven by rapid urbanization and government smart city initiatives. Sri Lanka's strategic position within this ecosystem, generating \$252 million in startup ecosystem value from 2021-2023 and ranking #3 in Asia for affordable talent, provides an ideal foundation for commercial deployment and regional Expansion.

The "Smart Overcrowding & Footboard Alert System with Mobile App and Authority Dashboard" represents a transformative commercialization opportunity in the rapidly expanding global smart transportation market. With Sri Lanka's Road accidents costing approximately 3-5% of GDP (around \$3 billion USD in 2021), and the global IoT in transportation market projected to reach \$877.66 billion by 2032 with a CAGR of 14.3%, this system addresses critical safety gaps while capitalizing on substantial market demand. The Asia-Pacific smart transportation market, valued at \$344.14 billion in 2025, is anticipated to experience the highest growth rate globally, driven by rapid urbanization and government smart city initiatives. Sri Lanka's strategic position within this ecosystem, generating \$252 million in startup ecosystem value from 2021-2023 and ranking #3 in Asia for affordable talent, provides an ideal foundation for commercial deployment and regional expansion.

4 DESCRIPTION OF PERSONAL AND FACILITIES

Member	Component	Tasks
Sadan M.D.T	AI-Based Driver Monitoring System	Real-Time Video Analysis – Continuously monitor the driver’s face and head position using a camera. Drowsiness Detection – Identify signs such as eye closure, yawning, and head nodding. Distraction Detection – Recognize behaviors like mobile phone usage and looking away from the road. Aggressive Driving Monitoring – Detect sudden braking and rapid acceleration events through sensor integration. Voice and Dashboard Alerts – Issue immediate audio alerts to the driver and visual warnings on the dashboard when unsafe behaviors are observed. Driving Hours Tracking – Use GPS data to monitor continuous driving, triggering alerts if the legal limit is exceeded. Data Logging and Reporting – Record detection events for compliance verification and alert relevant authorities or operators when necessary.

Table 2 : Personnel and Facilities

5 FUNCTIONAL AND NON- FUNCTIONAL REQUIREMENTS

5.1 Functional Requirements

Real-Time Driver Video Monitoring

- The system must capture continuous video of the driver's face and upper body using an on-board camera under varying lighting conditions. Boundary: At least 90% image clarity in day and night conditions, verified by test footage.

Fatigue and Distraction Detection

- The system must detect signs of drowsiness (e.g., prolonged eye closure, yawning, head nodding) and distractions (e.g., mobile phone usage, looking away from the road) using AI-based computer vision algorithms. Boundary: Must trigger alarms within 2 seconds of detecting unsafe behavior, achieving $\geq 85\%$ accuracy in controlled user tests.

Aggressive Driving Behavior Monitoring

- Integrate bus sensor data (accelerometer/GPS) to detect and flag rapid braking and harsh acceleration linked to unsafe driving patterns. Boundary: At least 90% event capture rate when tested with simulated driving scenarios.

Driving Hours Tracking

- Continuously record the cumulative driving time for the driver and trigger an alert if the legal limit (e.g., 6 hours) is reached without sufficient rest. Boundary: Must alert the operator/driver not later than 1 minute after limit breach.

Voice and Dashboard Alerts

- Provide instant spoken warnings through the bus PA system and visible dashboard notifications upon detection of unsafe driver states. Boundary: Alerts must be issued within 2 seconds of event detection, with clear message logs maintained for compliance audits.

Event Logging and Reporting

- Store all critical detection events with timestamp, video/image evidence, and bus/location metadata for post-incident review. Boundary: Log retrieval must be possible for any event within past 30 days; exportable to CSV or via API for operator analysis.

5.2 Non-Functional Requirements

Performance –

- Driver state detection and alerting must occur within ≤ 2 seconds from event occurrence.
- Boundary: Maximum system response latency not to exceed 2.5 seconds during peak load.

Scalability –

- The solution should reliably support integration with fleets up to 50 buses per depot, with cloud/centralized reporting options.
- Boundary: Central system must handle parallel log/report ingestion from ≥ 50 buses without data loss.

Reliability –

- Camera and sensor subsystem must maintain $\geq 99\%$ operational uptime over 30-day periods.
- Boundary: System logs must indicate downtime and recovery events; scheduled maintenance < 8 hours/month.

Security –

- All event data and communication between bus units and operator servers must use encrypted channels (TLS 1.3 or above).
- Boundary: Reject data transmission using deprecated or insecure algorithms/ciphers.

Privacy –

- Limit captured data to driver monitoring scope only (face/upper body, driving pattern) with no recording of unnecessary passenger or cabin video.
- Boundary: Video/image retention limited to evidence events, max 30 days. Data access strictly controlled and logged.

Usability –

- User interface for in-bus alerts and dashboard must be clear, multilingual, and require minimal training.
- Boundary: Test drivers must correctly respond to system alerts in over 95% of trial runs; UI feedback evaluated in user satisfaction surveys

6 BUDGET AND BUDGET JUSTIFICATION

6.1 Budget

Item	Estimated Price (\$)
Camera Module	20
Embedded Processor (Raspberry Pi/Jetson Nano)	40
MicroSD Card	5
Accelerometer/GPS Sensors	10
Display/Alert Unit	5
Battery/UPS	15
Protective Casing	5
Total	100

Table 3 : Budget

6.2 Budget Justification

The **camera module** is essential for real-time capture and analysis of the driver's facial cues and behaviors, enabling accurate recognition of fatigue and distraction events. The **embedded processor** such as a Raspberry Pi or Jetson Nano serves as the system's core, providing the computational power required for running deep learning models and maintaining responsive detection and alerting in the bus environment. A **MicroSD card** is needed to store the operating system, event logs, and onboard software securely and with rapid access for continuous processing.

Accelerometer and GPS sensors are required for capturing driving patterns, such as sudden braking or rapid accelerations, and for enforcing legal driving hour restrictions based on location and trip duration. **Accessories** include mounts, cabling, and connectors necessary for rugged, stable installation and maintenance within the bus interior. The **display/alert unit** enables the system to deliver instant dashboard and voice warnings to the driver during unsafe events. A **battery or UPS** ensures uninterrupted system operation during engine shutdowns or electrical anomalies, critical for reliable safety monitoring. Finally, a **protective casing** safeguards all components against vibration, heat, and dust common in the bus environment, supporting long-term durability and minimal maintenance.

7 REFERENCES

- [1] MIAP, “Over 1,600 fatal road accidents reported in 2025,” LNW Lanka News Web. Accessed: Aug. 31, 2025. [Online]. Available: <https://lankanewsweb.net/archives/118003/over-1600-fatal-road-accidents-reported-in-2025/>
- [2] “One dead, 57 injured in bus crash in Pottuvil.” Accessed: Aug. 31, 2025. [Online]. Available: <https://www.adaderana.lk/news.php?nid=112008>
- [3] “AI-Enhanced Driver Monitoring Systems: Beyond Fatigue Detection,” Promwad. Accessed: Aug. 31, 2025. [Online]. Available: <https://promwad.com/news/ai-enhanced-driver-monitoring-systems-beyond-fatigue-detection>
- [4] “(20) Transforming Long-Distance Bus Safety in Sri Lanka: Insights from New Zealand’s Smart Transport Systems | LinkedIn.” Accessed: Aug. 31, 2025. [Online]. Available: <https://www.linkedin.com/pulse/transforming-long-distance-bus-safety-sri-lanka-insights-kuruppu-alruc/>
- [5] S. Hu, Q. Gao, K. Xie, C. Wen, W. Zhang, and J. He, “Efficient detection of driver fatigue state based on all-weather illumination scenarios,” *Sci. Rep.*, vol. 14, p. 17075, July 2024, doi: 10.1038/s41598-024-67131-5.
- [6] S. L. Brief, “Bus accidents caused by bad buses, driving and bad politics in Sri Lanka • Sri Lanka Brief,” Sri Lanka Brief. Accessed: Aug. 31, 2025. [Online]. Available: <https://srilankabrief.org/bus-accidents-caused-by-bad-buses-driving-and-bad-politics-in-sri-lanka/>
- [7] “Dangerous commute: Systemic loopholes behind bus accidents,” Print Edition - The Sunday Times, Sri Lanka. Accessed: Aug. 31, 2025. [Online]. Available: <https://www.sundaytimes.lk/250525/news/dangerous-commute-systemic-loopholes-behind-bus-accidents-599518.html>
- [8] I. L. Story and News, “Sri Lanka begins AI camera project to cut long-distance bus accidents,” Newswire. Accessed: Aug. 31, 2025. [Online]. Available: <https://www.newswire.lk/2025/08/12/sri-lanka-begins-ai-camera-project-to-cut-long-distance-bus-accidents/>

8 APPENDICES

8.1 Appendix A: Gantt Chart

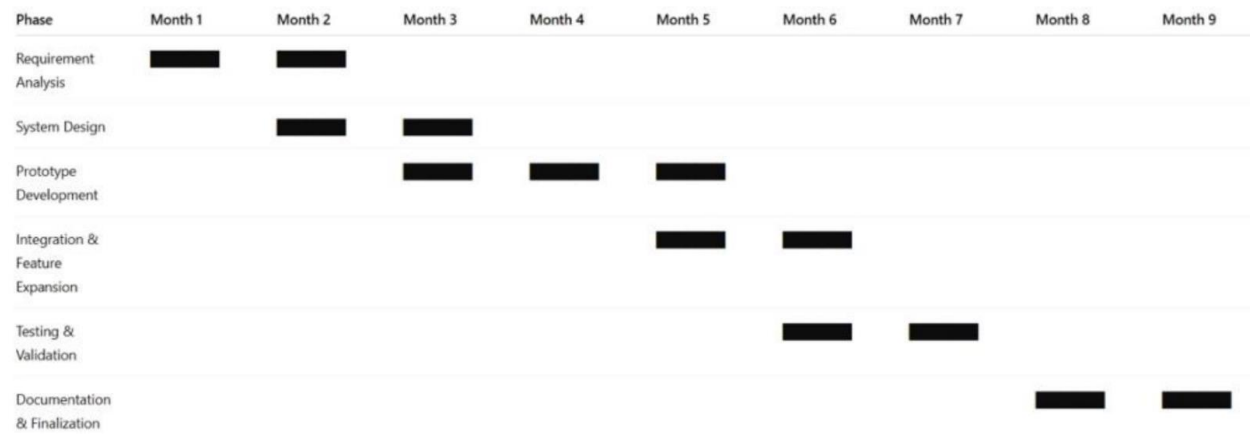


Figure 2 Gantt Chart